HOLDER POLICY OPTIMISATION

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ABSTRACT

Group Relative Policy Optimisation (GRPO) enhances large language models by estimating advantages across a group of sampled trajectories. However, mapping these trajectory-level advantages to policy updates requires aggregating token-level importance ratios within each sequence. A fixed aggregation mechanism limits the algorithm's adaptability across tasks and training stages. We propose **HolderPO**, a generalised policy optimisation framework unifying token-level ratio aggregation via the Holder mean. By modulating its order p, our framework provides continuous control over gradient concentration and variance bounds. Theoretically, positive orders concentrate gradient weight toward high-ratio tokens, whereas decreasing p monotonically tightens an upper bound on the policy-gradient variance under the stated assumptions. These complementary effects motivate a dynamic annealing algorithm that progressively decreases p during training. Empirical signal-density diagnostics characterise the availability of successful rollout feedback on AIME24 and MATH500, complementing their different static-p preferences. Evaluations compare static and dynamic orders across mathematical and agentic benchmarks. Our approach achieves an average accuracy of 55.2% across five mathematical benchmarks on Qwen2.5-Math-7B and a 95.3% success rate on ALFWorld.

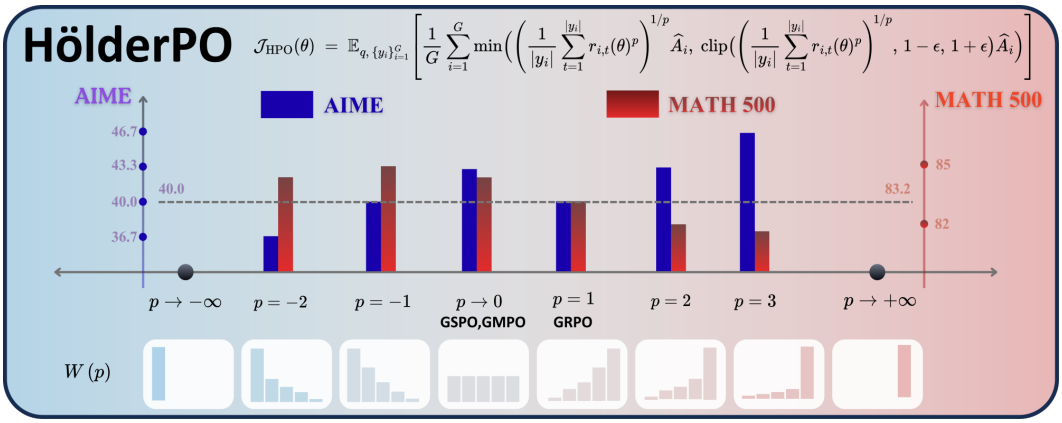


Figure 1: **HolderPO unifies token-level aggregation under a single parameter p.** The framework generalises arithmetic and geometric aggregation of token-level importance ratios through the Holder mean of order $p \in \mathbb{R}$. Under the corresponding clipping and normalisation conventions, HolderPO exactly recovers GRPO at $p = 1$ and GSPO at $p = 0$. The bar chart illustrates the static-p comparison on AIME24 (blue) and MATH500 (red), with dashed lines marking GRPO baselines. Bottom: the token weight distribution $W(p)$, with tokens ordered from small (left) to large (right) importance ratio.

1 INTRODUCTION

Reinforcement Learning (RL) has emerged as a key technique for advancing the alignment and complex reasoning capabilities of Large Language Models (LLMs) (Ouyang et al., 2022; Schulman et al., 2017). Recently, Group Relative Policy Optimisation (GRPO) has emerged as a highly effective and compute-efficient algorithm, largely driving the success of reasoning models like

DeepSeek-R1 (Shao et al., 2024). GRPO operates by estimating advantages across a group of sampled trajectories, substantially reducing training overhead by eliminating the need for an external critic model. However, mapping these trajectory-level advantages to policy updates requires aggregating token-level probabilities within each sequence. As the demand for solving long-horizon reasoning tasks grows, the fundamental mechanics of this fixed aggregation step have come under scrutiny (Liu et al., 2025). Existing algorithms rigidly rely on static aggregation functions: standard GRPO ($p = 1$) defaults to the Arithmetic Mean, while recent variants like GMPO (Zhao et al., 2025) and GSPO (Zheng et al., 2025) ($p \to 0$) attempt to mitigate variance by employing the Geometric Mean.

Despite their empirical success, fixed aggregation mechanisms may favour different tasks and training stages. Our static-order comparison illustrates this sensitivity: among the tested orders, the highest mean AIME24 accuracy occurs at $p = 3$ and the highest mean MATH500 accuracy occurs at $p = -1$. Figure 1 illustrates how the underlying token weight distribution $W(p)$ changes across the p-axis. To ground the accompanying signal-density interpretation, we additionally examine successful rollouts and zero-positive groups on the two benchmarks.

To address this task-dependent trade-off, we propose **HolderPO**, a generalised policy optimisation framework unifying token-level importance-ratio aggregation via the adaptable Holder mean. By explicitly modulating p, the framework provides continuous control over gradient concentration and variance bounds. Increasing positive p concentrates gradient weight on high-ratio tokens, which can strengthen useful contributions when these are aligned with objective improvement. Conversely, decreasing p tightens an upper bound on the policy-gradient variance under the stated assumptions, while weakening or redirecting the concentration. Because the preferred balance may change during training, we instantiate HolderPO with a dynamic annealing algorithm that progressively decreases p from a positive to a negative value.

Our experiments compare static orders and dynamic schedules across mathematical reasoning and agentic decision-making benchmarks. On the **Qwen2.5-Math-7B** base (Yang et al., 2024), our multi-seed results show a five-benchmark average of 55.2% for linear decay, compared with 51.4% for GRPO and 52.6% for the selected static HolderPO configuration. The benchmarks include AIME, AMC, MATH, Minerva (Lewkowycz et al., 2022), and OlympiadBench (He et al., 2024). The comparison also covers other model scales and the ALFWorld benchmark (Shridhar et al., 2020), where HolderPO achieves 95.3% for the $1 \to -1$ schedule.

In summary, our main contributions are as follows:

- **The HolderPO Framework:** We propose HolderPO, a generalised policy optimisation framework that dynamically unifies various mean-based probability aggregations through the adaptable Holder parameter p.
- **Theoretical Foundation:** We characterise how p controls the token-weight distribution and a policy-gradient variance bound. Additional analyses connect order selection to useful update directions, convergence under stated conditions, and the ordering of prescribed schedule values.
- **Empirical Evaluation:** We combine signal-density diagnostics, static-order comparisons, and dynamic-schedule evaluations across mathematical and agentic tasks, with multi-seed reporting to assess performance variability.

2 Related Work

Reinforcement Learning for Complex Reasoning. Reinforcement Learning (RL) has become the cornerstone of LLM post-training. While foundational work used RLHF for behavioural alignment (Ouyang et al., 2022; Stiennon et al., 2020), recent advances focus on complex reasoning via RLVR (Wen et al., 2025), pioneered by OpenAI o-series (Jaech et al., 2024) and DeepSeek-R1 (Guo et al., 2025; Shao et al., 2024), inspiring both proprietary (Comanici et al., 2025; Yang et al., 2025a) and open-source successors. GRPO (Shao et al., 2024) has emerged as the dominant algorithm; its broader ecosystem of refinements is surveyed in Appendix A.

Token-Level Aggregation. The aggregation operator that maps token-level importance ratios to a sequence-level signal is the most direct analogue of our framework. GRPO uses the arithmetic mean, while GMPO (Zhao et al., 2025) and GSPO (Zheng et al., 2025) adopt the geometric mean

to mitigate outlier variance. Concurrent PMPO (Zhao et al., 2026) parameterises a power-mean exponent $p \in [0, 1]$, adapted per-trajectory via clip-aware ESS matching. Our framework differs in two key respects: (i) we extend p to the *full real range*, identifying $p < 0$ as a qualitatively distinct *inverse-concentration* phase unexplored by prior work; and (ii) we adapt p along the *temporal* axis (across training steps) rather than per trajectory, enabling complementary roles for early high-ratio concentration and later variance-bound contraction.

Token Reweighting via Auxiliary Signals. A parallel line reweights tokens within each rollout using signals *external* to the importance ratio: token entropy (Wang et al., 2025a; Yu & Li, 2026; Simoni et al., 2025), token probability (Yang et al., 2025b), hidden contributions to response confidence (Deng et al., 2025), or selective KL masking (Lin et al., 2025). These approaches are orthogonal to ours and could in principle be combined with HolderPO's power-mean aggregatio

3 HolderPO: A Generalised Aggregation Framework

When adapting PPO for LLMs, particularly for training long-horizon reasoning tasks, group-based variants like GRPO (Shao et al., 2024) formulate the unclipped surrogate objective as

$$\mathcal{J}(\theta) = \mathbb{E}_{x,\{y_i\}} \left[\frac{1}{G} \sum_{i=1}^{G} \rho_i(\theta) \widehat{A}_i \right].$$

Here, $\rho_i(\theta)$ is an *aggregation operator*— a functional projection that compresses the full sequence of token-level importance ratios $\{r_{i,t}(\theta)\}_{t=1}^{|y_i|}$ into a well-behaved sequence-level scalar. While GRPO uses the arithmetic mean, GMPO (Zhao et al., 2025) and GSPO (Zheng et al., 2025) use geometric mean. However, these methods only represent static, isolated points within a broader, continuous spectrum of aggregation operators.

In Section 3.1, we propose **Holder Policy Optimisation**, a generalised framework that parameterises aggregation operators by a single scalar $p \in \mathbb{R}$ through the Holder mean. This parameter controls both **gradient concentration** (Section 3.2), which redistributes weight across token contributions, and the **variance bound** (Section 3.3), which bounds gradient variability under the stated assumptions. The interplay between these effects motivates our **dynamic scheduling strategy** in Section 3.4.

3.1 Aggregation via the Holder Mean

Given a prompt context x and a rollout y_i sampled from $\pi_{\theta_{\text{old}}}$, the token-level importance ratio for t-th token is $r_{i,t}(\theta) = \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})}$. Rather than relying on a fixed operator, HolderPO generalises the token-level aggregation by the Holder mean of order p:

$$\rho_{i,p}(\theta) = \begin{cases} \left(\frac{1}{|y_i|} \sum_{t=1}^{|y_i|} r_{i,t}(\theta)^p \right)^{1/p}, & \text{if } p \neq 0, \\ \exp\left(\frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \log r_{i,t}(\theta) \right), & \text{if } p = 0. \end{cases} \tag{1}$$

Due to the limit for $p \to 0$, we take the geometric mean for $p = 0$ branch (see Appendix H.4). The HolderPO objective then takes the standard PPO-style form with sequence-level clipping:

$$\mathcal{J}_{H_s}(\theta) = \mathbb{E}_{x,\,\{y_i\}_{i=1}^{G}} \left[\frac{1}{G} \sum_{i=1}^{G} \min\Big(\rho_{i,p}(\theta) \widehat{A}_i,\ \text{clip}\big(\rho_{i,p}(\theta),\, 1-\epsilon,\, 1+\epsilon\big) \widehat{A}_i \Big) \right]. \tag{2}$$

Here $\widehat{A}_i$ is the advantage estimator and ϵ is the clipping threshold. The reason we choose sequence-level clipping is to control gradient variance (see Appendices E and K.2). HolderPO supports both token-level and sequence-level clipping. With matching normalisation conventions, the token-clipped objective at $p = 1$ is exactly GRPO (Appendix H.2), while the sequence-clipped objective at $p = 0$ is exactly GSPO (Appendix H.3). To analyse how p shapes the optimisation, we study $\nabla_\theta \rho_{i,p}(\theta)$, which governs the direction of the policy gradients (see Eq. (10), (13), (16)). A direct calculation (Appendix H.1) yields

$$\nabla_\theta \rho_{i,p}(\theta) = \rho_{i,p}(\theta) \sum_{t=1}^{|y_i|} W_{i,t}(p) \cdot \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t}) \quad W_{i,t}(p) := \frac{r_{i,t}(\theta)^p}{\sum_{k=1}^{|y_i|} r_{i,k}(\theta)^p}, \tag{3}$$

where the per-token gradient weights $W_{i,t}(p)$ form a probability distribution denoted by W_i^p. Crucially, varying p does not alter the per-token log-gradient directions; instead, it changes their relative weights and the common scale $\rho_{i,p}(\theta)$. At the old policy, all token ratios equal one, so every p gives the same first update direction. As the policy moves away, differences between fixed orders first appear in the local curvature; Theorem A1 in Appendix I characterises this local effect.

3.2 Distributional Deformation and Gradient Concentration

We formalise the gradient concentration by analysing W_i^p through two complementary lenses. Locally, Theorem 5 (Appendix J.1) shows an increasingly strict token-level weight allocation: as p grows, maximal-ratio tokens monotonically dominate. Non-maximal ones may briefly gain weight before strictly decaying to zero once the rising threshold $\mu_i(p)$ surpasses their log-ratios. Globally, our next result (Appendix J.2) captures the dispersion of the entire weight distribution by Shannon entropy.

Theorem 1. *Assume the sequence y_i contains at least two tokens with distinct importance ratios. Then Shannon entropy of the weight distribution attains its global maximum at $p = 0$, where $W_{i,t}(0) = 1/|y_i|$ for every token, and strictly decreases as p moves away from zero along either half-axis. Moreover, as $p \to \pm\infty$, W_i^p concentrates uniformly on the subset $\mathcal{T}^+ = \arg\max_t r_{i,t}(\theta)$ and $\mathcal{T}^- = \arg\min_t r_{i,t}(\theta)$, respectively.*

Together, these results formally characterise gradient concentration—the skewing of the weight distribution toward a specific subset of tokens, yielding three regimes:

Upward Concentration ($p > 0$). A positive p drives the gradient concentration toward tokens with relatively high importance ratios. A prevailing view suggests that RL for reasoning primarily acts to sharpen the pre-existing knowledge distribution of the base model (e.g., Zhou et al. (2023); Li et al. (2024); Yue et al. (2025)). Under this view, an importance ratio > 1 serves as a confidence signal that, ideally, highlights the critical bottleneck tokens within reasoning steps. In long-horizon tasks, where such high-confidence tokens are sparse (Zelikman et al., 2022; Lightman et al., 2023; Yao et al., 2023), setting $p > 0$ explicitly amplifies their weight to prevent their gradients from being diluted.

Uniform Dispersion ($p \to 0$). As p approaches zero from either side, the token weights become more uniform. At $p = 0$, every token receives the same weight, although its gradient vector may differ in magnitude and direction.

Downward Concentration ($p < 0$). A negative p reverses the relative allocation, favouring tokens with smaller importance ratios. These include tokens whose probabilities have decreased relative to the old policy, but low ratios do not necessarily identify effective alternative reasoning paths. Whether this redistribution helps depends on the resulting update direction. Further discussion of its possible relation to exploration and exploitation appears in Appendix J.4.

For a fixed change in order, similar within-sequence ratios limit the resulting weight change, while greater heterogeneity permits a larger change (Appendix J.3).

Whether reweighting helps depends on its alignment with objective improvement. Appendix P shows that the derivative of the weighted directional contribution equals the weighted covariance between token log-ratios and their signed contributions along a common objective gradient. A positive covariance therefore favours increasing p locally.

Incorrect feedback can also be amplified. Appendix Q bounds token-level corruption amplification and reward-induced changes to the clipped group update, and shows that changing p alone cannot distinguish errors that leave the observable data unchanged.

3.3 Policy Gradient Variance Bound

Next, we analyse the variance of the policy gradient estimator induced by (2). In long-horizon reasoning, while concentration enables the amplification of targeted signals, it risks magnifying gradient variance. The following theorem bounds gradient variability and shows how the bound changes with p (proof is in Appendix K.1).

Theorem 2. *Let $\widehat{\nabla}_\theta \mathcal{J}_{H_s}$ (Eq. (17)) denote the unbiased mini-batch estimator induced by (2). Assume $\|\nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t})\| \le M$ for all tokens within the batch, the variance admits the bound*

$$\|\mathrm{Var}(\widehat{\nabla}_\theta \mathcal{J}_{H_s})\| \;\le\; \frac{M^2}{B}\,\mathbb{E}\Big[\widehat{A}_i^2\,\rho_{i,p}^2(\theta)\Big]\,, \tag{4}$$

which is monotonically increasing in p for all $p \in \mathbb{R}$, where B is the batch size.

Selecting an order. Appendix L compares useful mean progress against the update's total second moment. A finite exact optimum need not exist, but finite orders approximate the best efficiency under the stated non-degeneracy condition. In its common-direction, uncorrelated-noise model, the optimum is $p^* = \lambda_s - \lambda_v$, where λ_s and λ_v are the power-law exponents relating signal strength and noise variance to token ratios. Negative orders are therefore optimal in this model when noise variance grows faster than signal. For a shared learning rate, Appendix M instead compares complete updates, including clipping, by a lower bound on expected one-step improvement and bounds the loss from estimating this criterion.

Trade-off with concentration. Theorems 1 and 2 identify a structural trade-off controlled by p: increasing positive p concentrates the token weights toward high-ratio tokens but loosens the variance bound, whereas decreasing p tightens this bound but weakens or redirects the concentration. The preferred fixed order therefore depends on the underlying signal–noise structure. As this structure may vary across tasks and training stages, a single fixed p need not remain optimal throughout training, motivating the dynamic schedule introduced next.

3.4 A Dynamic p-Scheduling Strategy

Motivated by this trade-off, we linearly decrease p from $p_{\mathrm{high}} > 0$ to p_{low}, which may be negative. Larger initial orders favour high-ratio tokens, while lower terminal orders tighten the variance bound. The implementation is given in Appendix C.

Theorem 3. *Let $V(p) \coloneqq \mathbb{E}[G^{-1}\sum_{i=1}^{G} \widehat{A}_i^2\,\rho_{i,p}^2(\theta)]$ denote the term in the bound in Eq.(4), and let $p_{stat} \in [p_{low}, p_{high}]$ be any fixed parameter. Given a y_i of length n, at a fixed policy state and rollout distribution, the endpoint orders satisfy:*

1. High-ratio weight amplification: *Suppose y_i has a high-ratio token t^* with $r_{i,t^*} \gg 1$, while the other tokens have ratios bounded above and away from zero by constants independent of n, and the orders lie in a fixed bounded interval. Under the pre-saturation condition $r_{i,t^*}^{p_{high}} \ll n - 1$, shifting from p_{stat} to p_{high} gives the following weight-amplification bound: there exists a constant $C > 0$ such that*

$$\frac{W_{i,t^*}(p_{high})}{W_{i,t^*}(p_{stat})} \;\ge\; C \cdot r_{i,t^*}^{\,p_{high}-p_{stat}}. \tag{5}$$

2. Variance-bound contraction: *The terminal variance bound is contracted:*

$$V(p_{\mathrm{low}}) \le V(p_{\mathrm{stat}}). \tag{6}$$

Theorem 3 (Appendix R) compares endpoint orders at a fixed policy state. The evolution of the training updates is analysed separately. Appendix N relates the normalised update to a fixed objective and accounts for length-normalisation bias and buffer-reuse drift. With bounded buffer reuse and under its alignment, moment, bias, drift, and step-size conditions, dynamic orders admit an $O(T^{-1/2})$ bound on the average squared objective gradient. Appendix O gives conditions under which decreasing order minimises terminal model error for fixed order values and usage counts. Linear interpolation also minimises squared step-to-step order changes; both criteria select linear decay on a fixed equally spaced grid used once. Observed training dynamics and schedule comparisons are reported in Appendices B and F.

4 Experiment

We evaluate HolderPO on mathematical reasoning and agentic benchmarks. The experiments first characterise successful-feedback density and static-p sensitivity, then compare dynamic scheduling with matched baselines, and finally examine schedule design and transfer to ALFWorld.

4.1 Implementation Details

Model. We evaluate our framework on two task families: mathematical reasoning and agentic decision-making. For mathematical reasoning, following Dr.GRPO (Liu et al., 2025), we cover a broad spectrum of base models ranging from 1.5B to 8B parameters, including the Qwen2.5-Math series (1.5B and 7B) (Yang et al., 2024), DeepSeek-R1-Distill-Qwen-7B (Guo et al., 2025), and the Qwen3 series (4B and 8B) (Yang et al., 2025a). For agentic tasks, we adopt Qwen2.5-1.5B-Instruct (Qwen et al., 2025) as the policy backbone.

Training. Our training pipeline follows two established protocols depending on the task. For mathematical reasoning, we adopt the recipe of Dr.GRPO (Liu et al., 2025): training data consists of 8,523 problems from MATH (Hendrycks et al., 2021) (Levels 3–5), and each prompt is paired with 8 sampled rollouts capped at 3,000 tokens. Within each RL round, $\pi_{\theta_{\text{old}}}$ produces 1,024 trajectories, after which the current policy π_θ is refreshed 8 times using a mini-batch size of 128. For agentic tasks, we adhere to the GiGPO protocol (Feng et al., 2025) for both training and evaluation on ALFWorld. In terms of compute, all models are trained on 4×H100 GPUs. The main mathematical comparison includes GRPO (Shao et al., 2024), GMPO (Zhao et al., 2025), and PMPO (Zhao et al., 2026) under matched configurations.

Evaluation. We report mathematical performance on five benchmarks that span a wide difficulty range. AIME24 contains 30 olympiad-level problems drawn from the 2024 American Invitational Mathematics Examination, while AMC provides 83 competition problems of intermediate difficulty. MATH500 is a 500-problem subset of MATH covering algebra, geometry, and number theory. Minerva (Lewkowycz et al., 2022) consists of 272 graduate-level problems that demand multi-step derivations, and OlympiadBench (Oly.) (He et al., 2024) collects 675 high-difficulty olympiad problems. For agentic evaluation, we use the six ALFWorld (Shridhar et al., 2020) sub-task categories, namely Pick, Look, Clean, Heat, Cool, and Pick Two. Following Dr.GRPO (Liu et al., 2025), we adopt Pass@1 as the primary metric for mathematical tasks and decode greedily with temperature 0.0, generating one sample per question. For ALFWorld, we report task success rate under the given standard evaluation protocol.

Multi-seed reporting. We report mean±sample standard deviation over three independent training seeds. For mathematical benchmarks, Avg. is the unweighted five-benchmark mean for each seed, summarised across seeds; for ALFWorld, it is the overall success rate over the 134 games, summarised across seeds.

4.2 Signal-Density Diagnostics and Task-Specific Sensitivity of p

Empirical signal density. We use signal density as an operational description of the availability of successful feedback across sampled rollouts and groups. We compute all diagnostics from the initial Qwen2.5-Math-7B checkpoint under a matched sampling protocol: for every evaluation prompt, we independently sample one group of $G = 8$ responses at temperature 1.0, with a maximum of $3{,}000$ generated tokens. This gives 240 rollouts on AIME24 and $4{,}000$ rollouts on MATH500. A rollout is positive when it receives a positive correctness reward.

To measure how broadly the group-relative update signal is distributed, for each group g with $\sum_{j=1}^{G} |\widehat{A}_{g,j}| > 0$ we define

$$q_{g,i}^{|A|} = \frac{|\widehat{A}_{g,i}|}{\sum_{j=1}^{G} |\widehat{A}_{g,j}|}, \qquad \widetilde{H}_g^{|A|} = -\frac{1}{\log G} \sum_{i=1}^{G} q_{g,i}^{|A|} \log q_{g,i}^{|A|}. \tag{7}$$

The reported normalised advantage entropy is the mean of $\widetilde{H}_g^{|A|}$ over groups with nonzero advantage; groups with zero advantage are excluded from this mean. Zero-positive groups are reported separately. Table 1 summarises the results. AIME24 has fewer positive rollouts (4.6% versus 30.9%) and more groups without a positive rollout (76.7% versus 20.6%). The positive-rollout gap persists after excluding truncated outputs (5.5% versus 33.9%), while its normalised advantage entropy is substantially lower (0.158 versus 0.631). Together, these measurements verify that AIME24 operates in a substantially sparser-signal regime than MATH500.

The advantage-entropy diagnostic measures the dispersion of group-relative feedback, complementing the within-sequence token-weight entropy characterised in Theorem 1.

Diagnostic	AIME24	MATH500
Positive rollout rate	4.6%	30.9%
Zero-positive groups	76.7%	20.6%
Normalized advantage entropy	0.158	0.631
Positive among untruncated rollouts	5.5%	33.9%

Table 1: Signal-density diagnostics from the initial Qwen2.5-Math-7B checkpoint. Each prompt uses one independently sampled group of $G = 8$ responses at temperature 1.0 with a 3,000-token generation cap. Normalised advantage entropy is defined in Eq. (7).

Method	AIME24	MATH500
Baselines		
GRPO ($p = 1$)	41.1 $\pm$5.1	83.5 $\pm$0.1
GMPO ($p \to 0$)	42.2 $\pm$1.9	81.9 $\pm$0.1
Static HolderPO		
HolderPO ($p = -2$)	37.8 $\pm$3.8	84.6 $\pm$0.4
HolderPO ($p = -1$)	36.7 $\pm$5.8	**85.0** $\pm$0.8
HolderPO ($p \to 0$)	43.3 $\pm$3.3	84.6 $\pm$0.3
HolderPO ($p = 1$)	38.9 $\pm$7.7	83.2 $\pm$1.2
HolderPO ($p = 2$)	44.4 $\pm$6.9	82.0 $\pm$1.4
HolderPO ($p = 3$)	**46.7** $\pm$6.7	81.8 $\pm$1.6

Table 2: Static-p comparison on AIME24 and MATH500, reported as mean$\pm$standard deviation over three independent training seeds. The best result on each benchmark is shown in bold.

Static-order comparison. We next compare fixed orders on these two benchmarks to examine how their distinct signal-density profiles affect the preferred aggregation order.

The optimal aggregation order diverges across the two signal-density regimes: AIME24 achieves the highest accuracy at $p = 3$, whereas MATH500 performs best at $p = -1$.

AIME24 and positive orders. The sparse-signal AIME24 regime favours larger positive orders, which concentrate gradient weight on rare high-ratio tokens according to Theorem 1. This prevents the informative reasoning steps in successful trajectories from being diluted by the remainder of the sequence.

MATH500 and lower orders. The denser-signal MATH500 regime favours lower orders. Decreasing p tightens the gradient-variance bound in Theorem 2, suppressing unstable updates when useful feedback is distributed more broadly across sampled trajectories.

4.3 Main Performance and Dynamic Scheduling

The distinct static-order preferences motivate a dynamic schedule that combines early high-ratio concentration with late-stage variance control. We evaluate linear decay from 2 to -2 against static HolderPO and matched baselines.

Table 3 presents the multi-seed results. Linear decay achieves a five-benchmark average of 55.2% on Qwen2.5-Math-7B, compared with 52.6% for the selected static configuration and 51.4% for GRPO. The corresponding averages on Qwen2.5-Math-1.5B and DeepSeek-R1-Distill-Qwen-7B are 44.6% and 66.5%, demonstrating consistent gains across model scales and base architectures.

4.4 Schedule Direction, Shape, and Range

Direction and shape. Theorem 3 characterises the complementary roles of the endpoint orders at a fixed policy state. Appendix O proves the optimality of linear decay for a prescribed equally spaced order grid under its exchange condition: among all permutations of the same order values, the decreasing arrangement minimises terminal model error and simultaneously minimises squared

Method	AIME24	AMC	MATH500	Minerva	Oly.	Avg.
Qwen2.5-Math-1.5B						
GMPO ($p \to 0$) (Zhao et al., 2025)	20.0 $\pm$3.3	53.0 $\pm$1.2	77.6 $\pm$0.2	30.1 $\pm$0.4	38.7 $\pm$0.1	43.9 $\pm$1.1
HolderPO (Linear Dec: $2 \to -2$, Ours)	**31.1** $\pm$1.9	47.8 $\pm$0.7	77.0 $\pm$0.4	28.1 $\pm$0.6	**39.1** $\pm$0.2	**44.6** $\pm$0.7
Qwen2.5-Math-7B						
GRPO ($p = 1$) (Shao et al., 2024)	41.1 $\pm$5.1	58.6 $\pm$1.8	83.5 $\pm$0.1	32.2 $\pm$0.2	41.4 $\pm$0.1	51.4 $\pm$1.4
GMPO ($p \to 0$) (Zhao et al., 2025)	42.2 $\pm$1.9	61.8 $\pm$0.7	81.9 $\pm$0.1	33.5 $\pm$0.7	43.6 $\pm$0.3	52.6 $\pm$0.7
PMPO (Zhao et al., 2026)	35.6 $\pm$5.1	68.7 $\pm$2.1	83.7 $\pm$0.3	**35.0** $\pm$0.2	**46.6** $\pm$0.1	53.9 $\pm$1.5
HolderPO (Ours)						
HolderPO (Best Static: $p \to 0$)	43.3 $\pm$3.3	57.8 $\pm$1.2	**84.6** $\pm$0.3	31.6 $\pm$0.6	45.5 $\pm$0.3	52.6 $\pm$1.1
HolderPO (Linear Dec: $2 \to -2$)	**44.4** $\pm$1.9	**69.1** $\pm$1.8	82.3 $\pm$0.3	34.8 $\pm$0.6	45.4 $\pm$0.2	**55.2** $\pm$0.9
DeepSeek-R1-Distill-Qwen-7B						
GRPO ($p = 1$) (Shao et al., 2024)	43.3 $\pm$5.8	67.5 $\pm$2.1	89.0 $\pm$0.4	39.7 $\pm$0.4	56.7 $\pm$0.1	59.2 $\pm$1.7
GMPO ($p \to 0$) (Zhao et al., 2025)	45.6 $\pm$1.9	77.9 $\pm$0.7	91.4 $\pm$0.2	37.9 $\pm$0.6	62.5 $\pm$0.3	63.1 $\pm$0.7
PMPO (Zhao et al., 2026)	46.7 $\pm$5.8	79.5 $\pm$2.4	**93.5** $\pm$0.1	39.5 $\pm$0.2	**64.2** $\pm$0.1	64.7 $\pm$1.7
HolderPO (Linear Dec: $2 \to -2$, Ours)	**53.3** $\pm$3.3	**79.9** $\pm$0.7	92.6 $\pm$0.3	**42.4** $\pm$0.6	64.1 $\pm$0.3	**66.5** $\pm$1.0

Table 3: Multi-seed comparison of HolderPO against matched policy-optimization baselines across different model scales and base architectures. Results are reported as mean$\pm$standard deviation over three independent training seeds. The static HolderPO row uses the best fixed aggregation order by five-benchmark average, while the dynamic variant linearly decreases p from 2 to -2 throughout training. Additional comparisons with previously published RL post-trained models are deferred to Appendix D.

step-to-step order changes. Linear interpolation is also the unique minimiser of the squared-change cost among all schedules with fixed endpoints and step count. Appendix F complements this result by comparing increasing and decreasing schedules with linear, square, cube, and sinusoidal shapes.

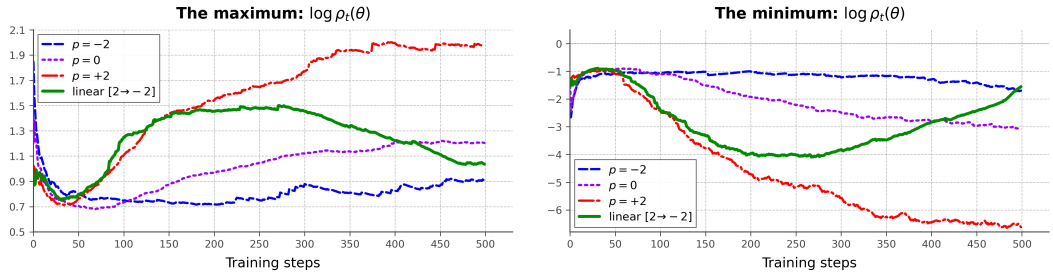


Figure 2: Token-level log importance ratios during training. **Left** and **right** show the per-step upper and lower envelopes, respectively, comparing linear decay $p : 2 \to -2$ (solid green) with constant orders $p \in \{+2, 0, -2\}$. As p decreases, the upper envelope drops and the lower envelope rises, progressively tightening the ratio range during training.

Figure 2 complements the performance comparisons by tracking token-ratio dynamics. Additional entropy and gradient-norm trajectories appear in Appendix B.

Choosing the endpoints. We use $2 \to -2$ as the default mathematical schedule, spanning positive, uniform, and negative weighting regimes. This moderate range provides strong early signal concentration while avoiding the excessive negative-order concentration that emerges as $p \to -\infty$. The order-selection analysis in Appendix L further characterises the balance between useful progress, second moments, and conditional variance.

The endpoint magnitude is selected according to the model's prior familiarity with the task domain. When the pretrained model is likely to have encountered related problem structures, as with a mathematics-specialised backbone on mathematical reasoning tasks, we use the more aggressive

Method	Pick	Look	Clean	Heat	Cool	Pick2	Avg.
Qwen2.5-Instruct-1.5B							
GRPO ($p = 1$) (Shao et al., 2024)	84.7 $\pm$8.7	53.7 $\pm$3.2	84.9 $\pm$1.9	78.3 $\pm$4.3	60.3 $\pm$2.7	52.9 $\pm$5.9	71.6 $\pm$0.7
GMPO ($p \to 0$) (Zhao et al., 2025)	93.1 $\pm$6.4	77.8 $\pm$9.6	80.6 $\pm$3.2	88.4 $\pm$5.0	82.5 $\pm$5.5	**90.2** $\pm$6.8	85.3 $\pm$0.9
GiGPO (Feng et al., 2025)	94.4 $\pm$9.6	66.7 $\pm$5.6	94.6 $\pm$3.7	94.2 $\pm$2.5	79.4 $\pm$9.9	76.5 $\pm$0.0	86.1 $\pm$0.9
HolderPO (Linear Dec: $2 \to -2$)	**97.2** $\pm$2.4	85.2 $\pm$6.4	87.1 $\pm$5.6	91.3 $\pm$7.5	79.4 $\pm$7.3	80.4 $\pm$9.0	87.3 $\pm$0.7
HolderPO (Linear Dec: $1 \to -1$)	**97.2** $\pm$4.8	**100.0** $\pm$0.0	**100.0** $\pm$0.0	**100.0** $\pm$0.0	**85.7** $\pm$4.8	84.3 $\pm$3.4	**95.3** $\pm$0.9

Table 4: Success rates (%) on the 134 unseen ALFWorld games. Results are reported as mean$\pm$standard deviation over three independent training seeds. Avg. denotes overall success across all 134 games. The best result in each column is shown **in bold**.

$2 \to -2$ range to amplify the model's emerging high-ratio signals. For a substantially less familiar domain, we use the more conservative $1 \to -1$ range to avoid excessive concentration on unreliable early updates. We therefore use $2 \to -2$ for mathematical reasoning and $1 \to -1$ for ALFWorld (Section 4.5), where the general-purpose instruction model has not received comparable domain-specific pretraining for embodied interaction.

4.5 Generalisation to Agentic Reasoning

To demonstrate that the advantages of HolderPO extend beyond pure mathematical domains to broader sequential decision-making scenarios, we evaluate our framework on the ALFWorld benchmark (Shridhar et al., 2020). ALFWorld is a challenging embodied agentic environment that requires models to complete multi-step, open-ended tasks (e.g., finding, cleaning, or heating objects) based entirely on textual observations and actions. Unlike mathematical proofs, where reasoning is largely self-contained, agentic tasks suffer from compounding errors over long horizons, making stable policy optimisation crucial for success. Following established setups, we employ Qwen2.5-Instruct-1.5B as our base model for this agentic reasoning task. Table 4 presents the success rates across the six distinct sub-task categories in the ALFWorld evaluation suite.

The $1 \to -1$ schedule outperforms $2 \to -2$ in overall success rate (95.3% versus 87.3%), while GMPO has the highest listed mean on Pick Two. This result supports our endpoint-selection principle. For mathematical reasoning, domain-specific pretraining provides a stronger prior over relevant solution structures, allowing the more aggressive $2 \to -2$ range to amplify emerging high-ratio signals. In ALFWorld, the policy encounters embodied interaction patterns that are less familiar to the general-purpose instruction model, so the conservative $1 \to -1$ range reduces the risk of over-concentrating on unreliable early updates. Appendix Q further analyses how concentrated weighting can amplify incorrect feedback.

5 Conclusion

We introduced **Holder Policy Optimisation (HolderPO)**, a generalised framework for controlling the trade-off between token-weight concentration and a policy-gradient variance bound. The Holder mean provides a continuous aggregation order: positive orders favour high-ratio tokens, zero gives uniform token weights, and negative orders favour low-ratio tokens. Decreasing p tightens the variance upper bound under the stated assumptions. This structure motivates a decreasing schedule, evaluated through signal-density diagnostics, static-order comparisons, and dynamic-schedule experiments across mathematical and agentic tasks. HolderPO achieves a five-benchmark average accuracy of 55.2% on Qwen2.5-Math-7B and an overall success rate of 95.3% on ALFWorld.

Limitations. Our analysis of linear decay applies to the prescribed schedule class and conditions in Appendix O. Endpoint selection remains task-dependent, and adaptive selection is left for future work.

Future Work. A primary direction is an *adaptive* scheduler that adjusts p from real-time metrics (e.g., batch-level gradient variance or token-ratio dispersion), removing the need for manual tuning.

AI USAGE STATEMENT

Large language models are the subject of this work: the teachers and students in all experiments are language models. Separately, we used Large language models were used as writing and editing assistants to improve the organization, terminology, and presentation of this manuscript. The authors defined the method, conducted all experiments, verified the claims and numerical results, and take full responsibility for the manuscript. Experimental evidence comes from the verified runs reported in this paper.

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APPENDIX OUTLINE

Appendices A–G provide related work, training dynamics, implementation, and additional experiments. Appendices H–K derive the objectives and core geometry, concentration, and variance results. Appendices L and M study parameter selection; Appendices N and O analyse convergence and schedule design. Appendices P and Q cover update alignment and imperfect rewards, followed by the endpoint comparison and broader impacts.

A EXTENDED RELATED WORK

We expand on the broader GRPO ecosystem briefly mentioned in Section 2. The variants below address aspects of the RL pipeline orthogonal to token-level aggregation.

Surrogate Loss and Critic-Free Variants. GPG (Chu et al., 2025) simplifies the GRPO objective by removing surrogate losses entirely, while DAPO (Yu et al., 2025) introduces dynamic sampling and decoupled clipping bounds. Dr.GRPO (Liu et al., 2025) mitigates length bias by removing the per-sequence length normalisation, λ-GRPO (Wang et al., 2025b) learns the length preference via a trainable parameter. These methods modify the loss normalisation rather than the aggregation function and are complementary to our framework.

Advantage Estimation and Reward Shaping. AAPO (Xiong et al., 2025) introduces advantage momentum to mitigate zero-gradient situations; BNPO (Xiao et al., 2025) adaptively normalises rewards via a Beta distribution; OPO (Hao et al., 2025) provides a variance-minimising baseline; and Seed-GRPO (Chen et al., 2025) scales updates by question difficulty. These contributions modify the advantage signal rather than how token-level ratios are aggregated.

Value-Model-Based Variants. To circumvent GRPO's variance issues, some approaches revert to PPO with pre-trained value models, including VC-PPO (Yuan et al., 2025) and T-PPO (Ouyang et al., 2025). While effective, the external value model introduces confounding factors and computational overhead that the critic-free GRPO family, including ours, deliberately avoids.

Data-Centric and Curriculum-Based Approaches. Open-Reasoner-Zero (Hu et al., 2025), PRIME (Cui et al., 2025), and SimpleRL-Zero (Zeng et al., 2025) democratise scalable RL training through curated training data, curriculum learning, and clean base models. These contributions are at the data and pipeline level, complementary to algorithmic refinements such as ours.

B TRAINING DYNAMICS: ENTROPY AND GRADIENT-NORM

C PSEUDOCODE

HolderPO is a single-line modification of the GRPO loss. To preserve numerical stability for large $|p|$, all power operations are evaluated in log-space via the log-sum-exp identity. Algorithm 1 summarises the full computation. The aggregation operator is applicable at any granularity: in our experiments we use a single sequence-level $\rho_{i,p}$, but token-level or block-level aggregation can be substituted without changing the algorithm or theory.

D ADDITIONAL BASELINE COMPARISONS

For completeness, Table 5 reports additional previously published baselines that are omitted from the main comparison for clarity. Unlike the matched multi-seed experiments in Table 3, these results are taken directly from the corresponding prior works and are reported using their published point estimates.

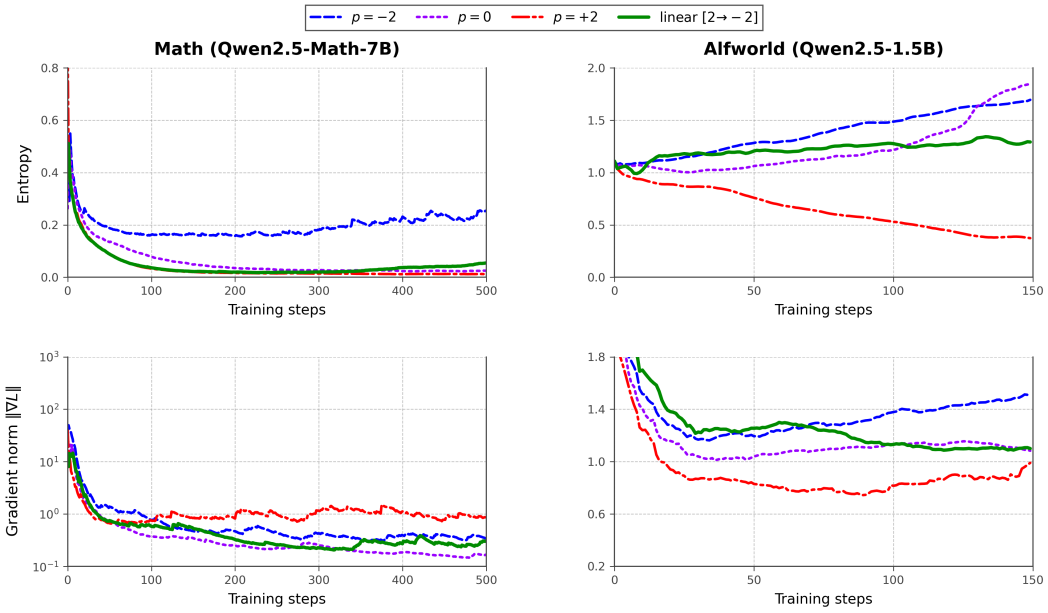


Figure 3: **Entropy and gradient-norm dynamics under different Holder exponents p. Columns:** Math (Qwen2.5-Math-7B on MATH-12k) and ALFWorld (Qwen2.5-Instruct-1.5B). **Rows:** per-step policy entropy and gradient norm $\|\nabla\mathcal{L}\|$ (log scale on Math, linear on ALFWorld). Constant-p baselines ($p \in \{+2, 0, -2\}$, dashed/dotted/dash-dotted) are compared with a linearly-decaying $p\!:2\to-2$ schedule (solid green) in both columns, providing a matched view of the full concentration–dispersion transition. This ALFWorld curve is a separate $2\to-2$ dynamics run; the 95.3% ALFWorld result in Table 4 is obtained by the $1\to-1$ schedule. Positive p concentrates mass on high-likelihood tokens and pushes entropy down; negative p disperses mass and pushes it up. The schedule inherits both regimes in sequence and keeps the gradient norm in a tighter band than any constant choice.

Algorithm 1 HolderPO Loss Computation

Require: current policy π_θ, reference policy $\pi_{\theta_{\text{old}}}$
Input: sequence y of length T, valid-token mask M, advantage $\widehat{A}$, parameter p, clip range ϵ
// Step 1: log-ratio computation
$\Delta_t \leftarrow \log \pi_\theta(y_t \mid x, y_{<t}) - \log \pi_{\theta_{\text{old}}}(y_t \mid x, y_{<t})$
$|y| \leftarrow \sum_{t=1}^{T} M_t$
// Step 2: Holder-mean aggregation in log-space
if $|p| < 10^{-6}$ **then**
  $\rho \leftarrow \exp\Big(\frac{1}{|y|}\sum_t M_t\,\Delta_t\Big)$ // limit $p \to 0$ (geometric mean)
else
  $\rho \leftarrow \Big(\frac{1}{|y|}\sum_t M_t\,\exp(p\,\Delta_t)\Big)^{1/p}$
end if
// Step 3: PPO-style clipping and loss
$\rho_{\text{clip}} \leftarrow \text{clip}(\rho,\, 1-\epsilon,\, 1+\epsilon)$
$\mathcal{L}_{\text{unclip}} \leftarrow -\widehat{A}\cdot\rho$
$\mathcal{L}_{\text{clip}} \leftarrow -\widehat{A}\cdot\rho_{\text{clip}}$
$\mathcal{L}_{\text{HPO}} \leftarrow \max\big(\mathcal{L}_{\text{unclip}},\ \mathcal{L}_{\text{clip}}\big)$ // minimised via SGD
return $\mathcal{L}_{\text{HPO}}$

Training Objectives	AIME24	AMC	MATH500	Minerva	Oly.	Avg.
Qwen2.5-Math-1.5B						
Base & Instruct Models						
Qwen2.5-Math-1.5B (Yang et al., 2024)	16.7	43.4	61.8	15.1	28.4	33.1
Qwen2.5-Math-1.5B-Instruct (Yang et al., 2024)	10.0	48.2	74.2	26.5	40.2	39.8
Additional RL Post-Trained Models						
Oat-Zero-1.5B (Liu et al., 2025)	20.0	53.0	74.2	25.7	37.6	42.1
Qwen2.5-Math-7B						
Base Models						
Qwen2.5-Math-7B (Yang et al., 2024)	16.7	38.6	50.6	9.9	16.6	26.5
Additional RL Post-Trained Models						
SimpleRL-Zero-7B (Zeng et al., 2025)	26.7	60.2	78.2	27.6	40.3	46.6
PRIME-Zero-7B (Cui et al., 2025)	16.7	62.7	83.8	36.0	40.9	48.0
OpenReasoner-Zero-7B @ 3k (Hu et al., 2025)	13.3	47.0	79.2	31.6	44.0	43.0
OpenReasoner-Zero-7B @ 8k (Hu et al., 2025)	13.3	54.2	82.4	31.6	47.9	45.9
Eurus-7B (Yuan et al., 2024)	16.7	62.7	83.8	36.0	40.9	48.0
GPG-7B (Chu et al., 2025)	33.3	65.0	80.0	34.2	42.4	51.0
Oat-Zero-7B (Liu et al., 2025)	43.3	62.7	80.0	30.1	41.0	51.4
DeepSeek-R1-Distill-Qwen-7B						
Additional RL Post-Trained Models						
Dr.GRPO (Liu et al., 2025)	50.0	74.7	89.6	37.5	55.7	61.5

Table 5: Additional comparisons with previously published base, instruct, and RL post-trained models across five mathematical reasoning benchmarks. These methods are omitted from the main table to keep the primary comparison focused on matched multi-seed experiments. Values in this table are taken from the corresponding prior works and are reported as originally published; they are therefore not directly comparable to the uncertainty estimates reported for our multi-seed experiments.

E TOKEN-LEVEL CLIPPING OF HOLDERPO

Method	AIME24	AMC	MATH500	Minerva	Oly.	Avg.
Token-Level Clip HolderPO						
HolderPO ($p = -2$)	36.7 $\pm$3.3	61.4 $\pm$1.2	81.6 $\pm$0.2	**35.7** $\pm$0.4	43.6 $\pm$0.1	51.8 $\pm$0.7
HolderPO ($p = -1$)	40.0 $\pm$5.8	62.7 $\pm$2.1	**81.6** $\pm$0.3	33.5 $\pm$0.6	44.5 $\pm$0.2	52.4 $\pm$1.0
HolderPO ($p \to 0$)	43.3 $\pm$6.7	61.4 $\pm$2.4	**81.6** $\pm$0.4	34.6 $\pm$0.7	42.7 $\pm$0.3	**52.7** $\pm$1.3
HolderPO ($p = 1$)	43.3 $\pm$8.8	**63.9** $\pm$3.2	80.8 $\pm$0.5	31.6 $\pm$1.0	42.8 $\pm$0.4	52.5 $\pm$1.6
HolderPO ($p = 2$)	**43.3** $\pm$10.0	60.2 $\pm$3.6	81.2 $\pm$0.6	33.5 $\pm$1.1	**44.9** $\pm$0.5	52.6 $\pm$1.9

Table 6: HolderPO under *token-level* clipping (Eq. 12) on Qwen2.5-Math-7B across five mathematical benchmarks. Results are reported as mean$\pm$standard deviation over three independent training seeds. Compared with the sequence-level clipping setting reported in Table 3, the token-level variant produces a noticeably narrower performance spread across p, consistent with our discussion in Section 3.3: token-level clipping breaks the algebraic structure underlying the variance bound's monotonicity in p, weakening the controlled trade-off that motivates dynamic scheduling.

F SCHEDULE-SHAPE ABLATION

Method	AIME24	AMC	MATH500	Minerva	Oly.	Avg.
Dynamic HolderPO Variants						
HolderPO (Square Asc: $-2 \to 2$)	33.3 $\pm$3.3	62.7 $\pm$1.2	82.6 $\pm$0.2	33.8 $\pm$0.4	44.4 $\pm$0.1	51.4 $\pm$1.0
HolderPO (Cube Asc: $-2 \to 2$)	36.7 $\pm$5.8	62.7 $\pm$2.4	**82.8** $\pm$0.3	32.0 $\pm$0.6	43.1 $\pm$0.3	51.4 $\pm$1.8
HolderPO (Sin Asc: $-2 \to 2$)	**42.2** $\pm$1.9	59.0 $\pm$0.0	81.6 $\pm$0.4	**35.8** $\pm$0.2	45.8 $\pm$0.4	**52.9** $\pm$0.5
HolderPO (Linear Asc: $-2 \to 2$)	40.0 $\pm$6.7	**66.3** $\pm$2.1	81.2 $\pm$0.5	32.7 $\pm$0.7	42.7 $\pm$0.5	52.6 $\pm$2.1
HolderPO (Square Dec: $2 \to -2$)	36.7 $\pm$8.8	61.4 $\pm$3.2	81.6 $\pm$0.6	33.1 $\pm$1.0	44.0 $\pm$0.6	51.4 $\pm$2.8
HolderPO (Cube Dec: $2 \to -2$)	36.7 $\pm$10.0	60.2 $\pm$4.3	80.8 $\pm$0.7	31.6 $\pm$1.1	46.1 $\pm$0.7	51.1 $\pm$3.3
HolderPO (Sin Dec: $2 \to -2$)	37.8 $\pm$7.7	61.4 $\pm$3.6	81.6 $\pm$0.8	35.3 $\pm$1.3	**46.8** $\pm$0.8	52.6 $\pm$2.7

Table 7: Comparison of alternative annealing shapes for the dynamic schedule on Qwen2.5-Math-7B. Results are reported as mean$\pm$standard deviation over three independent training seeds. We sweep four monotonic interpolation families (linear, square, cube, sinusoidal) in both ascending ($-2 \to 2$) and descending ($2 \to -2$) directions, holding the endpoints fixed at $\{-2, +2\}$. Among the seven variants listed here, none surpasses the descending linear schedule of 55.2% reported in Table 3, supporting our choice of linear decay as the default.

G GENERALISATION TO QWEN3 BASE MODELS

To verify that HolderPO transfers beyond the Qwen2.5-Math-7B setting, we additionally evaluate on the Qwen3-Base series and compare against three strong token-aggregation baselines (GRPO, GSPO, DAPO) under matched configurations.

Method	MATH500	AIME25†	AMC23	Minerva	Oly.	Avg.
			Qwen3-4B-Base			
Qwen3-4B-Base (Yang et al., 2025a)	58.2	7.4	45.0	14.0	28.6	30.6
GRPO (Shao et al., 2024)	79.3 ±0.2	18.5 ±1.5	60.0 ±2.5	21.0 ±0.4	40.5 ±0.2	43.8 ±1.0
GSPO (Zheng et al., 2025)	78.5 ±0.3	18.5 ±2.0	62.5 ±4.3	23.2 ±0.6	39.8 ±0.3	44.5 ±1.4
DAPO (Yu et al., 2025)	81.3 ±0.4	22.2 ±2.6	**65.0** ±5.0	21.7 ±0.7	**41.8** ±0.4	46.4 ±1.8
HolderPO (Linear Dec: $2 \to -2$, Ours)	**88.0** ±0.5	**30.0** ±3.3	60.0 ±6.6	**39.0** ±1.0	40.6 ±0.5	**51.5** ±2.3

Table 8: Results on **Qwen3-4B-Base**. We report Pass@1 accuracy (%) for MATH500, AMC23, Minerva, and Olympiad, and Pass@8 accuracy (%) for AIME25 (†). Results for RL post-trained models are reported as mean±standard deviation over three independent training seeds. HolderPO with our default linear annealing schedule (p: $2 \to -2$) achieves 51.5% average accuracy, compared with 46.4% for DAPO and 43.8% for GRPO.

Method	MATH500	AIME25†	AMC23	Minerva	Oly.	Avg.
			Qwen3-8B-Base			
Qwen3-8B-Base (Yang et al., 2025a)	65.0	11.1	45.0	17.3	31.1	33.9
GRPO (Shao et al., 2024)	80.1 ±0.2	22.2 ±1.5	67.5 ±2.5	27.6 ±0.4	42.4 ±0.2	48.0 ±1.0
GSPO (Zheng et al., 2025)	81.7 ±0.3	22.2 ±2.0	67.5 ±4.3	26.8 ±0.6	45.5 ±0.3	48.7 ±1.5
DAPO (Yu et al., 2025)	85.3 ±0.4	25.9 ±2.6	75.0 ±5.0	27.8 ±0.8	48.7 ±0.4	52.5 ±1.8
HolderPO (Linear Dec: $2 \to -2$, Ours)	**88.4** ±0.5	**33.3** ±3.3	**75.0** ±6.6	**37.5** ±1.0	**50.3** ±0.5	**56.9** ±2.4

Table 9: Results on **Qwen3-8B-Base**. Same evaluation protocol as Table 8. Results for RL post-trained models are reported as mean±standard deviation over three independent training seeds. HolderPO reaches 56.9% average accuracy, compared with 52.5% for DAPO and 48.0% for GRPO, indicating that the improvement persists at the larger model scale.

H Formulas and Derivation

In this section, we derive the formulas involved in our theory. These formulas can be divided into three parts according to different clipping mechanisms: the original unclipped, token-level clipping, and sequence-level clipping. Finally, we discuss the definition of Holder p-norm when $p = 0$.

We define some notation used throughout this chapter. Let $\mathcal{D}$ denote the dataset of input prompts, from which a query q (or context x) is sampled. For each prompt, we sample a group of G responses, denoted as $\{y_i\}_{i=1}^{G}$, from the reference or old policy $\pi_{\theta_{old}}$. For the i-th response y_i, let $|y_i|$ represent its total token length, where $y_{i,t}$ is the t-th token and $y_{i,<t}$ denotes the prefix. The current policy parameterised by θ is denoted as π_θ. Finally, $\widehat{A}_i$ represents the estimated advantage for the i-th response, defined as

$$\widehat{A}_i = \frac{r\left(x, y_i\right) - \operatorname{mean}\left(\left\{r\left(x, y_i\right)\right\}_{i=1}^{G}\right)}{\operatorname{std}\left(\left\{r\left(x, y_i\right)\right\}_{i=1}^{G}\right)},$$

where $r(x, y_i)$ denotes the absolute reward score assigned to the i-th generated response y_i conditioned on the input x. Here $\operatorname{mean}(\cdot)$ and $\operatorname{std}(\cdot)$ represent the arithmetic mean and standard deviation. For simplicity, in this section we omit the KL regularisation term from all PPO-style objective function formulas.

H.1 No Clipping Formulas

As we know, the simplest unclipped GRPO objective function formula is

$$\mathbb{E}_{q\sim\mathcal{D},\{y_i\}_{i=1}^{G}\sim\pi_{\theta_{old}}}\left[\frac{1}{G}\sum_{i=1}^{G}\left(\frac{1}{|y_i|}\sum_{t=1}^{|y_i|}\frac{\pi_\theta(y_{i,t}|x,y_{i,<t})}{\pi_{\theta_{old}}(y_{i,t}|x,y_{i,<t})}\right)\widehat{A}_i\right],$$

which can be regarded as a special case of the objective function given by Schulman et al. (2015) with the surrogate term $\frac{1}{|y_i|}\sum_{t=1}^{|y_i|}\frac{\pi_\theta(y_{i,t}|x,y_{i,<t})}{\pi_{\theta_{old}}(y_{i,t}|x,y_{i,<t})}$. Obviously it is the arithmetic mean value of importance sampling ratios $r_{i,t}=\frac{\pi_\theta(y_{i,t}|x,y_{i,<t})}{\pi_{\theta_{old}}(y_{i,t}|x,y_{i,<t})}$ with respect to tokens in the sequence y_i. By Holder-p norm, we extend the arithmetic mean value of ratios to $\rho_{i,p}(\theta)$, which is defined as

$$\rho_{i,p}(\theta) \;=\; \left(\frac{1}{|y_i|}\sum_{t=1}^{|y_i|} r_{i,t}(\theta)^p\right)^{1/p}, \qquad p\in\mathbb{R}\setminus\{0\}. \tag{1}$$

Later we discuss the case for $p=0$ in H.4. The unclipped objective function of HolderPO is

$$\mathcal{J}_{\mathrm{H}}(\theta)=\mathbb{E}_{x\sim\mathcal{D},\{y_i\}_{i=1}^{G}\sim\pi_{\theta_{\mathrm{old}}}(\cdot|x)}\left[\frac{1}{G}\sum_{i=1}^{\mathrm{G}}\rho_{i,p}(\theta)\widehat{A}_i\right]. \tag{8}$$

To calculate the policy gradient of this objective function, we first prove

$$\nabla_\theta\rho_{i,p}(\theta)=\frac{\rho_{i,p}(\theta)^{1-p}}{|y_i|}\sum_{t=1}^{|y_i|} r_{i,t}(\theta)^p\nabla_\theta\log\pi_\theta\left(y_{i,t}\mid x,y_{i,<t}\right). \tag{9}$$

Logarithmic differentiation gives

$$\nabla_\theta\log\rho_{i,p}=\frac{\sum_t r_{i,t}^p g_{i,t}}{\sum_t r_{i,t}^p},\qquad g_{i,t}=\nabla_\theta\log\pi_\theta(y_{i,t}\mid x,y_{i,<t}),$$

because the old policy is fixed and $\nabla_\theta r_{i,t}=r_{i,t}g_{i,t}$. Multiplication by $\rho_{i,p}$ proves Eq. (9).

Then the policy gradient of the unclipped HolderPO is

$$\nabla_\theta\mathcal{J}_{\mathrm{H}}(\theta)=\mathbb{E}_{x\sim\mathcal{D},\{y_i\}_{i=1}^{G}\sim\pi_{\theta_{\mathrm{old}}}(\cdot|x)}\left[\frac{1}{G}\sum_{i=1}^{\mathrm{G}}\nabla_\theta\rho_{i,p}(\theta)\widehat{A}_i\right], \tag{10}$$

whose unbiased mini-batch estimator is denoted by $\widehat{\nabla}_\theta\mathcal{J}_{\mathrm{H}}(\theta)$

$$\widehat{\nabla}_\theta\mathcal{J}_{\mathrm{H}}(\theta)=\frac{1}{B}\sum_{b=1}^{B}\left[\frac{1}{G}\sum_{i=1}^{\mathrm{G}}\nabla_\theta\rho_{i,p}(\theta)\widehat{A}_i\right].$$

where B denotes the batch size. By Eq. (9), we know

$$\widehat{\nabla}_\theta\mathcal{J}_{\mathrm{H}}(\theta)=\frac{1}{B}\sum_{b=1}^{B}\frac{1}{G}\sum_{i=1}^{\mathrm{G}}\left(\widehat{A}_i\frac{\rho_{i,p}(\theta)^{1-p}}{|y_i|}\right)\sum_{t=1}^{|y_i|} r_{i,t}(\theta)^p\nabla_\theta\log\pi_\theta(y_{i,t}\mid x,y_{i,<t}). \tag{11}$$

H.2 Token-Level Clipping Formulas

There are many PPO extensions adopting token-level clipping mechanisms to ensure training stability and prevent policy collapse. For instance, Group Relative Policy Optimisation (GRPO), Geometric-Mean Policy Optimisation (GMPO) (Zhao et al., 2025) and Dynamic Sampling Policy Optimisation (DAPO) (Yu et al., 2025). With token-level clipping, the objective function (8) of our HolderPO becomes

$$\mathcal{J}_{\mathrm{H}_t}(\theta)=\mathbb{E}_{x\sim\mathcal{D},\{y_i\}_{i=1}^{G}\sim\pi_{\theta_{\mathrm{old}}}(\cdot|x)}\left[\frac{1}{G}\left(\sum_{\widehat{A}_i>0}C_{i,p}\widehat{A}_i+\sum_{\widehat{A}_i<0}D_{i,p}\widehat{A}_i\right)\right], \tag{12}$$

$$C_{i,p}=\left(\frac{1}{|y_i|}\sum_{t=1}^{|y_i|}\min\left(r_{i,t}(\theta),\mathrm{clip}\left(r_{i,t}(\theta),1-\varepsilon,1+\varepsilon\right)\right)^p\right)^{1/p},$$

$$D_{i,p}=\left(\frac{1}{|y_i|}\sum_{t=1}^{|y_i|}\max\left(r_{i,t}(\theta),\mathrm{clip}\left(r_{i,t}(\theta),1-\varepsilon,1+\varepsilon\right)\right)^p\right)^{1/p},$$

For $0 < \epsilon < 1$, the clipped token ratio is $\min\{r_{i,t}, 1+\epsilon\}$ for a positive advantage and $\max\{r_{i,t}, 1-\epsilon\}$ for a negative advantage. This follows by factoring the sign of $\widehat{A}_i$ out of the PPO minimum. At $p = 1$ the expression recovers token-level clipped GRPO; its geometric extension is discussed in Appendix H.4.

Next we deduce the policy gradient formula of token-level clipping objective function. It is obvious that

$$\nabla_\theta \mathcal{J}_{\mathrm{H}_t}(\theta) = \mathbb{E}_{x\sim\mathcal{D},\{y_i\}_{i=1}^{G}\sim\pi_{\theta_{\mathrm{old}}}(\cdot|x)} \left[\frac{1}{G} \left(\sum_{\widehat{A}_i>0} \nabla_\theta C_{i,p} \widehat{A}_i + \sum_{\widehat{A}_i<0} \nabla_\theta D_{i,p} \widehat{A}_i \right) \right].$$

An active token ratio has derivative $r_{i,t} g_{i,t}$, while a clipped ratio is locally constant. Applying the chain rule to $C_{i,p}$ and $D_{i,p}$ therefore gives the corresponding active-token masks. Derivatives are taken away from clipping thresholds; use a consistent subgradient at a threshold. We summarise all the cases in the following formula

$$\nabla_\theta \mathcal{J}_{\mathrm{H}_t}(\theta) = \mathbb{E}_{x,\{y_i\}} \left[\frac{1}{G} \sum_{i=1}^{G} \widehat{A}_i \cdot \frac{H_{i,p}(\theta)^{1-p}}{|y_i|} \sum_{t=1}^{|y_i|} \mathbb{I}_{i,t}(\theta) \cdot r_{i,t}(\theta)^p \cdot \nabla_\theta \log \pi_\theta \left(y_{i,t} \mid x, y_{i,<t} \right) \right], \tag{13}$$

$$H_{i,p}(\theta) = \begin{cases} C_{i,p}, & \text{if } \widehat{A}_i \geq 0 \\ D_{i,p}, & \text{if } \widehat{A}_i < 0, \end{cases}$$

$$\mathbb{I}_{i,t}(\theta) = \begin{cases} 0, & \text{if } \widehat{A}_i > 0 \text{ and } r_{i,t}(\theta) > 1+\epsilon, \\ & \text{or, if } \widehat{A}_i < 0 \text{ and } r_{i,t}(\theta) < 1-\epsilon \\ 1, & \text{otherwise.} \end{cases}$$

The unbiased mini-batch estimator is

$$\widehat{\nabla}_\theta \mathcal{J}_{\mathrm{H}_t}(\theta) = \frac{1}{B} \sum_{b=1}^{B} \frac{1}{G} \sum_{i=1}^{G} \widehat{A}_i \cdot \frac{H_{i,p}(\theta)^{1-p}}{|y_i|} \sum_{t=1}^{|y_i|} \mathbb{I}_{i,t}(\theta) \cdot r_{i,t}(\theta)^p \cdot \nabla_\theta \log \pi_\theta \left(y_{i,t} \mid x, y_{i,<t} \right). \tag{14}$$

H.3 Sequence-Level Clipping Formulas

Notably, alongside the widespread adoption of token-level clipping, several recent studies have shifted towards sequence-level clipping strategies, such as Group Sequence Policy Optimisation (GSPO) (Zheng et al., 2025), whose objective function is

$$\mathcal{J}_{\mathrm{GSPO}}(\theta) = \mathbb{E}_{x\sim\mathcal{D},\{y_i\}_{i=1}^{G}\sim\pi_{\theta_{\mathrm{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^{G} \min \left(s_i(\theta) \widehat{A}_i, \mathrm{clip} \left(s_i(\theta), 1-\varepsilon, 1+\varepsilon \right) \widehat{A}_i \right) \right],$$

where $s_i(\theta) = \left(\prod_{t=1}^{|y_i|} r_{i,t} \right)^{\frac{1}{|y_i|}} = \exp \left(\frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \log \frac{\pi_\theta(y_{i,t}|x,y_{i,<t})}{\pi_{\theta_{\mathrm{old}}}(y_{i,t}|x,y_{i,<t})} \right)$ is the geometric mean of ratio of each token. Actually it is a special case of our HolderPO objective function with sequence-level clipping

$$\mathcal{J}_{\mathrm{H}_s}(\theta) = \mathbb{E}_{x\sim\mathcal{D},\{y_i\}_{i=1}^{G}\sim\pi_{\theta_{\mathrm{old}}}(\cdot|x)} \left\{ \frac{1}{G} \sum_{i=1}^{G} \min \left[\rho_{i,p}(\theta) \widehat{A}_i, \mathrm{clip} \left(\rho_{i,p}(\theta), 1-\epsilon, 1+\epsilon \right) \widehat{A}_i \right] \right\}. \tag{15}$$

In Lemma 1, we will show $s_i(\theta)$ in GSPO is equal to $\rho_{i,0}(\theta)$. By a similar discussion for $\rho_{i,p}$, we can obtain the policy gradient

$$\nabla_\theta \mathcal{J}_{\mathrm{H}_s}(\theta) = \mathbb{E}_{x,\{y_i\}} \left[\frac{1}{G} \sum_{i=1}^{G} \mathbb{I}_i(\theta) \cdot \widehat{A}_i \cdot \frac{\rho_{i,p}(\theta)^{1-p}}{|y_i|} \sum_{t=1}^{|y_i|} r_{i,t}(\theta)^p \nabla_\theta \log \pi_\theta \left(y_{i,t} \mid x, y_{i,<t} \right) \right], \tag{16}$$

$$\mathbb{I}_i(\theta) = \begin{cases} 0, & \text{if } \widehat{A}_i > 0 \text{ and } \rho_{i,p}(\theta) > 1+\epsilon, \text{ or, if } \widehat{A}_i < 0 \text{ and } \rho_{i,p}(\theta) < 1-\epsilon \\ 1, & \text{otherwise.} \end{cases}$$

The unbiased mini-batch estimator is

$$\widehat{\nabla}_\theta \mathcal{J}_{\mathrm{H}_s}(\theta) = \frac{1}{B}\sum_{b=1}^{B}\frac{1}{G}\sum_{i=1}^{G}\mathbb{I}_i(\theta)\cdot\widehat{A}_i\cdot\frac{\rho_{i,p}(\theta)^{1-p}}{|y_i|}\sum_{t=1}^{|y_i|} r_{i,t}(\theta)^p \nabla_\theta \log \pi_\theta\left(y_{i,t} \mid x, y_{i,<t}\right). \tag{17}$$

H.4 $p = 0$ FORMULAS

Lemma 1. *For positive $x_1, \ldots, x_n$, the Holder mean extends at $p = 0$ to the geometric mean* $\exp(n^{-1}\sum_t \log x_t)$.

Proof. Let $K(p) = \log(n^{-1}\sum_t e^{p\log x_t})$. Since $K(0) = 0$,

$$\lim_{p\to 0}\log M_p = \lim_{p\to 0}\frac{K(p)}{p} = K'(0) = \frac{1}{n}\sum_t \log x_t.$$

Exponentiating gives the claim. □

This extends both unclipped and clipped aggregates, since the selected token-clipped ratios remain positive for $0 < \epsilon < 1$.

Theorem 4 (Gradient extension at zero)**.** *For a fixed finite rollout, assume positive, continuously differentiable token ratios in a neighbourhood of the parameter being considered. Then $\rho_{i,p}$ and its parameter gradient extend continuously to $p = 0$, with*

$$\rho_{i,0} = \exp\left(\frac{1}{n_i}\sum_t \log r_{i,t}\right), \qquad \nabla_\theta \rho_{i,0} = \frac{\rho_{i,0}}{n_i}\sum_t \nabla_\theta \log r_{i,t}.$$

Proof. Set $K(p,\theta) = \log(n_i^{-1}\sum_t e^{p\log r_{i,t}(\theta)})$. The identity

$$\frac{K(p,\theta)}{p} = \int_0^1 \partial_p K(up,\theta)\, du$$

removes the singularity at zero. The finite sums and positive denominators make the integrand and its parameter gradient continuous locally. For $p \neq 0$, $\nabla_\theta \rho_{i,p} = \rho_{i,p}\sum_t W_{i,t}(p)\nabla_\theta \log r_{i,t}$; using $W_{i,t}(0) = 1/n_i$ proves the extension. □

The clipped formulas have the same extension on regions where their masks are fixed; at a clipping threshold use a consistent subgradient. Passing a parameter derivative through a population expectation additionally requires the usual integrability or domination conditions. The argument above establishes the finite-rollout formulas used in the estimators.

I LOCAL GEOMETRY AT THE OLD POLICY

At the old policy, every token ratio is one, so fixed orders share the same first update. The theorem below identifies their curvature difference on the same rollout and with the same advantage.

Theorem A1 (Fixed-order local geometry at the old policy)**.** *Assume that the token log-probabilities are twice continuously differentiable in a neighbourhood of θ_{old}. Write $n_i = |y_i|$ and, at $\theta = \theta_{\mathrm{old}}$, let*

$$g_{i,t} \coloneqq \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t}), \qquad \bar{g}_i \coloneqq \frac{1}{n_i}\sum_{t=1}^{n_i} g_{i,t}, \qquad C_i \coloneqq \frac{1}{n_i}\sum_{t=1}^{n_i}(g_{i,t}-\bar{g}_i)(g_{i,t}-\bar{g}_i)^\top. \tag{18}$$

Then, for any finite $p, q \in \mathbb{R}$,

$$\nabla_\theta^2 \rho_{i,p}(\theta_{\mathrm{old}}) - \nabla_\theta^2 \rho_{i,q}(\theta_{\mathrm{old}}) = (p-q)C_i. \tag{19}$$

The token score $g_{i,t}$ is the gradient of token t's log probability, and $\bar{g}_i$ is its average across the sequence. For a small parameter displacement d, the scalar $g_{i,t}^\top d$ is the first-order change in that token's log-ratio. Accordingly, $d^\top C_i d$ is the variance of these first-order changes across token positions. It identifies the directions in which choosing a different order has the largest local effect: the curvature difference is greater when the token log-ratios move more unevenly along that direction.

Proof of Theorem A1. For this proof only, write $z_{i,t}(\theta) = \log r_{i,t}(\theta)$ and

$$m_{i,p}(\theta) = \sum_t W_{i,t}(p) \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t}).$$

Differentiating Eq. (1) gives

$$\nabla_\theta \log \rho_{i,p}(\theta) = m_{i,p}(\theta), \qquad \nabla_\theta W_{i,t}(p) = p W_{i,t}(p) \big(g_{i,t}(\theta) - m_{i,p}(\theta)\big), \tag{20}$$

where

$$g_{i,t}(\theta) = \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t}).$$

Consequently,

$$\begin{aligned} \nabla^2_\theta \log \rho_{i,p}(\theta) = & \sum_t W_{i,t}(p) \nabla^2_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t}) \\ & + p \left(\sum_t W_{i,t}(p) g_{i,t}(\theta) g_{i,t}(\theta)^\top - m_{i,p}(\theta) m_{i,p}(\theta)^\top \right). \end{aligned} \tag{21}$$

Using

$$\nabla^2 \rho = \rho \left\{ \nabla^2 \log \rho + (\nabla \log \rho)(\nabla \log \rho)^\top \right\}$$

gives the Hessian of $\rho_{i,p}$.

At $\theta = \theta_{\text{old}}$, every $r_{i,t} = 1$, hence $\rho_{i,p} = 1$, $W_{i,t}(p) = 1/n_i$, and $m_{i,p} = \bar{g}_i$, independently of p. Thus the aggregates have the common value 1 and gradient $\bar{g}_i$ at the old policy. All terms in the Hessian are also independent of p except the covariance term in Eq. (21); at the old policy this covariance is exactly C_i. Subtracting the Hessians for p and q proves the last identity. The $p = 0$ case follows by the removable extension established in Appendix H.4.

Finally, for any displacement d,

$$d^\top C_i d = \frac{1}{n_i} \sum_t \big((g_{i,t} - \bar{g}_i)^\top d\big)^2, \tag{22}$$

while Taylor expansion gives

$$\log r_{i,t}(\theta_{\text{old}} + d) = g_{i,t}^\top d + \mathcal{O}(\|d\|^2).$$

Thus $d^\top C_i d$ is precisely the leading token-wise variance of the log-ratio drift along d. □

J Distribution Deformation

This appendix supplements Section 3.2 by providing formal proofs and broader theoretical contexts for our gradient concentration mechanism. Specifically, J.1 and J.2 present the proofs for the local weight allocation (Theorem 5) and the global distributional deformation (Theorem 1), respectively. Furthermore, J.4 discusses the profound connection between our three gradient concentration regimes and the traditional exploration-exploitation trade-off in reinforcement learning.

J.1 Local Property

In this section, we prove the following theorem, which is mentioned in Section 3.2 as the local property of the token-level weight allocation $W_{i,t}(p)$ induced by the aggregation parameter p.

Theorem 5. *Assume at least two token ratios differ, and let $\mathcal{T}^* = \arg\max_t r_{i,t}(\theta)$. Each maximal-ratio token's weight increases with p and tends to $1/|\mathcal{T}^*|$ as $p \to \infty$. Every non-maximal token's weight tends to zero and eventually decreases. Starting from p_0, an interior maximum occurs only if the increasing weighted mean $\mu_{y_i}(p)$ crosses that token's log-ratio; a token already below this mean need not gain weight first.*

To prove Theorem 5, we first establish two fundamental lemmas regarding the dynamic weight allocation mechanism controlled by p. Let $\mu_{y_i}(p)$ denote the $W_{i,t}$-weighted mean of the log-ratios across the sequence:

$$\mu_{y_i}(p) := \sum_{t=1}^{|y_i|} W_{i,t}(p) \log r_{i,t}(\theta). \tag{23}$$

Lemma 2. *The partial derivative of the gradient weight $W_{i,t}(p)$ with respect to the Holder parameter p is strictly governed by its log-ratio relative to the sequence mean $\mu_{y_i}(p)$:*

$$\frac{\partial W_{i,t}(p)}{\partial p} = W_{i,t}(p)\Big(\log r_{i,t}(\theta) - \mu_{y_i}(p)\Big). \tag{24}$$

Proof. To provide a complete calculation, we begin by rewriting the gradient weight definition in its exponential form. By expanding the base $r_{i,t}(\theta)^p$, the weight can be expressed as

$$W_{i,t}(p) = \frac{\exp(p \log r_{i,t}(\theta))}{\sum_{k=1}^{|y_i|} \exp(p \log r_{i,k}(\theta))}.$$

Let $u = \exp(p \log r_{i,t}(\theta))$ and $v = \sum_{k=1}^{|y_i|} \exp(p \log r_{i,k}(\theta))$. Using the chain rule, the derivative of the numerator is simply

$$\frac{\partial u}{\partial p} = \exp(p \log r_{i,t}(\theta)) \log r_{i,t}(\theta),$$

while the derivative of the denominator is

$$\frac{\partial v}{\partial p} = \sum_{k=1}^{|y_i|} \exp(p \log r_{i,k}(\theta)) \log r_{i,k}(\theta).$$

The quotient rule formula is

$$\frac{d}{dp}\left[\frac{u}{v}\right] = \frac{u'v - uv'}{v^2}.$$

Now, we substitute these components back into the quotient rule formula

$$\frac{\partial W_{i,t}(p)}{\partial p} = \frac{\exp(p \log r_{i,t}(\theta)) \log r_{i,t}(\theta)}{\sum_k \exp(p \log r_{i,k}(\theta))} - \frac{\exp(p \log r_{i,t}(\theta)) \left[\sum_k \exp(p \log r_{i,k}(\theta)) \log r_{i,k}(\theta)\right]}{\left[\sum_k \exp(p \log r_{i,k}(\theta))\right]^2}.$$

Looking closely at the first term, we can isolate the definition of the original weight $W_{i,t}(p)$, leaving us with $W_{i,t}(p) \log r_{i,t}(\theta)$. For the second term, we can factor the fraction into the product of two separate fractions. The first fraction is exactly $W_{i,t}(p)$, and the second fraction represents the weighted sum over all tokens

$$\frac{\partial W_{i,t}(p)}{\partial p} = W_{i,t}(p) \log r_{i,t}(\theta) - W_{i,t}(p) \left[\sum_{k=1}^{|y_i|} \frac{\exp(p \log r_{i,k}(\theta))}{\sum_j \exp(p \log r_{i,j}(\theta))} \log r_{i,k}(\theta)\right].$$

We know that the term inside the summation is simply $W_{i,k}(p) \log r_{i,k}(\theta)$, and the entire bracketed sum represents $\mu_{y_i}(p) = \sum_k W_{i,k}(p) \log r_{i,k}(\theta)$. Substituting this notation into our equation gives

$$\frac{\partial W_{i,t}(p)}{\partial p} = W_{i,t}(p) \log r_{i,t}(\theta) - W_{i,t}(p)\mu_{y_i}(p) = W_{i,t}(p)\Big(\log r_{i,t}(\theta) - \mu_{y_i}(p)\Big).$$

□

Lemma 3. *Assuming the sequence contains at least two tokens with differing importance ratios, the weighted sequence mean $\mu_{y_i}(p)$ is strictly monotonically increasing with respect to p.*

Proof. Taking the derivative of $\mu_{y_i}(p)$ with respect to p, we have

$$\frac{\partial \mu_{y_i}(p)}{\partial p} = \sum_{t=1}^{|y_i|} \frac{\partial W_{i,t}(p)}{\partial p} \log r_{i,t} = \sum_{t=1}^{|y_i|} W_{i,t}(p)\Big(\log r_{i,t} - \mu_{y_i}(p)\Big) \log r_{i,t}.$$

Since $\sum_{t=1}^{|y_i|} W_{i,t}(p) = 1$ and by the definition of the mean $\mu_{y_i}(p) = \sum_{t=1}^{|y_i|} W_{i,t}(p) \log r_{i,t}$, we have

$$\sum_{t=1}^{|y_i|} W_{i,t}(p)\Big(\log r_{i,t} - \mu_{y_i}(p)\Big) = \mu_{y_i}(p) - \mu_{y_i}(p) = 0.$$

Multiplying this entire zero-sum by the constant $\mu_{y_i}(p)$ yields

$$\sum_{t=1}^{|y_i|} W_{i,t}(p)\mu_{y_i}(p)\Big(\log r_{i,t} - \mu_{y_i}(p)\Big) = 0.$$

We can subtract this identically zero term from our derivative equation without changing its value. By grouping the common factor $W_{i,t}(p)\big(\log r_{i,t} - \mu_{y_i}(p)\big)$, the equation collapses into a squared difference

$$\begin{aligned}\frac{\partial \mu_{y_i}(p)}{\partial p} &= \sum_{t=1}^{|y_i|} W_{i,t}(p)\Big(\log r_{i,t} - \mu_{y_i}(p)\Big)\log r_{i,t} - \sum_{t=1}^{|y_i|} W_{i,t}(p)\mu_{y_i}(p)\Big(\log r_{i,t} - \mu_{y_i}(p)\Big) \\ &= \sum_{t=1}^{|y_i|} W_{i,t}(p)\Big(\log r_{i,t} - \mu_{y_i}(p)\Big)^2.\end{aligned}$$

This final expression is exactly the definition of the variance of $\log r_{i,t}$ under the weight distribution W_i^p, denoted as $\mathrm{Var}_{W_i^p}(\log r_{i,t})$. Given our assumption that the sequence contains at least two tokens with differing importance ratios, this variance is strictly positive. □

Proof of Theorem 5. By definition, $\mathcal{T}^*$ contains the tokens with the strictly maximum importance ratio. Since the weighted sequence mean $\mu_{y_i}(p)$ is a convex combination of all token log-ratios (with $W_{i,k}(p) > 0$ for all finite p), it must be strictly bounded by the maximum value: $\mu_{y_i}(p) < \log r_{i,t^*}$ for any finite p. By Lemma 2, the derivative of the weight is governed by its deviation from this mean: $\frac{\partial W_{i,t^*}(p)}{\partial p} = W_{i,t^*}(p)(\log r_{i,t^*} - \mu_{y_i}(p))$. Because both the weight and the deviation are strictly positive, the weight of any optimal token increases monotonically as p grows.

Furthermore, Lemma 3 establishes that $\mu_{y_i}(p)$ is strictly monotonically increasing with p. Since it is continuously increasing and bounded above by the maximum log-ratio, it is convergent as $p \to +\infty$. By dividing the numerator and the denominator of $W_{i,t}(p)$ by r_{i,t^*}^p, where r_{i,t^*} is the maximum ratio, we rewrite the weight as

$$W_{i,t}(p) = \frac{\left(\frac{r_{i,t}}{r_{i,t^*}}\right)^p}{\sum_{k=1}^{|y_i|}\left(\frac{r_{i,k}}{r_{i,t^*}}\right)^p}.$$

For any optimal token $t^* \in \mathcal{T}^*$, the base is 1. For any sub-optimal token $k \notin \mathcal{T}^*$, the base is strictly less than 1, causing $(r_{i,k}/r_{i,t^*})^p \to 0$ as $p \to \infty$. Consequently, the denominator converges exactly to $|\mathcal{T}^*|$, the total number of optimal tokens. Thus, the weight distribution concentrates entirely on the optimal subset: $\lim_{p\to\infty} W_{i,t^*}(p) = 1/|\mathcal{T}^*|$ and $\lim_{p\to\infty} W_{i,k}(p) = 0$. Since the sequence mean is defined as $\mu_{y_i}(p) = \sum_t W_{i,t}(p)\log r_{i,t}$, taking the limit yields:

$$\lim_{p\to\infty} \mu_{y_i}(p) = \sum_{t\in\mathcal{T}^*}\left(\frac{1}{|\mathcal{T}^*|}\log r_{i,t^*}\right) + \sum_{k\notin\mathcal{T}^*}(0\cdot\log r_{i,k}) = \log r_{i,t^*}.$$

Combining this exact limit with Lemma 3, we establish that $\mu_{y_i}(p)$ strictly and monotonically approaches the maximum log-ratio $\log r_{i,t^*}$.

For any non-maximal token, its log-ratio is below the limiting mean $\max_t \log r_{i,t}$. Its derivative is therefore eventually negative. Because $\mu_{y_i}(p)$ is strictly increasing, it can cross this log-ratio at most once. If the crossing occurs after p_0, it is the unique maximum on that interval; otherwise the weight is already decreasing. The limit calculation above gives convergence to zero in either case. □

J.2 Global Property

In this section, we prove the following Theorem 1, which is mentioned in Section 3.2 as the global property of the sequence-level distributional deformation induced by the aggregation parameter p.

Theorem. *Assume the sequence y_i contains at least two tokens with distinct importance ratios. Then the Shannon entropy of the weight distribution attains its global maximum at $p = 0$, where $W_{i,t}(0) = 1/|y_i|$ for each token, and strictly decreases as p moves away from zero along either ray. Moreover, as $p \to \pm\infty$, W_i^p concentrates uniformly on the subset $\mathcal{T}^+ = \arg\max_t r_{i,t}(\theta)$ and $\mathcal{T}^- = \arg\min_t r_{i,t}(\theta)$, respectively.*

Proof. The Shannon entropy of the weight distribution is defined as

$$\mathcal{H}(W_i^p) := -\sum_t W_{i,t}(p) \ln W_{i,t}(p).$$

To analyse its monotonicity, we compute the derivative of $\mathcal{H}$ with respect to p. First, we compute the derivative of the token weight $W_{i,t}$. Let

$$\mathbb{E}_{W_i^p}[\ln r_{i,t}] \coloneqq \sum_{k=1}^{|y_i|} W_{i,k} \ln r_{i,k}$$

denote the expected log-ratio under the current weight distribution. The derivative of the weight is given by

$$\frac{\partial W_{i,t}}{\partial p} = W_{i,t} \left(\ln r_{i,t} - \sum_k W_{i,k} \ln r_{i,k} \right) = W_{i,t}(\ln r_{i,t} - \mathbb{E}_{W_i^p}[\ln r_{i,t}]).$$

Next, taking the derivative of the entropy yields

$$\frac{\partial \mathcal{H}}{\partial p} = -\sum_t \left(\frac{\partial W_{i,t}}{\partial p} \ln W_{i,t} + W_{i,t} \frac{\partial \ln W_{i,t}}{\partial p} \right).$$

Notice that

$$\sum_t W_{i,t} \frac{\partial \ln W_{i,t}}{\partial p} = \sum_t \frac{\partial W_{i,t}}{\partial p} = \frac{\partial}{\partial p} \left(\sum_t W_{i,t} \right) = 0.$$

Thus, the entropy derivative simplifies to

$$\frac{\partial \mathcal{H}}{\partial p} = -\sum_t \frac{\partial W_{i,t}}{\partial p} \ln W_{i,t}.$$

To proceed, we explicitly write out $\ln W_{i,t}$. Let $Z(p) \coloneqq \sum_k r_{i,k}^p$. Since $W_{i,t} = r_{i,t}^p / Z(p)$, taking the natural logarithm gives

$$\ln W_{i,t} = p \ln r_{i,t} - \ln \sum_k r_{i,k}^p = p \ln r_{i,t} - \ln Z(p).$$

Substituting both $\frac{\partial W_{i,t}}{\partial p}$ and $\ln W_{i,t}$ into the simplified entropy derivative, we obtain

$$\frac{\partial \mathcal{H}}{\partial p} = -\sum_t \Big[W_{i,t} \big(\ln r_{i,t} - \mathbb{E}_{W_i^p}[\ln r_{i,t}] \big) \Big] \Big[p \ln r_{i,t} - \ln Z(p) \Big].$$

We can expand this product into two separate sums. Notice that the expected deviation from the mean is identically zero. Specifically, because $\mathbb{E}_{W_i^p}[\ln r_{i,t}]$ is a constant with respect to the summation index t, we have

$$\sum_t W_{i,t} \big(\ln r_{i,t} - \mathbb{E}_{W_i^p}[\ln r_{i,t}] \big) = \sum_t W_{i,t} \ln r_{i,t} - \mathbb{E}_{W_i^p}[\ln r_{i,t}] \sum_t W_{i,t} = 0.$$

Because this term is zero, any constant multiplier distributed into it vanishes. When we distribute the expanded brackets, the term multiplied by the constant $\ln Z(p)$ completely drops out

$$\sum_t W_{i,t} \big(\ln r_{i,t} - \mathbb{E}_{W_i^p}[\ln r_{i,t}] \big) \ln Z(p) = 0.$$

This leaves only the term

$$\frac{\partial \mathcal{H}}{\partial p} = -p \sum_t W_{i,t} \big(\ln r_{i,t} - \mathbb{E}_{W_i^p}[\ln r_{i,t}] \big) \ln r_{i,t}.$$

Finally, we expand the remaining summation by distributing $\ln r_{i,t}$

$$\frac{\partial \mathcal{H}}{\partial p} = -p \left(\sum_t W_{i,t} (\ln r_{i,t})^2 - \mathbb{E}_{W_i^p}[\ln r_{i,t}] \sum_t W_{i,t} \ln r_{i,t} \right).$$

Recognising that $\sum_t W_{i,t} \ln r_{i,t}$ is exactly $\mathbb{E}_{W_i^p}[\ln r_{i,t}]$, this equation collapses into the definition of variance

$$\frac{\partial \mathcal{H}}{\partial p} = -p \left(\mathbb{E}_{W_i^p}[(\ln r_{i,t})^2] - \left(\mathbb{E}_{W_i^p}[\ln r_{i,t}] \right)^2 \right) = -p \cdot \mathrm{Var}_{W_i^p}(\ln r_{i,t}).$$

Since the sequence contains non-uniform importance ratios, the variance is strictly positive ($\mathrm{Var}_{W_i^p}(\ln r_{i,t}) > 0$). Therefore for $p > 0$, $\frac{\partial \mathcal{H}}{\partial p} < 0$, meaning $\mathcal{H}$ strictly decreases. For $p < 0$, $\frac{\partial \mathcal{H}}{\partial p} > 0$, meaning $\mathcal{H}$ strictly increases towards $p = 0$ (or decreases as $p \to -\infty$). At $p = 0$, W_i^0 becomes a uniform distribution where each token is assigned an identical weight of $1/|y_i|$, and $\mathcal{H}$ reaches its global maximum $\ln |y_i|$.

Finally, we evaluate the limits as $p \to \pm\infty$. Let $r_{\max} = \max_t r_{i,t}$ and $\mathcal{M}^* = \{k \mid r_{i,k} = r_{\max}\}$. We can rewrite $W_{i,t}(p)$ by dividing the numerator and denominator by $r_{\max}^p$:

$$W_{i,t}(p) = \frac{(r_{i,t}/r_{\max})^p}{\sum_{k \in \mathcal{M}^*} 1 + \sum_{j \notin \mathcal{M}^*} (r_{i,j}/r_{\max})^p}.$$

For $j \notin \mathcal{M}^*$, $r_{i,j}/r_{\max} < 1$, so $(r_{i,j}/r_{\max})^p \to 0$ as $p \to \infty$. Thus, the denominator converges to $|\mathcal{M}^*|$. For the numerator, if $t \in \mathcal{M}^*$, $(r_{i,t}/r_{\max})^p = 1$. If $t \notin \mathcal{M}^*$, $(r_{i,t}/r_{\max})^p \to 0$. Therefore, $\lim_{p\to\infty} W_{i,t}(p) = \frac{1}{|\mathcal{M}^*|}$ for $t \in \mathcal{M}^*$, and 0 otherwise.

Let $r_{\min} = \min_t r_{i,t}$ and $\mathcal{M}_{\min} = \{k \mid r_{i,k} = r_{\min}\}$. Let $q = -p$, so $q \to \infty$. We can rewrite the weight as:

$$W_{i,t}(-q) = \frac{(1/r_{i,t})^q}{\sum_k (1/r_{i,k})^q} = \frac{(r_{\min}/r_{i,t})^q}{\sum_{k \in \mathcal{M}_{\min}} 1 + \sum_{j \notin \mathcal{M}_{\min}} (r_{\min}/r_{i,j})^q}.$$

Since $r_{\min}/r_{i,j} < 1$ for $j \notin \mathcal{M}_{\min}$, the term $(r_{\min}/r_{i,j})^q \to 0$ as $q \to \infty$. Following the exact same logic, the distribution collapses to a uniform distribution over $\mathcal{M}_{\min}$. □

J.3 Finite-Order Sensitivity Bounds

We compare orders at the same rollout and policy state. The log-ratio range controls changes in weights, aggregate scale, and gradient contribution.

Theorem A2 (Finite-order sensitivity)**.** *Fix a rollout y_i and policy parameter θ, write $n_i = |y_i|$, and define*

$$D_i = \max_t \log r_{i,t}(\theta) - \min_t \log r_{i,t}(\theta). \tag{25}$$

For finite p, q,

$$\|W_i^p - W_i^q\|_1 \le \min\{2, |p-q| D_i/2\}, \tag{26}$$

$$0 \le \partial_p \log \rho_{i,p} \le D_i^2/8, \tag{27}$$

$$\frac{\mathrm{ESS}(W_i^p)}{n_i} \ge \mathrm{sech}^2(|p| D_i/2), \quad \mathrm{ESS}(W_i^p) = \left(\sum_t W_{i,t}(p)^2 \right)^{-1}. \tag{28}$$

Moreover,

$$\log n_i - \mathcal{H}(W_i^p) = p^2 \partial_p \log \rho_{i,p} \le p^2 D_i^2/8. \tag{29}$$

If all token score norms are bounded by M, then, holding the advantage fixed,

$$\left\| \partial_p [\widehat{A}_i \nabla_\theta \rho_{i,p}] \right\| \le |\widehat{A}_i| \rho_{i,p} M (D_i/2 + D_i^2/8). \tag{30}$$

At zero, derivatives use the extension in Appendix H.4.

Proof. Suppress i and let $z_t = \log r_{i,t}$, $K(p) = \log(n^{-1} \sum_t e^{p z_t})$, and $\mu(p) = \sum_t W_t(p) z_t$. Differentiation gives

$$K' = \mu, \qquad K'' = \mathrm{Var}_{W(p)}(z), \qquad W_t' = W_t(z_t - \mu). \tag{31}$$

An interval of length D has variance at most $D^2/4$, because $\mathrm{Var}(z) \le \mathbb{E}(z - (\min z + \max z)/2)^2$. Thus Cauchy–Schwarz yields

$$\sum_t |W_t'| = \mathbb{E}_W |z - \mu| \le \sqrt{\mathrm{Var}_W(z)} \le D/2. \tag{32}$$

Integration from p to q, together with the maximal distance 2 between probability vectors, proves Eq. (26).

Since $\log \rho_p = K(p)/p$ and $K(0) = 0$,

$$\partial_p \log \rho_p = \frac{pK'(p) - K(p)}{p^2} = \int_0^1 u \operatorname{Var}_{W(up)}(z)\, du \in [0, D^2/8]. \tag{33}$$

The integral also gives the extension at zero. This proves the aggregate bound, while $\mathcal{H}(W(p)) = K(p) + \log n - pK'(p)$ gives the entropy identity.

For ESS, put $b_t = e^{pz_t}$, $a = \min_t b_t$, $b = \max_t b_t$, and $\bar{b} = n^{-1} \sum_t b_t$. From $(b_t - a)(b - b_t) \geq 0$,

$$\frac{n^{-1} \sum_t b_t^2}{\bar{b}^2} \leq \frac{(a+b)\bar{b} - ab}{\bar{b}^2} \leq \frac{(a+b)^2}{4ab}.$$

The second inequality follows by expanding $((a+b)\bar{b} - 2ab)^2 \geq 0$. Taking reciprocals and using $b/a = e^{|p|D}$ proves Eq. (28).

Finally let $s_t = \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t})$, $m_p = \sum_t W_t(p) s_t$, and $\alpha_p = \partial_p \log \rho_p$. The product rule yields

$$\partial_p[\widehat{A}_i \nabla_\theta \rho_{i,p}] = \widehat{A}_i \rho_{i,p} \left(\alpha_p m_p + \sum_t W_t'(p) s_t \right).$$

Use $\|m_p\| \leq M$, $\alpha_p \leq D^2/8$, and $\| \sum_t W_t' s_t \| \leq M \sum_t |W_t'| \leq MD/2$. □

ESS is an effective token count for this weight vector, not a sample-size guarantee for independent data. The bounds limit sensitivity when within-sequence ratios are similar; a larger range permits, but does not force, a larger change. They do not identify which direction improves the objective. The alignment analysis is in Appendix P, and buffer drift during training is treated in Appendix N.5.

Relation to the experiments. Figures 2 and 3 motivate studying changing ratio profiles and gradient behaviour. Applying these bounds quantitatively requires within-sequence D_i and the stated score bounds; the figures alone do not estimate all of these quantities.

J.4 Gradient Concentration vs. Exploration-Exploitation Trade-off

As established in Section 3.2, the aggregation parameter p induces three distinct gradient concentration regimes: upward concentration ($p > 0$) strictly allocates the gradient weights onto high-ratio tokens, uniform dispersion ($p \to 0$) distributes the gradient equally across all tokens, and downward concentration ($p < 0$) upweights the gradient contributions of low-ratio, hesitant tokens. Conceptually, dynamically shifting between these regimes closely mirrors the classical exploration-exploitation dilemma in reinforcement learning (Sutton & Barto, 2018). However, the exact nature of this mechanism in the context of LLMs requires careful theoretical contextualisation.

Is a large p considered "Exploitation"? When $p \gg 0$, the algorithm hyper-focuses the gradient updates on tokens where the current policy has already shown the most aggressive improvement relative to the reference policy (i.e., maximal importance ratios). In traditional RL, exploitation implies a behavioural shift—acting greedily according to the current value function during environmental interaction. In our framework, however, a large p acts as a form of post-hoc exploitation that precisely targets two urgent algorithmic crises recently identified in LLM reasoning: the distribution sharpening trap and spurious rewards.

Recent work by (He et al., 2025) reveals that standard GRPO is fundamentally constrained by a distribution sharpening effect, predominantly rewarding tokens that are already likely while failing to amplify sparse, unlikely, yet correct reasoning leaps. Furthermore, (Shao et al., 2025) demonstrate that Reinforcement Learning with Verifiable Rewards (RLVR) is heavily plagued by spurious signals, where flawed intermediate logic coincidentally yields a correct final answer. When sequence-level advantages are distributed uniformly across the entire trajectory, the optimiser inevitably reinforces this uninformative or even toxic pseudo-logic.

Our upward concentration mechanism ($p > 0$) provides a mathematically principled resolution to both vulnerabilities. It does not exploit by altering the sampling trajectory, but by aggressively filtering the learning signal. By exponentially amplifying the gradient weights of rare, high-ratio tokens, it bypasses the sharpening trap to successfully obtain genuine "aha moments" (He et al., 2025). Simultaneously, by starving the gradient from the bulk of unremarkable tokens, it naturally defends the policy against the integration of spurious background noise (Shao et al., 2025). This theoretical intuition is vividly corroborated by our empirical results in Table 2: on the AIME24 benchmark, where correct reasoning steps are exceptionally sparse, a highly aggressive static configuration of $p = 3$ achieves the peak accuracy of 46.7%. Furthermore, as demonstrated in Figure 3, setting $p = +2$ rapidly drives the policy entropy down during the early training stages, visually confirming this intense knowledge-sharpening and mass-concentration effect.

Is a negative p considered "Exploration"? Conversely, when $p < 0$, the gradient concentration shifts toward tokens where the model exhibits hesitation or deviation from previously confident paths. In standard continuous-control RL, exploration is typically enforced via entropy bonuses in the objective (Haarnoja et al., 2018; Schulman et al., 2017) or noise injection during sampling. While $p < 0$ empirically preserves reasoning diversity, it is not an exploration mechanism in the active sense. Instead, it serves as retrospective diversity preservation. By upweighting less-confident tokens within successful trajectories, it forces the optimiser to consolidate alternative, unconventional reasoning pathways rather than collapsing into a single, greedy solution. We observe this exact dynamic in Figure 3, where a static $p = -2$ sustains significantly higher entropy levels across the entire training trajectory compared to positive p values, actively resisting mode collapse. Moreover, Figure 2 illustrates that decreasing p systematically tightens the gap between the upper and lower envelopes of token-level ratios, redistributing credit to underemphasised tokens. This variance-controlling mechanism proves exceptionally beneficial for dense-signal tasks like MATH500, which strictly favours lower p values (peaking at $p = -1$) to maintain stable optimisation.

To formalise this critical boundary between our gradient mechanisms and traditional RL terminology, we state the following remark.

Remark 1. *While our concentration mechanism conceptually echoes the exploration-exploitation tradeoff, with $p < 0$ preserving diversity and $p > 0$ sharpening known knowledge, it must not be conflated with classical exploration. In standard RL, exploration refers to actively altering the trajectory sampling distribution (the behavioural policy) to visit unseen states. In contrast, our parameter p operates entirely on the hindsight aggregation of already-sampled trajectories. It functions strictly as a gradient reweighting mechanism, reshaping how the optimisation priority is distributed across a fixed rollout without intervening in how the rollouts are generated.*

K Variance Behaviours

This appendix provides analysis of the policy gradient variance under the HolderPO framework. We first establish a monotonic upper bound for the variance of the unclipped estimator in Section K.1, and then extend it to the sequence-level clipping case in Section K.2, proving Theorem 2.

K.1 An upper bound with monotonicity

In this section, we prove another version of Theorem 2 for the unclipped gradient estimator (11).

Theorem 6. *Let $\widehat{\nabla}_\theta \mathcal{J}_H$ (Eq. (11)) denote the estimator. Assume $\|\nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t})\| \le M$ for all tokens within the batch, the variance admits the bound*

$$\|\mathrm{Var}(\widehat{\nabla}_\theta \mathcal{J}_H)\| \;\le\; \frac{M^2}{B}\,\mathbb{E}\Big[\widehat{A}_i^2\,\rho_{i,p}^2(\theta)\Big], \tag{34}$$

which is monotonically increasing in p for all $p \in \mathbb{R}$, where B is the batch size.

Proof. We compute the unbiased estimator of the policy gradient by sampling a mini-batch of size B, denoted as $\widehat{\nabla}_\theta \mathcal{J}_{\mathrm{H}}$. For a mini-batch containing B i.i.d. sampled trajectories, we have

$$\widehat{\nabla}_\theta \mathcal{J}_{\mathrm{H}} = \frac{1}{B}\sum_{i=1}^{B} \hat{g}_i(p),$$

where the unclipped single-step stochastic gradient $\hat{g}_i(p)$ is

$$\hat{g}_i(p) = \widehat{A}_i \cdot \nabla_\theta \rho_{i,p}(\theta) = \widehat{A}_i \cdot \left[\frac{\rho_{i,p}(\theta)^{1-p}}{|y_i|} \sum_{t=1}^{|y_i|} r_{i,t}(\theta)^p \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t}) \right].$$

Since every rollout in the mini-batch is independent, we obtain

$$\begin{aligned} \mathrm{Var}\Big(\widehat{\nabla}_\theta \mathcal{J}_{\mathrm{H}}\Big) &= \mathrm{Var}\left(\frac{1}{B} \sum_{i=1}^{B} \hat{g}_i(p) \right) \\ &= \frac{1}{B^2} \sum_{i=1}^{B} \mathrm{Var}(\hat{g}_i(p)) = \frac{1}{B} \mathrm{Var}(\hat{g}_1(p)). \end{aligned}$$

For any stochastic gradient, its variance $\mathrm{Var}(\hat{g}_i(p))$ is strictly bounded by its second moment

$$\mathrm{Var}(\hat{g}_i(p)) = \mathbb{E}\big[\|\hat{g}_i(p)\|^2\big] - \|\mathbb{E}[\hat{g}_i(p)]\|^2 \le \mathbb{E}\big[\|\hat{g}_i(p)\|^2\big].$$

By applying the triangle inequality and $\|\nabla_\theta \log \pi_\theta(y_{i,t})\| \le M$, we can obtain the upper bound

$$\begin{aligned} \|\nabla_\theta \rho_{i,p}(\theta)\| &= \left\| \frac{\rho_{i,p}(\theta)^{1-p}}{|y_i|} \sum_{t=1}^{|y_i|} r_{i,t}(\theta)^p \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t}) \right\| \\ &\le \frac{\rho_{i,p}(\theta)^{1-p}}{|y_i|} \sum_{t=1}^{|y_i|} r_{i,t}(\theta)^p \left\| \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t}) \right\| \\ &\le M \cdot \frac{\rho_{i,p}(\theta)^{1-p}}{|y_i|} \sum_{t=1}^{|y_i|} r_{i,t}(\theta)^p = M \cdot \rho_{i,p}(\theta). \end{aligned}$$

Thus we can bound the squared norm of $\hat{g}_i(p)$ by

$$\begin{aligned} \|\hat{g}_i(p)\|^2 &= \widehat{A}_i^2 \cdot \|\nabla_\theta \rho_{i,p}(\theta)\|^2 \\ &\le \widehat{A}_i^2 \cdot \big(M \cdot \rho_{i,p}(\theta)\big)^2 = M^2 \cdot \widehat{A}_i^2 \cdot \rho_{i,p}(\theta)^2, \end{aligned}$$

which implies the upper bound of variance of $\widehat{\nabla}_\theta \mathcal{J}_{\mathrm{H}}$

$$\mathrm{Var}\Big(\widehat{\nabla}_\theta \mathcal{J}_{\mathrm{H}}\Big) \le \frac{1}{B} \mathbb{E}\big[\|\hat{g}_i(p)\|^2\big] \le \frac{M^2}{B} \cdot \mathbb{E}_{q,\{y_i\}}\Big[\widehat{A}_i^2 \cdot \rho_{i,p}(\theta)^2\Big].$$

According to the Generalised Mean Inequality, for any non-uniform sequence of importance ratios, the p-mean $\rho_{i,p}(\theta)$ is a strictly monotonically increasing function of p. Thus we obtain the conclusion. □

K.2 Variance and Sequence-level Clipping

To ensure training stability, policy optimisation algorithms typically employ clipping mechanisms to prevent destructively large updates. In the HolderPO framework, we explicitly adopt a *sequence-level clipping* strategy rather than the standard token-level clipping. In this section, we formalise the mathematical necessity of clipping itself, and justify why sequence-level clipping is structurally required to preserve the variance properties established in Theorem 6.

Why Clipping is Necessary. To understand why the unclipped case is susceptible to exponential explosion, we can analyse its gradient dynamics through an ordinary differential equation. Recall

$$\nabla_\theta \rho_{i,p}(\theta) = \rho_{i,p}(\theta) \sum_{t=1}^{|y_i|} W_{i,t}(p) \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t}).$$

By abstracting the weighted sum of token-level log-gradients into a single vector $g_w(\theta)$, the gradient equation simplifies to a form

$$\nabla_\theta \rho_{i,p}(\theta) = \rho_{i,p}(\theta) g_w(\theta).$$

During neural network training, the parameters θ are not static; they continuously evolve along an optimisation trajectory parameterised by a continuous virtual time τ. Applying the chain rule, the rate of change of the ratio with respect to this optimisation time is given by the derivative

$$\frac{d\rho_{i,p}}{d\tau} = \big(\nabla_\theta \rho_{i,p}(\theta)\big)^T \frac{d\theta}{d\tau}.$$

Assuming a standard gradient ascent step aimed at maximising a trajectory with a positive advantage, the parameter update direction $\frac{d\theta}{d\tau}$ naturally aligns with the gradient. Substituting our simplified gradient expression into the derivative yields

$$\frac{d\rho_{i,p}}{d\tau} = \big(\rho_{i,p} g_w(\theta)\big)^T \frac{d\theta}{d\tau} = \rho_{i,p} \left(g_w(\theta)^T \frac{d\theta}{d\tau} \right).$$

Because the optimiser attempts to increase the likelihood of these correct tokens, the update direction $\frac{d\theta}{d\tau}$ forms an acute angle with the composite gradient direction $g_w(\theta)$. This implies that their inner product is a strictly positive scalar, which we can denote as $k(\tau) > 0$. Substituting this scalar reduces the complex optimisation dynamics into a canonical ODE for exponential growth

$$\frac{d\rho_{i,p}}{d\tau} = k(\tau)\rho_{i,p}.$$

Integrating this differential equation over the time interval $[0, \tau]$ provides the exact analytical solution

$$\rho_{i,p}(\tau) = \rho_{i,p}(0) \exp\left(\int_0^\tau k(t)\, dt \right).$$

This mathematically dictates that without an explicit clipping mechanism to interrupt the ODE, the scaling factor will inevitably suffer from an unbounded exponential explosion. Therefore, explicitly bounding the surrogate ratio via a clipping mechanism is an absolute prerequisite.

Proof of Theorem 2. The clipping operator acts as a binary sequence-level mask $\mathbb{I}_i(\theta) \in \{0, 1\}$ applied directly to the aggregated ratio $\rho_{i,p}(\theta)$ and its advantage $\widehat{A}_i$ (see Eq. (16)). Consequently, the clipped stochastic gradient $\hat{g}_i^{\text{clip}}(p)$ is either preserved in full (when $\mathbb{I}_i = 1$) or completely zeroed out (when $\mathbb{I}_i = 0$). Mathematically, this guarantees that the squared norm of the clipped gradient is universally bounded by the unclipped one:

$$\|\hat{g}_i^{\text{clip}}(p)\|^2 = \mathbb{I}_i(\theta) \cdot \|\hat{g}_i(p)\|^2 \leq \|\hat{g}_i(p)\|^2.$$

Because the variance is bounded by the second moment, the upper bound we derived in Theorem 6 carries over:

$$\|\mathrm{Var}(\widehat{\nabla}_\theta \mathcal{J}_{H_s})\| \leq \frac{1}{B} \mathbb{E}\Big[\|\hat{g}_i^{\text{clip}}(p)\|^2 \Big] \leq \frac{M^2}{B}\, \mathbb{E}\Big[\widehat{A}_i^2\, \rho_{i,p}^2(\theta) \Big].$$

Thus, the monotonic relationship between the variance bound and the parameter p remains intact. □

In contrast, *token-level clipping* applies the clipping operator *inside* the summation over individual tokens. It unpredictably alters specific token ratios, destroying the correspondence between the outer multiplier $\rho_{i,p}(\theta)^{1-p}$ and the inner weights $W_{i,t}(p)$ (see Eq. (14)). This structural fracture voids the monotonic upper bound derived above, making the variance highly uncontrolled. Our empirical results in Appendix E (Table 6) corroborate this: token-level clipping narrows the performance spread across p, confirming that the parameter p loses its tight, predictable control over gradient variance.

L Finite-Order Optimality and Approximation

This appendix compares fixed orders through useful mean progress and the update's total second moment. We retain the directional-efficiency criterion, a sufficient condition for a finite optimum, a finite-grid approximation, and the interpretable signal–noise optimum. All comparisons hold the training state fixed; the assumptions specify when the conclusions apply.

L.1 Framework for Selecting a Fixed Holder Order

Fix the current policy and the positive token ratios in one sequence of length n. Write $z_t = \log r_{i,t}$, $w_t(p) = e^{pz_t} / \sum_k e^{pz_k}$, and $\rho_p = \rho_{i,p}$. Conditional on these ratios and other fixed information, let $Z_t \in \mathbb{R}^d$ be the token update before weighting, such as $\widehat{A}_i \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t})$. Its joint distribution is shared by every candidate order. Assume finite second moments and define

$$m_t = \mathbb{E}Z_t, \qquad \Sigma_{tk} = \mathbb{E}[(Z_t - m_t)^\top (Z_k - m_k)], \qquad Q_{tk} = \mathbb{E}[Z_t^\top Z_k] = \Sigma_{tk} + m_t^\top m_k. \quad (35)$$

Both matrices are positive semidefinite: for example, $b^\top Q b = \mathbb{E}\| \sum_t b_t Z_t\|^2 \geq 0$. This allows correlated token noise and different mean directions. If the conditioning fixes all computations, Z_t is deterministic and $\Sigma = 0$.

Fix a differentiable evaluation objective F to maximise, with $v = \nabla F(\theta)$ and $a_t = v^\top m_t$. For w in the probability simplex $\Delta_n = \{w \geq 0 : \sum_t w_t = 1\}$, define

$$\mathcal{R}(w) = \frac{[a^\top w]_+^2}{w^\top Q w}, \qquad [s]_+ = \max\{s, 0\}, \qquad \mathcal{R}(p) = \mathcal{R}(w(p)). \quad (36)$$

The denominator must be positive. The numerator measures the useful component of the mean update along v, while the denominator is the total second moment, including both squared mean and noise variance.

Suppose F has an L-Lipschitz gradient, $L > 0$, throughout the candidate update segments. Write $H_p = c_p \sum_t w_t(p) Z_t$, where $c_p > 0$ is deterministic under the conditioning and common across tokens. For example, $c_p = \rho_p$. Smoothness yields

$$\mathbb{E}[F(\theta + \eta H_p) - F(\theta)] \geq \eta c_p a^\top w(p) - \frac{L\eta^2 c_p^2}{2} w(p)^\top Q w(p). \quad (37)$$

Maximising the quadratic lower bound over $\eta \geq 0$ gives

$$\eta_p^* = \frac{[a^\top w(p)]_+}{L c_p\, w(p)^\top Q w(p)}, \qquad \max_{\eta \geq 0} \left\{ \eta c_p a^\top w(p) - \frac{L\eta^2 c_p^2}{2} w(p)^\top Q w(p) \right\} = \frac{\mathcal{R}(p)}{2L}. \quad (38)$$

Thus the criterion ranks best one-step lower bounds when each order can use its own step size, provided those steps stay in the smoothness region. For a shared learning rate and clipping, use the complete-update utility in Appendix M instead.

Assume at least two log-ratios differ; otherwise all weights are uniform. Let $I_\pm$ be the sets of maximal/minimal log-ratios, $b_\pm$ their indicator vectors, $n_\pm = |I_\pm|$, and $w_\pm = b_\pm / n_\pm$. The extended weight curve $\mathcal{W} = \{w(p) : p \in \mathbb{R}\} \cup \{w_+, w_-\}$ is compact: it is bounded, continuous in finite p, and includes both limits. We require

$$q_{\text{curve}} := \min_{w \in \mathcal{W}} w^\top Q w > 0. \quad (39)$$

This controls the denominator only on the Holder curve; Q may be singular and unrelated simplex weights may have zero second moment.

L.2 A sufficient condition for a finite exact optimum

Exact attainment at a finite order requires a condition beyond continuity on the unbounded real line. We retain a sufficient endpoint comparison; when it fails, the finite-grid approximation in the next subsection remains the relevant value guarantee.

Theorem B1 (Sufficient conditions for a finite optimum)**.** *Assume Eq.* (39)*. Then*

$$L_\pm = \lim_{p \to \pm\infty} \mathcal{R}(p) = \frac{[a^\top b_\pm]_+^2}{b_\pm^\top Q b_\pm}. \quad (40)$$

If a finite p_0 satisfies

$$\mathcal{R}(p_0) > \max\{L_+, L_-\}, \quad (41)$$

then a finite global maximiser exists. A sufficient test at a better endpoint $L_\sigma = \max\{L_+, L_-\} > 0$ is as follows. Let c_σ indicate the second-largest distinct log-ratio layer for $\sigma = +$ or second-smallest for $\sigma = -$, and set

$$A_0 = a^\top b_\sigma, \quad A_1 = a^\top c_\sigma, \quad V_0 = b_\sigma^\top Q b_\sigma, \quad V_{01} = b_\sigma^\top Q c_\sigma.$$

If

$$A_1 V_0 - A_0 V_{01} > 0, \tag{42}$$

a finite maximiser exists. If $L_+ = L_- = 0$, any finite order with $a^\top w(p) > 0$ suffices. These are sufficient conditions, not uniqueness claims.

Proof. The endpoint values follow from $w(p) \to w_\pm$ and homogeneity. Under Eq. (41), choose $P > |p_0|$ so both tails have values below $\mathcal{R}(p_0)$. Continuity on $[-P, P]$ gives a maximum there, which is global.

For the layer test at $+$, let $\delta_+ > 0$ be the gap between the largest and next distinct log-ratio. Dividing the unnormalised weights by their largest exponential gives $b_+ + tc_+ + o(t)$, with $t = e^{-p\delta_+}$. Since $A_0 > 0$, expansion yields

$$\mathcal{R}(p) = \frac{A_0^2}{V_0} + \frac{2A_0(A_1V_0 - A_0V_{01})}{V_0^2} t + o(t).$$

The stated condition therefore gives a finite order better than both limits. At the negative endpoint use $t = e^{p\delta_-}$ and the same argument. If both limits are zero, any positive finite value gives the interior witness. □

The test asks whether adding the neighbouring ratio layer improves aligned mean progress enough to offset its second moment. A failed test leaves exact attainment open; it does not invalidate finite approximation of the best value.

L.3 Finite approximation without a finite exact optimum

A finite exact maximiser is not needed to obtain a nearly optimal efficiency value. We search a bounded interval on a finite grid and bound the loss from omitting the tails and the gaps between grid points. The target is a finite order p_ε satisfying

$$\sup_{p \in \mathbb{R}} \mathcal{R}(p) - \mathcal{R}(p_\varepsilon) \le \varepsilon.$$

The following estimate translates an error in token weights into an error in efficiency. Define

$$A_\infty = \max_t |a_t|, \qquad Q_\infty = \max_{t,k} |Q_{tk}|, \qquad L_R(q) = \frac{2A_\infty^2}{q} + \frac{2A_\infty^2 Q_\infty}{q^2}, \qquad q > 0.$$

Here A_∞ and Q_∞ bound the magnitudes of the entries of a and Q, while $L_R(q)$ is the sensitivity bound obtained when both compared second moments are at least $q > 0$. For such weights $w, \widetilde{w} \in \Delta_n$,

$$|\mathcal{R}(w) - \mathcal{R}(\widetilde{w})| \le L_R(q) \|w - \widetilde{w}\|_1. \tag{43}$$

Indeed,

$$\left|[a^\top w]_+^2 - [a^\top \widetilde{w}]_+^2\right| \le 2A_\infty^2 \|w - \widetilde{w}\|_1,$$

and

$$|w^\top Q w - \widetilde{w}^\top Q \widetilde{w}| \le 2Q_\infty \|w - \widetilde{w}\|_1.$$

Subtracting the two fractions, using the lower bounds on their denominators and the bound $[a^\top w]_+^2 \le A_\infty^2$, proves Eq. (43).

The denominator bound is needed only at the two weights being compared. Since the grid search compares Holder weights, q_{curve} supplies the required bound.

Choose a search interval $[-P, P]$ and a finite grid $\mathcal{G}$ inside it. The number h below is the greatest distance from any point of the interval to its nearest grid point. Increasing P covers more of the tails; decreasing h gives a finer search inside the interval.

Theorem B2 (A finite grid gives a finite near-optimal order)**.** *Assume Eq.* (39)*, and write*

$$L_R = L_R(q_{\mathrm{curve}}), \qquad D = \max_t z_t - \min_t z_t.$$

Let δ_+ and δ_- be the gaps from the maximum and minimum log-ratio layers to their neighbouring distinct layers, and define

$$\delta = \min\{\delta_+, \delta_-\}, \qquad c_z = \max\left\{\frac{n-n_+}{n_+}, \frac{n-n_-}{n_-}\right\}.$$

For $P \geq 0$, let $\mathcal{G} \subset [-P, P]$ be a finite grid containing both endpoints, with covering radius at most h:

$$\sup_{p\in[-P,P]} \min_{q\in\mathcal{G}} |p-q| \leq h.$$

If

$$p^*_{\mathcal{G}} \in \arg\max_{p\in\mathcal{G}} \mathcal{R}(p),$$

then

$$\sup_{p\in\mathbb{R}} \mathcal{R}(p) - \mathcal{R}(p^*_{\mathcal{G}}) \leq 4L_R c_z e^{-P\delta} + \frac{L_R D}{2} h. \tag{44}$$

In particular, if $L_R > 0$, choosing

$$P \geq \max\left\{0, \frac{1}{\delta}\log\frac{8L_R c_z}{\varepsilon}\right\}, \qquad h \leq \frac{\varepsilon}{L_R D}$$

gives a finite ε-optimal order for every $\varepsilon > 0$. If $L_R = 0$, the criterion is identically zero and any finite order suffices.

Proof. For $p \geq 0$, divide the unnormalised weights by $e^{p\max_t z_t}$. The total mass outside I_+ is at most $(n-n_+)e^{-p\delta_+}$, while the unnormalised mass on I_+ is n_+. Consequently,

$$\|w(p) - w_+\|_1 \leq 2\frac{n-n_+}{n_+}e^{-p\delta_+}.$$

For $p \geq P$, the triangle inequality therefore gives

$$\|w(p) - w(P)\|_1 \leq 4c_z e^{-P\delta}.$$

Equation (43) implies

$$\mathcal{R}(p) \leq \mathcal{R}(P) + 4L_R c_z e^{-P\delta}.$$

The same argument on the negative tail gives

$$\mathcal{R}(p) \leq \mathcal{R}(-P) + 4L_R c_z e^{-P\delta}, \qquad p \leq -P.$$

For $p \in [-P, P]$, choose $q \in \mathcal{G}$ with $|p-q| \leq h$. Theorem A2 gives

$$\|w(p) - w(q)\|_1 \leq \frac{D}{2}|p-q| \leq \frac{D}{2}h.$$

Thus,

$$\mathcal{R}(p) \leq \mathcal{R}(q) + \frac{L_R D}{2}h.$$

Since the grid contains $\pm P$ and $p^*_{\mathcal{G}}$ maximises $\mathcal{R}$ over the grid, combining the tail and interior bounds proves Eq. (44). The stated choices make each displayed error term at most $\varepsilon/2$. □

The first term in Eq. (44) bounds the loss from excluding orders outside $[-P, P]$; the second bounds the loss from using the nearest grid point inside the interval. For a uniform grid with spacing Δp, take $h = \Delta p/2$. The result approximates the best efficiency value, whether or not that value is attained at a finite order. It does not measure distance to a particular optimal parameter.

Applying the bound requires the conditional moments and a valid value of q_{curve}. Small gaps between extreme ratio layers slow convergence towards the limiting weights, while a small denominator bound increases L_R; either can increase the grid size needed for the stated accuracy.

A uniform estimation error $\sup_{p\in\mathcal{G}} |\widehat{\mathcal{R}}(p) - \mathcal{R}(p)| \leq \zeta$ adds at most 2ζ to Eq. (44), by comparing the true and estimated grid maximisers. This requires valid moment estimates; the grid theorem itself concerns the fixed-state population criterion.

L.4 AN INTERPRETABLE SIGNAL–NOISE SPECIAL CASE

We now give a model in which the best order has an explicit formula: it is the difference between the rates at which signal strength and noise variance change with the token log-ratio. For this result, all mean token updates point in one beneficial direction, and noise from different tokens has zero expected inner product. These are additional assumptions for this subsection.

Recall that z_t is the token log-ratio and Z_t is its update vector, for example $\widehat{A}_i \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t})$. Assume

$$Z_t = s_t e + \varepsilon_t, \qquad \|e\| = 1, \qquad v^\top e > 0, \qquad s_t > 0, \tag{45}$$

where

$$\mathbb{E}[\varepsilon_t] = 0, \qquad \mathbb{E}[\varepsilon_t^\top \varepsilon_k] = 0 \quad (t \neq k), \qquad \mathbb{E}\|\varepsilon_t\|^2 = \tau_t^2 > 0.$$

The vector e is a unit direction and $s_t > 0$ is the signal magnitude of token t. This is a common-direction model because

$$\mathbb{E}[Z_t] = s_t e.$$

Every mean update points along e, which has a positive component along the objective gradient. The realised updates also contain zero-mean noise ε_t, with noise energy τ_t^2. The cross-inner-product condition makes the cross-token noise terms vanish in expectation; it does not require independence. For weights $w \in \Delta_n$, the combined update is

$$\sum_t w_t Z_t = \left(\sum_t w_t s_t \right) e + \sum_t w_t \varepsilon_t.$$

Its mean signal has squared magnitude

$$\left\| \mathbb{E}\left[\sum_t w_t Z_t \right] \right\|^2 = \left(\sum_t w_t s_t \right)^2,$$

while the cross-inner-product condition gives the conditional noise energy

$$\mathbb{E}\left\| \sum_t w_t \varepsilon_t \right\|^2 = \sum_t w_t^2 \tau_t^2.$$

The conditional signal-to-noise ratio compares these two quantities:

$$\mathrm{SNR}(w) = \frac{\left(\sum_t w_t s_t \right)^2}{\sum_t w_t^2 \tau_t^2}. \tag{46}$$

A larger SNR means that the weighted update has a stronger mean signal relative to its random fluctuations. In this model, the same comparison also ranks directional efficiency.

Under this model,

$$a_t = v^\top \mathbb{E}[Z_t] = (v^\top e) s_t$$

and

$$w^\top Q w = \left(\sum_t w_t s_t \right)^2 + \sum_t w_t^2 \tau_t^2.$$

Consequently,

$$\mathcal{R}(w) = (v^\top e)^2 \frac{\mathrm{SNR}(w)}{1 + \mathrm{SNR}(w)}. \tag{47}$$

Because $x \mapsto x/(1+x)$ is strictly increasing for $x \geq 0$, maximising the conditional SNR is equivalent to maximising directional efficiency in this special model.

In the next theorem, λ_s is the rate of change of log signal strength with z_t, and λ_v is the corresponding rate for log noise variance. Since z_t is itself a log-ratio, these rates are also called elasticities. The theorem gives the exact optimum under exponential relationships and an efficiency bound when those relationships hold approximately.

Theorem B3 (Signal and noise elasticities determine the order)**.** *Assume Eq.* (45) *and the moment conditions above. Suppose further that*

$$s_t = s_0 e^{\lambda_s z_t}, \qquad \tau_t^2 = \tau_0^2 e^{\lambda_v z_t}, \qquad s_0, \tau_0^2 > 0. \tag{48}$$

If the log-ratios are not all equal, then both $\mathrm{SNR}(w(p))$ *and* $\mathcal{R}(p)$ *have the unique global maximiser*

$$p^* = \lambda_s - \lambda_v. \tag{49}$$

More generally, suppose

$$s_t = s_0 e^{\lambda_s z_t + d_t}, \qquad |d_t| \le \epsilon_s,$$

and

$$\tau_t^2 = \tau_0^2 e^{\lambda_v z_t + f_t}, \qquad |f_t| \le \epsilon_v.$$

Then $p_0 = \lambda_s - \lambda_v$ *satisfies*

$$\frac{\mathrm{SNR}(w(p_0))}{\max_{w \in \Delta_n} \mathrm{SNR}(w)} \ge \operatorname{sech}^2(\epsilon_s + \epsilon_v). \tag{50}$$

The same lower bound holds for $\mathcal{R}(w(p_0))/\max_{w \in \Delta_n} \mathcal{R}(w)$.

Proof. Cauchy–Schwarz gives

$$\left(\sum_t w_t s_t\right)^2 = \left(\sum_t (w_t \tau_t)\frac{s_t}{\tau_t}\right)^2 \le \left(\sum_t w_t^2 \tau_t^2\right)\left(\sum_t \frac{s_t^2}{\tau_t^2}\right).$$

Equality holds exactly when

$$w_t \propto \frac{s_t}{\tau_t^2}.$$

Since every s_t and τ_t^2 is positive, these weights define the unique maximiser on the simplex. Under Eq. (48), they satisfy

$$w_t^* \propto e^{(\lambda_s - \lambda_v) z_t} = w_t(p^*) \times \text{a common constant}.$$

If $z_t \neq z_k$ for some pair, the ratio $w_t(p)/w_k(p) = e^{p(z_t - z_k)}$ uniquely determines p. The optimum in the Holder family is therefore unique. Equation (47) transfers the conclusion to $\mathcal{R}$.

For the approximate model, let w^* denote the exact unrestricted SNR optimiser and put

$$U_t = \frac{w_t(p_0)}{w_t^*}, \qquad \nu_t = \frac{s_t^2/\tau_t^2}{\sum_k s_k^2/\tau_k^2}.$$

A direct substitution using $w_t^* \propto s_t/\tau_t^2$ gives

$$\frac{\mathrm{SNR}(w(p_0))}{\mathrm{SNR}(w^*)} = \frac{(\mathbb{E}_\nu U)^2}{\mathbb{E}_\nu U^2}.$$

Moreover,

$$\frac{\max_t U_t}{\min_t U_t} \le e^{2(\epsilon_s + \epsilon_v)}.$$

If $0 < m \le U \le M$, then $(U - m)(M - U) \ge 0$, and hence

$$\mathbb{E}U^2 \le (m + M)\mathbb{E}U - mM.$$

Writing $\mu = \mathbb{E}U$, the elementary inequality

$$\frac{(m + M)\mu - mM}{\mu^2} \le \frac{(m + M)^2}{4mM}$$

implies

$$\frac{(\mathbb{E}U)^2}{\mathbb{E}U^2} \ge \frac{4mM}{(m + M)^2}.$$

Using the bound on M/m proves Eq. (50).

Finally, let $r = \mathrm{SNR}(w(p_0))/\mathrm{SNR}(w^*) \le 1$ and $S^* = \mathrm{SNR}(w^*)$. Then

$$\frac{\mathcal{R}(w(p_0))}{\mathcal{R}(w^*)} = r\frac{1 + S^*}{1 + rS^*} \ge r,$$

which proves the directional-efficiency guarantee. □

The optimum is positive when signal strength grows faster with the ratio than noise variance, and negative when noise variance grows faster. Thus, the preferred sign depends on the relation between ratios, useful updates, and noise. Two tasks with the same token ratios can prefer different orders if these signal and noise relationships differ.

L.5 Relation to the main theory and dynamic scheduling

The main-text variance bound describes one cost of changing p; this appendix adds a criterion for choosing p based on both the mean update and its second moment. Theorem A2 connects the two analyses by controlling how quickly token weights change with the order. Its log-ratio spread D appears directly in the grid error of Theorem B2: with the other constants fixed, a smaller spread permits a coarser grid for the same error bound. A larger spread permits faster variation but does not force it.

The preferred order depends on which token updates help the objective, their magnitudes and correlations, and how those quantities relate to the token ratios. The signal–noise model makes this dependence explicit through $p^* = \lambda_s - \lambda_v$. These relationships can change during training, so an order that is efficient at one state need not remain efficient later. This motivates adapting the order in Section 3.4; the present results provide the fixed-state comparison on which such a choice can be based.

Relation to the experiments. Table 2 shows that the best tested fixed order differs across benchmarks: $p = 3$ achieves 46.7% on AIME24, whereas $p = -1$ achieves 85.0% on MATH500. Supplementary repeated evaluations also show an overall preference for larger orders on AIME. These observations motivate choosing the order according to the update's useful contribution and total second moment, as captured by $\mathcal{R}(p)$.

M Utility-Based Selection of the Holder Order

For a shared learning rate, compare complete candidate updates through a lower bound on expected one-step improvement. This keeps scaling, clipping, and group aggregation in the criterion. We retain existence and consistency for finite candidate sets; this is a theoretical comparison, not a selector evaluated in the current experiments.

Fix the current parameter θ and a differentiable objective F to be maximised, such as expected return. Every candidate is evaluated against this same F, although its training surrogate may depend on p. Write

$$v = \nabla F(\theta),$$

Assume that F has an L-Lipschitz gradient on a region containing θ and the candidate one-step updates, so its gradient changes by at most L times the parameter displacement. Let $\mathcal{P}$ be the candidate orders, $\eta > 0$ their shared learning rate, and H_p the complete random update direction produced by order p at the current state. Its randomness comes from the data or other random choices used to construct the update. For example, a rollout-group update is

$$H_p = \frac{1}{G}\sum_{i=1}^{G} \mathbb{I}_{i,p}\widehat{A}_i \nabla_\theta \rho_{i,p}(\theta),$$

Here $\mathbb{I}_{i,p}$ is the sequence-level clipping mask: it records whether rollout i contributes an active gradient under order p. Because H_p includes the whole group, its moments retain the effects of advantage normalisation, negative advantages, and dependence between rollout or token contributions. The same definition can include minibatch aggregation. Define the one-step utility

$$U(p) \coloneqq \eta\langle v, \mathbb{E}H_p\rangle - \frac{L\eta^2}{2}\mathbb{E}\|H_p\|^2. \tag{51}$$

Smoothness makes this utility a lower bound on expected improvement:

$$\mathbb{E}[F(\theta + \eta H_p) - F(\theta)] \geq U(p). \tag{52}$$

In $U(p)$, the first term, $\eta\langle v, \mathbb{E}H_p\rangle$, measures the expected first-order improvement along the objective gradient $v = \nabla F(\theta)$. The second term, $(L\eta^2/2)\mathbb{E}\|H_p\|^2$, is subtracted to account for curvature and

the overall size of the update, including both its mean and its random fluctuations. Maximising $U(p)$ selects the largest one-step improvement lower bound among the candidates. This compares their lower bounds; the actual improvements may be larger.

M.1 Existence of an optimal candidate

For a finite list of orders, a best utility value exists whenever the candidate updates have finite second moments. Searching a whole closed bounded interval requires an additional continuity condition: nearby orders must produce nearby updates in mean square. Theorem C1 states both cases.

Theorem C1 (Existence for finite and compact candidate sets)**.** *Suppose first that $\mathcal{P}$ is finite and $\mathbb{E}\|H_p\|^2 < \infty$ for every $p \in \mathcal{P}$. Then*

$$\arg\max_{p\in\mathcal{P}} U(p) \neq \varnothing.$$

More generally, suppose that $\mathcal{P} \subset \mathbb{R}$ is compact and that $p \mapsto H_p$ is continuous in L^2, meaning

$$\|H_p - H_q\|_{L^2} \coloneqq \big(\mathbb{E}\|H_p - H_q\|^2\big)^{1/2} \longrightarrow 0 \quad \text{as } q \to p.$$

Then U is continuous on $\mathcal{P}$ and attains its maximum there.

Proof. For a finite candidate set, each $U(p)$ is finite and a finite collection of real numbers has a maximum.

For the compact case, Cauchy–Schwarz gives

$$|\mathbb{E}\langle v, H_p - H_q\rangle| \le \|v\|\,\|H_p - H_q\|_{L^2}.$$

Moreover,

$$\begin{aligned}\big|\mathbb{E}\|H_p\|^2 - \mathbb{E}\|H_q\|^2\big| &= |\mathbb{E}\langle H_p - H_q, H_p + H_q\rangle| \\ &\le (\|H_p\|_{L^2} + \|H_q\|_{L^2})\,\|H_p - H_q\|_{L^2}.\end{aligned}$$

Hence L^2 continuity of H_p implies continuity of both terms in $U(p)$. The extreme-value theorem then yields a maximiser on the compact set $\mathcal{P}$. □

A finite candidate list also covers hard clipping, which can make the update jump when an order crosses a clipping threshold.

M.2 Consistent selection from independent evaluation groups

We can estimate each candidate's utility by averaging its values over evaluation groups. With enough independent groups, the selected utility approaches the best value in a fixed finite candidate set, using only finite second moments of the updates.

Let $\mathcal{P}$ be finite, and let $\mathsf{Z}_1, \dots, \mathsf{Z}_m$ be independent, identically distributed evaluation units. A unit may be an entire rollout group; its members may be dependent. All candidates can use the same units, so only independence between units is required. Denote the complete update on one unit by $H_p(\mathsf{Z})$, and its observed utility by

$$Y_p(\mathsf{Z}) \coloneqq \eta\langle v, H_p(\mathsf{Z})\rangle - \frac{L\eta^2}{2}\|H_p(\mathsf{Z})\|^2. \tag{53}$$

Its expectation is $U(p)$. Average these values over the m units and choose the candidate with the largest average:

$$\widehat{U}_m(p) \coloneqq \frac{1}{m}\sum_{j=1}^{m} Y_p(\mathsf{Z}_j), \qquad \widehat{p}_m \in \arg\max_{p\in\mathcal{P}} \widehat{U}_m(p).$$

Theorem C2 (Consistency under finite second moments)**.** *If*

$$\mathbb{E}\|H_p(\mathsf{Z})\|^2 < \infty \qquad \text{for every } p \in \mathcal{P},$$

then

$$\max_{p\in\mathcal{P}} U(p) - U(\widehat{p}_m) \longrightarrow 0 \qquad \text{almost surely as } m \to \infty. \tag{54}$$

Proof. Finite second moments of H_p imply $\mathbb{E}|Y_p| < \infty$. The strong law of large numbers therefore gives $\widehat{U}_m(p) \to U(p)$ almost surely for every fixed $p \in \mathcal{P}$. Since $\mathcal{P}$ is finite,

$$\delta_m \coloneqq \max_{p \in \mathcal{P}} |\widehat{U}_m(p) - U(p)| \longrightarrow 0 \qquad \text{almost surely.}$$

Let

$$p^* \in \arg\max_{p \in \mathcal{P}} U(p).$$

Empirical maximality of $\widehat{p}_m$ gives

$$\begin{aligned} U(p^*) - U(\widehat{p}_m) &\le [U(p^*) - \widehat{U}_m(p^*)] + [\widehat{U}_m(\widehat{p}_m) - U(\widehat{p}_m)] \\ &\le 2\delta_m, \end{aligned}$$

which proves Eq. (54). □

The selected utility value converges to the best candidate value. If several orders are equally good, this can happen without the selection settling on a single order.

M.3 Relation to the other results

Appendix L compares token weighting through $\mathcal{R}(p)$ after optimising the step size separately for each order. The present criterion $U(p)$ keeps one learning rate and evaluates the complete update, including its scale and clipping. It therefore provides a selection rule for a finite list of orders at the current policy state. Appendix N takes the next step: it studies a sequence of such updates when the order changes during training.

N Convergence with Dynamic Holder Orders

We study updates with changing orders against a fixed objective F. The results retain conditional alignment, the original binary-reward update's length-normalisation bias, buffer reuse, and Holder-specific drift. Convergence means expected first-order stationarity under the stated conditions, not a guarantee of optimal return. The proof follows smooth stochastic approximation arguments Karimi et al. (2019).

N.1 Why the fixed objective must be specified

We first distinguish the training surrogate from the objective used to assess progress. Let $J(\theta) = \mathbb{E}_{x,y\sim\pi_\theta}[R(x,y)]$ be expected return under a fixed prompt distribution. At the old policy, Theorem A1 gives the same sample surrogate gradient for every finite order:

$$\frac{1}{G}\sum_{i=1}^{G} \widehat{A}_i \frac{1}{n_i}\sum_{t=1}^{n_i} \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t}). \tag{55}$$

Taking its expectation also averages the effects of sequence-length normalisation and group standardisation. Those operations determine whether the resulting direction agrees with ∇J. Length normalisation and gradient unbiasedness are studied in Ding et al. (2026). Here we retain the exact mean-update decomposition in Section N.3, which identifies a transformed objective and the bias relative to it. Alignment with raw expected return is not assumed automatically.

N.2 Convergence under conditional alignment

The convergence argument requires the mean update to point far enough along the objective gradient, with its squared size controlled. Let F be the fixed objective to maximise. Assume its gradient is L-Lipschitz along the update segments and F is bounded above by $F_{\max}$ on the training region. Starting from a fixed θ_0, consider the actual parameter updates

$$\theta_{k+1} = \theta_k + \eta_k H_k, \qquad \eta_k > 0, \tag{56}$$

The step sizes η_k are deterministic. The information $\mathcal{F}_k$ contains everything known before the random choices for update k, including the current parameter and any stored buffer. Conditioning on $\mathcal{F}_k$ means averaging only over randomness that remains at that point. The vector H_k includes the chosen order, clipping, aggregation, and any transformations applied before the parameter update.

An order may be selected from past information or from the current data. In the latter case, the conditional mean covers both order selection and the resulting update. In the explicit Holder formulas below, the selected order is held fixed when differentiating the surrogate. Assume all displayed expectations are finite, and denote the cumulative step size and maximum available objective increase by

$$\mathcal{S}_T = \sum_{k=0}^{T-1} \eta_k, \qquad \Delta_F = F_{\max} - F(\theta_0).$$

Theorem D0 (Stationarity under conditional alignment)**.** *Suppose that, almost surely, for some constant $\kappa > 0$ and nonnegative $\mathcal{F}_k$-measurable quantities ε_k, V_k,*

$$\langle \nabla F(\theta_k), \mathbb{E}[H_k \mid \mathcal{F}_k] \rangle \geq \kappa \|\nabla F(\theta_k)\|^2 - \varepsilon_k, \tag{57}$$

$$\mathbb{E}[\|H_k\|^2 \mid \mathcal{F}_k] \leq V_k. \tag{58}$$

Then

$$\frac{1}{\mathcal{S}_T} \sum_{k=0}^{T-1} \eta_k \mathbb{E}\|\nabla F(\theta_k)\|^2 \leq \frac{\Delta_F}{\kappa \mathcal{S}_T} + \frac{\sum_{k<T} \eta_k \mathbb{E}\varepsilon_k}{\kappa \mathcal{S}_T} + \frac{L \sum_{k<T} \eta_k^2 \mathbb{E} V_k}{2\kappa \mathcal{S}_T}. \tag{59}$$

Proof. Write $\mathbb{E}_k[\cdot] = \mathbb{E}[\cdot \mid \mathcal{F}_k]$. Smoothness gives

$$\mathbb{E}_k[F(\theta_{k+1}) - F(\theta_k)] \geq \eta_k \langle \nabla F(\theta_k), \mathbb{E}_k H_k \rangle - \frac{L\eta_k^2}{2} \mathbb{E}_k \|H_k\|^2.$$

Substituting the assumptions and taking expectations yields

$$\mathbb{E}[F(\theta_{k+1}) - F(\theta_k)] \geq \kappa \eta_k \mathbb{E}\|\nabla F(\theta_k)\|^2 - \eta_k \mathbb{E}\varepsilon_k - \frac{L\eta_k^2}{2} \mathbb{E} V_k.$$

Summing over $k < T$ telescopes the left-hand side. Using $\mathbb{E}F(\theta_T) - F(\theta_0) \leq \Delta_F$ and dividing by $\kappa \mathcal{S}_T$ proves the claim. □

For example, if $\mathbb{E}_k H_k = P_k \nabla F + b_k$ with $P_k = P_k^\top \succeq \lambda I$, Young's inequality permits $\kappa = \lambda/2$ and $\varepsilon_k = \|b_k\|^2/(2\lambda)$.

What the bound guarantees. If $\mathcal{S}_T \to \infty$ and the two error terms in Eq. (59) vanish, the step-size-weighted average of $\mathbb{E}\|\nabla F(\theta_k)\|^2$ tends to zero. Equivalently, draw an index K_T independently with $\Pr(K_T = k) = \eta_k / \mathcal{S}_T$; the selected iterate satisfies $\mathbb{E}\|\nabla F(\theta_{K_T})\|^2 \to 0$. This is expected first-order stationarity: it concerns a small objective gradient, rather than convergence of every parameter iterate or attainment of a global optimum.

With uniformly bounded $\mathbb{E}V_k$, zero alignment error, and a constant step size $\eta \asymp T^{-1/2}$ chosen for a run of T updates, the bound is $O(T^{-1/2})$. A persistent alignment error remains as an additional term.

N.3 Exact length bias of the original on-policy update

We now calculate the directional error for the original update, retaining both sequence-length normalisation and group standardisation. For binary rewards, the mean update separates into a success-probability gradient and a term caused by how success and length vary together.

Fix a prompt x and draw $G \geq 2$ independent rollouts from $\pi_\theta(\cdot \mid x)$. For a fixed trajectory y, the reward $R = R(x, y) \in \{0, 1\}$ and length $n = n(y) \geq 1$ have no explicit parameter dependence. The

full trajectory score is the gradient of its log-probability, including a sampled termination token when present. Suppress x and θ in the following quantities:

$$q = \mathbb{E}R, \qquad b = \mathbb{E}(1/n), \qquad a = \mathbb{E}(R/n), \qquad c = a - qb = \mathrm{Cov}(R, 1/n).$$

Here q is the success probability, b is the mean inverse length, and c measures the association between success and inverse length. Use advantages normalised by the population standard deviation, setting all advantages to zero when the group rewards are equal. For $0 < q < 1$, let K be the number of successful rollouts among the G independent draws and define

$$a_G(q) = \frac{\mathbb{E}_{K\sim\mathrm{Bin}(G,q)}\sqrt{K(G-K)}}{Gq(1-q)}, \qquad 0 < q < 1. \tag{60}$$

Lemma D1 (Exact decomposition with variable lengths)**.** *Let $h_x(\theta)$ be the conditional expected update in Eq.* (55)*. Under the assumptions above,*

$$h_x(\theta) = a_G(q)\big[b\nabla q + \nabla c\big]. \tag{61}$$

Proof. Write $S(y) = \nabla_\theta \log \pi_\theta(y \mid x)$ for the full trajectory score. The score identities give

$$\nabla a = \mathbb{E}[RS/n], \qquad \nabla b = \mathbb{E}[S/n].$$

Consequently,

$$\mathbb{E}[S/n \mid R = 1] = \frac{\nabla a}{q}, \qquad \mathbb{E}[S/n \mid R = 0] = \frac{\nabla b - \nabla a}{1-q}.$$

Conditional on $K = j$ successful rollouts, with $0 < j < G$, the positive and negative advantages are respectively $\sqrt{(G-j)/j}$ and $-\sqrt{j/(G-j)}$. Thus the conditional expected group update is

$$\frac{\sqrt{j(G-j)}}{G}\left(\mathbb{E}[S/n \mid R = 1] - \mathbb{E}[S/n \mid R = 0]\right) = \frac{\sqrt{j(G-j)}}{Gq(1-q)}(\nabla a - q\nabla b).$$

The groups $j = 0, G$ contribute zero. Averaging over K gives $a_G(q)(\nabla a - q\nabla b)$. Finally, differentiating $a = qb + c$ gives $\nabla a - q\nabla b = b\nabla q + \nabla c$. □

The positive factor $a_G(q)$ comes from group standardisation Bay & Yearick (2026). The additional ∇c term comes from the division by sequence length: it measures how the association between success and inverse length changes with the policy.

A fixed objective for the mean update. The decomposition suggests a fixed transformed-success objective. Restore the prompt subscript and choose constants $b_x^0 > 0$ with $\mathbb{E}_x b_x^0 < \infty$. These constants fix the inverse-length weights used in the objective and remain unchanged during training. Define

$$\Psi_G(q) = \int_0^q a_G(u)\,du, \qquad F_0(\theta) = \mathbb{E}_x[b_x^0 \Psi_G(q_x(\theta))]. \tag{62}$$

The transformation Ψ_G incorporates the success-dependent scaling caused by group standardisation. Although the expression for a_G in Eq. (60) has $q(1-q)$ in its denominator, it extends continuously to $q = 0, 1$:

$$a_G(q) = (G-1)\sum_{j=0}^{G-2}\binom{G-2}{j}\frac{q^j(1-q)^{G-2-j}}{\sqrt{(j+1)(G-j-1)}}.$$

This follows by cancelling $q(1-q)$ in the binomial expression and using $\binom{G}{j}j(G-j) = G(G-1)\binom{G-2}{j-1}$.

The extension is positive, continuous, and bounded on $[0, 1]$, so F_0 is bounded above. Its smoothness depends on the policy's regularity. Writing $h(\theta) = \mathbb{E}_x h_x(\theta)$, the difference between the mean update and this objective gradient is

$$h(\theta) = \nabla F_0(\theta) + e_{\mathrm{len}}(\theta),$$

$$e_{\mathrm{len}}(\theta) = \mathbb{E}_x a_G(q_x)\left[(b_x - b_x^0)\nabla q_x + \nabla c_x\right], \tag{63}$$

$$\|e_{\mathrm{len}}(\theta)\| \le \mathbb{E}_x a_G(q_x)\left[|b_x - b_x^0|\,\|\nabla q_x\| + \|\nabla c_x\|\right]. \tag{64}$$

The error has two sources: the current mean inverse length b_x may differ from its fixed weight b_x^0, and the success–length association may change through ∇c_x. If $b_x = b_x^0$ and $c_x \equiv 0$ throughout the analysis region, the mean update equals ∇F_0 even when individual lengths vary. More generally, if $\|e_{\text{len}}\| \le \gamma \|\nabla F_0\|$ with $\gamma < 1$, then

$$\langle \nabla F_0, h \rangle \ge (1-\gamma)\|\nabla F_0\|^2.$$

This gives the alignment required by Theorem D0. An absolute bound on $\|e_{\text{len}}\|$ gives an additive alignment error through Young's inequality instead.

For fixed length n, $b_x = b_x^0 = 1/n$ and $c_x = 0$, giving $F_0 = n^{-1}\mathbb{E}_x \Psi_G(q_x)$. Thus the fixed objective is a transformed, prompt-weighted success rate; raw expected return is $J = \mathbb{E}_x q_x$.

For a different advantage-normalisation denominator, the coefficient a_G and the resulting mean mismatch must be recalculated. The statements above use the specified population-standard-deviation convention.

N.4 Convergence while reusing rollout buffers

A rollout buffer is sampled once and may then influence several updates. Its sampling error is therefore shared across those updates and can become correlated with the current parameter. The next result includes that correlation in the convergence bound.

Partition the updates into deterministic epochs. At the start of epoch e, the old-policy parameter β_e is fixed before the buffer is collected. Let $\mathcal{F}_e^{\text{start}}$ be the information available at that point, and let $\bar{H}_e$ be the buffer's batch update evaluated at β_e. Decompose it into the target gradient, a mean mismatch e_e, and a centred sampling error Z_e:

$$\mathbb{E}[\bar{H}_e \mid \mathcal{F}_e^{\text{start}}] = \nabla F(\beta_e) + e_e, \qquad Z_e = \bar{H}_e - \nabla F(\beta_e) - e_e. \tag{65}$$

The mismatch e_e is determined before this buffer is sampled. For the original binary-reward update with target F_0, it is $e_{\text{len}}(\beta_e)$.

Within an epoch, $\mathcal{F}_k$ contains the stored buffer, current parameter, and previous updates. Any new minibatch randomness and order selection are included in H_k. Let $e(k)$ identify the epoch containing step k, and define

$$m_k = \mathbb{E}[H_k \mid \mathcal{F}_k], \qquad r_k = m_k - \bar{H}_{e(k)} + \nabla F(\beta_{e(k)}) + e_{e(k)} - \nabla F(\theta_k).$$

Here m_k is the mean current update. The residual r_k collects the change from the old-policy batch update, the change in the target gradient, and the old-policy mean mismatch. Theorem D2 uses bounds on four quantities: batch sampling error, this residual, distance travelled from the old policy, and update size.

Theorem D2 (A stationarity bound with buffer reuse)**.** *Under the smoothness, upper-bound, and integrability conditions above, suppose that deterministic nonnegative bounds satisfy*

$$\begin{aligned} \mathbb{E}\|Z_e\|^2 \le v_e, \qquad \mathbb{E}\|r_k\|^2 \le d_k^2, \\ \mathbb{E}\|\theta_k - \beta_{e(k)}\|^2 \le \ell_k^2, \qquad \mathbb{E}\|H_k\|^2 \le V_k. \end{aligned} \tag{66}$$

Then

$$\begin{aligned} \frac{1}{\mathcal{S}_T}\sum_{k<T} \eta_k \mathbb{E}\|\nabla F(\theta_k)\|^2 \le \frac{2\Delta_F}{\mathcal{S}_T} + \frac{\sum_{k<T}\eta_k d_k^2}{\mathcal{S}_T} \\ + \frac{2L\sum_{k<T}\eta_k \ell_k \sqrt{v_{e(k)}}}{\mathcal{S}_T} + \frac{L\sum_{k<T}\eta_k^2 V_k}{\mathcal{S}_T}. \end{aligned} \tag{67}$$

Proof. Abbreviate $g_k = \nabla F(\theta_k)$ and $g_e = \nabla F(\beta_e)$. The definitions imply $m_k = g_k + Z_e + r_k$ when $e = e(k)$. Although g_k generally depends on the reused error Z_e, g_e is measurable before the buffer is sampled and $\mathbb{E}[Z_e \mid \mathcal{F}_e^{\text{start}}] = 0$. Consequently,

$$\begin{aligned} \mathbb{E}\langle g_k, Z_e \rangle &= \mathbb{E}\langle g_k - g_e, Z_e \rangle \\ &\ge -\left(\mathbb{E}\|g_k - g_e\|^2\right)^{1/2} \left(\mathbb{E}\|Z_e\|^2\right)^{1/2} \\ &\ge -L\ell_k \sqrt{v_e}. \end{aligned}$$

Young's inequality also gives

$$\mathbb{E}\langle g_k, r_k\rangle \geq -\frac{1}{2}\mathbb{E}\|g_k\|^2 - \frac{1}{2}d_k^2.$$

Combining the two bounds yields

$$\mathbb{E}\langle g_k, m_k\rangle \geq \frac{1}{2}\mathbb{E}\|g_k\|^2 - \frac{1}{2}d_k^2 - L\ell_k\sqrt{v_e}.$$

Since θ_k is $\mathcal{F}_k$-measurable, smoothness implies

$$\mathbb{E}[F(\theta_{k+1}) - F(\theta_k)] \geq \frac{\eta_k}{2}\mathbb{E}\|g_k\|^2 - \frac{\eta_k}{2}d_k^2 - L\eta_k\ell_k\sqrt{v_e} - \frac{L\eta_k^2}{2}V_k.$$

Sum over $k < T$, use the upper bound on F, multiply by two, and divide by $\mathcal{S}_T$. □

The factor $\ell_k\sqrt{v_{e(k)}}$ combines the root-mean-square distance from the old policy with the size of the batch sampling error. It accounts for their possible correlation. If the batch averages B_e conditionally independent groups, each with centred conditional second moment at most σ_e^2, then $v_e \leq \sigma_e^2/B_e$. This averaging condition is across groups; tokens or rollouts within a group may remain dependent.

A rate for a bounded number of buffer reuses. Suppose epoch e starts at $\theta_{k_e} = \beta_e$, lasts at most K updates, and uses a constant step size η, with $\mathbb{E}\|H_k\|^2 \leq V$. The root-mean-square distance from β_e is bounded by the sum of the root-mean-square step sizes:

$$\ell_k \leq \sum_{j=k_e}^{k-1} \eta\left(\mathbb{E}\|H_j\|^2\right)^{1/2} \leq \eta K\sqrt{V}.$$

If the residual also satisfies $d_k \leq C\eta K$ and $v_e \leq \sigma^2/B$, then

$$\frac{1}{T}\sum_{k<T}\mathbb{E}\|\nabla F(\theta_k)\|^2 \leq \frac{2\Delta_F}{\eta T} + C^2\eta^2K^2 + 2L\eta K\sigma\sqrt{V/B} + L\eta V. \tag{68}$$

For fixed K and B, and constants independent of the run length T, choosing $\eta \asymp T^{-1/2}$ gives $O(T^{-1/2})$. When K grows, the additional $K/\sqrt{T}$ and K^2/T terms remain in the bound. The condition $d_k \leq C\eta K$ requires control of both update drift and mean mismatch; a persistent e_e can leave a residual. The next subsection supplies an explicit drift bound for HolderPO.

N.5 Bounding the drift using Holder quantities

We now bound how much the update changes on the same buffer after the policy moves. The bound separates four effects: a change in the sequence aggregate, token reweighting, movement of the token scores, and clipping. This connects the quantities in Theorems A1 and A2 to the residual in Theorem D2.

Fix the buffer, its advantages, and its old-policy parameter β. For one group, write its unclipped update and its value at the old policy as

$$h_p^{\mathrm{un}}(\theta;\beta) = \frac{1}{G}\sum_i \widehat{A}_i\rho_{i,p}(\theta;\beta)\sum_t W_{i,t}(p;\theta,\beta)s_{i,t}(\theta), \tag{69}$$

$$h_\beta = \frac{1}{G}\sum_i \widehat{A}_i\frac{1}{n_i}\sum_t s_{i,t}(\beta). \tag{70}$$

All ratios use β as the denominator policy. Let $d = \theta - \beta$ be the parameter displacement, and let

$$D_i = \max_t \log r_{i,t}(\theta;\beta) - \min_t \log r_{i,t}(\theta;\beta), \qquad M_i = \max_t \|s_{i,t}(\theta)\|.$$

Thus D_i is the range of token log-ratios within sequence i, and M_i is its largest token-score norm at the current parameter. Choose L_i so that $\|s_{i,t}(\theta) - s_{i,t}(\beta)\| \leq L_i\|d\|$ for every observed token. It controls how quickly those score vectors change as the parameter moves.

Lemma D3 (Pathwise drift of the Holder update)**.** *For every finite order p and the fixed buffer above,*

$$\|h_p^{\mathrm{un}}(\theta;\beta) - h_\beta\| \le \Delta_p(\theta;\beta) := \frac{1}{G}\sum_i |\widehat{A}_i| \Bigg\{ M_i \left[|\rho_{i,p} - 1| + \rho_{i,p} \min\left\{2, \frac{|p| D_i}{2}\right\}\right] + L_i \|d\| \Bigg\}. \tag{71}$$

If sequence clipping multiplies the i-th update by $I_{i,p} \in [0,1]$, then the clipped update h_p^{cl} satisfies

$$\|h_p^{\mathrm{cl}}(\theta;\beta) - h_\beta\| \le \Delta_p(\theta;\beta) + \frac{1}{G}\sum_i |\widehat{A}_i| \rho_{i,p} M_i |1 - I_{i,p}|. \tag{72}$$

Proof. Let $u_t = 1/n_i$. For one sequence, subtract the old-policy average score and decompose the difference as

$$\begin{aligned}
&\rho_{i,p} \sum_t W_{i,t}(p) s_{i,t}(\theta) - \sum_t u_t s_{i,t}(\beta) \\
&\quad = \rho_{i,p} \sum_t (W_{i,t}(p) - u_t) s_{i,t}(\theta) + (\rho_{i,p} - 1) \sum_t u_t s_{i,t}(\theta) + \sum_t u_t (s_{i,t}(\theta) - s_{i,t}(\beta)).
\end{aligned}$$

Its norm is at most

$$M_i \left[\rho_{i,p} \|W_i^p - W_i^0\|_1 + |\rho_{i,p} - 1| \right] + L_i \|d\|.$$

Theorem A2 bounds the weight difference by $\min\{2, |p| D_i/2\}$. Multiply by $|\widehat{A}_i|$, average over the group, and apply the triangle inequality. Clipping changes the i-th contribution by a vector of norm at most $|\widehat{A}_i| \rho_{i,p} M_i |1 - I_{i,p}|$, which proves the second bound. □

The bound applies to each realised buffer and each finite order, including an order selected using that buffer. Its three terms in Eq. (71) measure aggregate change, token-weight change, and score-vector change; Eq. (72) adds the contribution removed by clipping. The same bounds apply to a fixed average of groups.

Using the bound in the convergence theorem. For a full-buffer update, $\bar{H}_e$ averages the old-policy updates h_{β_e}. Let $\mathcal{D}_k$ bound the difference $\|m_k - \bar{H}_e\|$, using the preceding inequalities. The residual also includes the change in the target gradient and the mean mismatch:

$$\|r_k\| \le \mathcal{D}_k + L\|\theta_k - \beta_e\| + \|e_e\|.$$

Thus a sufficient choice in Eq. (66) is

$$d_k \ge \left(\mathbb{E}[(\mathcal{D}_k + L\|\theta_k - \beta_e\| + \|e_e\|)^2] \right)^{1/2}.$$

For a minibatch update, let H_k^0 be the old-policy update on the same sampled minibatch. If its conditional mean equals the full-buffer update, $\mathbb{E}[H_k^0 \mid \mathcal{F}_k] = \bar{H}_e$, then

$$m_k - \bar{H}_e = \mathbb{E}[H_k - H_k^0 \mid \mathcal{F}_k].$$

Conditional Jensen's inequality transfers the bound for each minibatch to its conditional mean, including when order selection uses that minibatch. Uniform sampling with replacement and correct averaging give the required old-policy mean. With another sampling scheme, recomputed advantages, or an optimizer transformation, $\mathcal{D}_k$ must also include the changes those operations make to the update.

A sufficient condition for drift proportional to distance. The residual required for Eq. (68) can be controlled locally. Suppose, on the buffer and the segments from β to θ, that $|\widehat{A}_i| \le A_0$, score norms are at most M, and score Lipschitz constants are at most L_s. Also suppose $|p| \le P$ and $\|\theta - \beta\| \le r$. Then

$$|\log r_{i,t}| \le M\|d\|, \qquad D_i \le 2M\|d\|, \qquad \rho_{i,p} \le e^{Mr}, \qquad |\rho_{i,p} - 1| \le e^{Mr} M \|d\|.$$

The bounds on $\rho_{i,p}$ follow because every power mean lies between the minimum and maximum of its positive inputs. Consequently,

$$\Delta_p(\theta;\beta) \le A_0 \left[L_s + M^2 e^{Mr}(1+P)\right] \|d\|. \tag{73}$$

For $0 < \epsilon < 1$, the condition

$$Mr < \min\{\log(1+\epsilon), -\log(1-\epsilon)\}$$

keeps every sequence aggregate strictly inside the clipping interval, so clipping adds no drift in this neighbourhood. With zero old-policy mean mismatch, the residual is then bounded by a constant times $\|\theta_k - \beta_e\|$. Combining this with $\ell_k \le \eta K\sqrt{V}$ gives $d_k = O(\eta K)$, as needed for Eq. (68). When clipping is active, its explicit term remains part of the second-moment bound. These local conditions provide one way to control the drift; the general bound can be used directly whenever its terms are controlled. In particular, the reweighting term depends on $|p|D_i$, so a shrinking ratio range must be considered together with the order.

N.6 Interpretation for the dynamic schedule

The convergence argument has a direct chain of inputs: Theorem A1 identifies the common old-policy update, Theorem A2 controls changes in token weights, and Lemma D3 bounds the resulting update drift. Theorem D2 then converts the drift and sampling bounds into an expected stationarity guarantee. Fixed, decreasing, nonmonotone, and data-adaptive orders can all satisfy these conditions; comparing their scheduling choices is the separate question studied in Appendix O.

For the original binary-reward update, the analysis supplies the fixed objective F_0 and the explicit mismatch e_{len}. Other rewards or normalisations require their corresponding objective and alignment calculation.

Throughout, the convergence conditions apply to the direction actually used to change the parameters. Thus an optimizer such as AdamW must be included in H_k, and any shared KL regulariser must be included consistently in both the fixed objective and the update. This keeps the guarantee tied to the implemented training rule.

Relation to the experiments. Figures 2 and 3 track ratio ranges, policy entropy, and training-gradient norms under fixed and decreasing orders. The later contraction of the ratio range provides empirical motivation for controlling update drift during training. This appendix identifies the additional quantities needed for a convergence guarantee: alignment with a fixed objective gradient, update second moments, and errors from length normalisation and buffer reuse. The plotted training-loss gradient norms describe the observed optimisation dynamics.

O Optimality of Linear Holder-Order Decay

Two questions underlie linear decay: how to order prescribed parameter values, and how smoothly to connect fixed endpoints. We retain an affine error model for the first question, the squared-change criterion for the second, and their joint consequence on a prescribed uniform grid. The optimality statements are limited to their stated comparison classes.

O.1 Two comparison classes

Each optimality statement specifies which schedules are being compared. Fix $T \ge 2$ updates and a high and low endpoint $p_{\mathrm{H}} > p_{\mathrm{L}}$. Write the endpoint difference and the linear decreasing schedule as

$$\Delta = p_{\mathrm{H}} - p_{\mathrm{L}}, \qquad p_k^{\mathrm{lin}} = p_{\mathrm{H}} - \frac{k\Delta}{T-1}, \quad k = 0, \dots, T-1.$$

There are T order values, $T-1$ transitions between them, and a terminal state indexed by T after all updates.

The first comparison keeps the order values and the number of times each is used fixed, and changes only their sequence. A collection with specified repetition counts is called a multiset. For linear

decay, the relevant collection is the equally spaced grid

$$\mathcal{P}_T = \left\{ p_{\mathrm{H}} - \frac{j\Delta}{T-1} : j = 0, \dots, T-1 \right\}, \tag{74}$$

Each value is used once, and $\mathrm{Perm}(\mathcal{P}_T)$ denotes all possible orderings of these values.

The second comparison fixes only the endpoints and step count. The intermediate values can be chosen anywhere in $[p_{\mathrm{L}}, p_{\mathrm{H}}]$, with $p_0 = p_{\mathrm{H}}$ and $p_{T-1} = p_{\mathrm{L}}$. Denote these schedules by $\mathcal{A}$, including those that are not monotone. This second comparison can change which values are used; a permutation of a prescribed grid cannot.

O.2 Optimal ordering of fixed choices

We begin with a model in which each update retains a fraction of the current error and adds a nonnegative error contribution:

$$v_{k+1} = T_{p_k}(v_k), \qquad T_p(v) = a(p)v + b(p), \qquad 0 \le a(p) < 1, \quad b(p) \ge 0. \tag{75}$$

The nonnegative scalar v_k measures error, so smaller values are preferred. For order p, $a(p)$ is the retained fraction and $b(p)$ is the added contribution. The same map applies whenever that order is used. Its fixed-order error floor is

$$\nu(p) = \frac{b(p)}{1 - a(p)}, \tag{76}$$

This is the limiting error when the same order is repeated: $v_k = \nu(p) + a(p)^k(v_0 - \nu(p))$. The functions a and b can depend nonlinearly on p. The next theorem shows that, for a fixed collection of updates, larger error floors should be used earlier. General affine composition ordering is studied in Kawase et al. (2016); Kubo et al. (2024); here a two-update comparison gives the required condition directly.

Theorem E1 (Optimal ordering of a prescribed multiset)**.** *Under Eq.* (75)*, sorting any finite multiset by nonincreasing $\nu(p)$ minimises the terminal state among its permutations, for every $v_0 \ge 0$. If $\nu(p)$ is nondecreasing on the selected order values, nonincreasing p is therefore optimal. In particular, for the grid in Eq.* (74)*, the linear sequence $\boldsymbol{p}^{\mathrm{lin}}$ minimises v_T over* $\mathrm{Perm}(\mathcal{P}_T)$.

Proof. For adjacent choices p and q, direct calculation gives

$$\begin{aligned} T_q(T_p(v)) - T_p(T_q(v)) &= b(q)(1 - a(p)) - b(p)(1 - a(q)) \\ &= (1 - a(p))(1 - a(q))(\nu(q) - \nu(p)). \end{aligned} \tag{77}$$

Thus a higher-floor choice followed by a lower-floor choice gives a state no larger than the reversed pair. Every subsequent map is nondecreasing, so the common suffix preserves this comparison. Repeatedly exchanging adjacent inversions produces a sorted sequence without increasing the terminal state. If the floors are nondecreasing in p, sorting p downwards also sorts the floors downwards. On the uniform grid, the result is exactly p_k^{lin}. □

To apply the result, check the ordering of $\nu(p)$ at the values actually used. Neither monotonicity of $a(p)$ nor a restriction on the initial error relative to the floors is needed. Because the floor combines contraction and added error, an increasing $b(p)$ or variance bound alone does not determine its ordering. Ties are possible: equal floors give equal two-update compositions, and a later map with zero slope can remove an earlier difference.

O.3 Why the interpolation is linear

Sorting determines how to use chosen values. To choose their spacing, consider a separate goal: spreading the total order change evenly across the available transitions. With equally spaced update indices, measure uneven changes by

$$\mathcal{E}(\boldsymbol{p}) = \sum_{k=0}^{T-2} (p_{k+1} - p_k)^2. \tag{78}$$

The square penalises a large jump more than several smaller jumps with the same total change. The next theorem shows that equal increments uniquely minimise this cost. It also covers unequal time intervals and any common convex cost of the rate of change.

Theorem E2 (Linear interpolation minimises convex rate costs)**.** *Fix times $t_0 < \cdots < t_{T-1}$, with $h_k = t_{k+1} - t_k > 0$ and $\mathcal{T} = t_{T-1} - t_0$. For any convex function $\varphi : \mathbb{R} \to \mathbb{R}$ and any schedule with the stated endpoints,*

$$\sum_{k=0}^{T-2} h_k \varphi\left(\frac{p_{k+1} - p_k}{h_k}\right) \geq \mathcal{T}\varphi\left(-\frac{\Delta}{\mathcal{T}}\right). \tag{79}$$

Linear interpolation in time, $p_k = p_{\mathrm{H}} - \Delta(t_k - t_0)/\mathcal{T}$, attains equality. If φ is strictly convex, it is the unique minimiser.

In particular, for $t_k = k$ and $\varphi(u) = u^2$,

$$\mathcal{E}(\boldsymbol{p}) = \frac{\Delta^2}{T-1} + \sum_{k=0}^{T-2}\left(p_{k+1} - p_k + \frac{\Delta}{T-1}\right)^2. \tag{80}$$

Thus $\boldsymbol{p}^{\mathrm{lin}}$ uniquely minimises $\mathcal{E}$ over the fixed-endpoint class $\mathcal{A}$. Moreover, for every $k = 0, \ldots, T-1$,

$$|p_k - p_k^{\mathrm{lin}}|^2 \leq \frac{k(T-1-k)}{T-1}\left(\mathcal{E}(\boldsymbol{p}) - \frac{\Delta^2}{T-1}\right). \tag{81}$$

Proof. The weighted average of the rates is fixed by the endpoints:

$$\sum_k \frac{h_k}{\mathcal{T}} \frac{p_{k+1} - p_k}{h_k} = -\frac{\Delta}{\mathcal{T}}.$$

Jensen's inequality gives Eq. (79). Equal rates attain the bound, and strict convexity makes equal rates necessary.

For the quadratic statement, put $N = T - 1$ and $d_k = p_{k+1} - p_k + \Delta/N$. The endpoints give $\sum_{k=0}^{N-1} d_k = 0$. Expanding $\sum_k (d_k - \Delta/N)^2$ proves Eq. (80). For a fixed interior index j,

$$p_j - p_j^{\mathrm{lin}} = \sum_{k<j} d_k = \sum_{k=0}^{N-1}\left(\mathbf{1}\{k < j\} - \frac{j}{N}\right) d_k.$$

The squared norm of the coefficient vector is $j(N-j)/N$. Cauchy–Schwarz and the decomposition give Eq. (81). The two endpoints satisfy the bound trivially. □

This conclusion holds for all schedules with the given endpoints, including nonmonotone schedules, without a training-dynamics model. Strict convexity selects a unique line: the absolute-change cost $\sum_k |p_{k+1} - p_k|$, for example, is minimised by every nonincreasing schedule with those endpoints.

The quadratic result also gives a quantitative form of closeness. If the squared-change cost exceeds its minimum by at most ε, Eq. (81) yields

$$\max_k |p_k - p_k^{\mathrm{lin}}| \leq \frac{1}{2}\sqrt{(T-1)\varepsilon}. \tag{82}$$

Thus nearly minimising the squared changes also keeps every order value close to the corresponding point on the line.

Equal increments reflect equal treatment of the transitions. For prescribed weights $\omega_k > 0$, minimising $\sum_k \omega_k (p_{k+1} - p_k)^2$ instead gives, by weighted Cauchy–Schwarz,

$$p_{k+1} - p_k = -\Delta \frac{\omega_k^{-1}}{\sum_j \omega_j^{-1}}.$$

Larger-cost transitions receive smaller changes. Equal weights give equal increments; weights $\omega_k = 1/h_k$ give a constant rate in physical time, where h_k is the duration of transition k.

What a smooth order change controls. Theorem A2 connects the change in order to the change in token weights. At transition k, hold the sequence and policy parameter fixed, denote the within-sequence log-ratio range by D_k, and write W_k^p for the weights at order p. Comparing the two orders on this same ratio profile gives

$$\|W_k^{p_{k+1}} - W_k^{p_k}\|_1 \le \frac{D_k}{2}|p_{k+1} - p_k|.$$

If $D_k \le \overline{D}$ in all comparisons, then

$$\sum_{k=0}^{T-2} \|W_k^{p_{k+1}} - W_k^{p_k}\|_1^2 \le \frac{\overline{D}^2}{4}\mathcal{E}(\boldsymbol{p}). \tag{83}$$

The linear schedule minimises this shared upper bound on weight changes caused by changing the order. Changes in the policy or buffer are additional effects, and the upper bound need not equal the observed weight changes. If the prescribed bounds D_k differ, they give a weighted squared-change cost, whose optimal increments can differ across transitions.

O.4 Joint optimality and the uniform-grid consequence

For a fixed collection of orders, the decreasing arrangement can minimise both the terminal error and the squared changes. Equal spacing is unnecessary for this joint result; it makes the decreasing arrangement a linear schedule. Write $J(\boldsymbol{p}) = v_T$ for the scalar terminal model error, a loss to minimise.

Theorem E3 (Joint optimality for prescribed order values)**.** *Fix any multiset of T orders, with prescribed multiplicities, and let $\boldsymbol{p}^{\downarrow}$ be its nonincreasing ordering. Assume the affine model and that $\nu(p)$ is nondecreasing on the selected values, as in Theorem E1. Then for every permutation $\boldsymbol{p}$ of that multiset,*

$$J(\boldsymbol{p}^{\downarrow}) \le J(\boldsymbol{p}), \qquad \mathcal{E}(\boldsymbol{p}^{\downarrow}) \le \mathcal{E}(\boldsymbol{p}). \tag{84}$$

Consequently, the decreasing ordering minimises every criterion $\Phi(J(\boldsymbol{p}), \mathcal{E}(\boldsymbol{p}))$ that is nondecreasing in each argument, including $J + \lambda\mathcal{E}$ for any $\lambda \ge 0$.

In particular, for the grid $\mathcal{P}_T$ in Eq. (74)*, using each value once, $\boldsymbol{p}^{\downarrow} = \boldsymbol{p}^{\mathrm{lin}}$. Thus linear decay minimises*

$$J(\boldsymbol{p}) + \lambda\mathcal{E}(\boldsymbol{p}) \quad \textit{over } \mathrm{Perm}(\mathcal{P}_T). \tag{85}$$

If permutations are also required to start at p_{H} and finish at p_{L}, the minimiser of (85) *is unique for $\lambda > 0$.*

Proof. The sorting theorem gives the first inequality in Eq. (84). For the second, list the distinct selected values as $\xi_0 < \cdots < \xi_m$ and put $g_j = \xi_{j+1} - \xi_j > 0$. A jump crossing some consecutive gaps has squared length at least the sum of those gaps' squares, since $(\sum_j g_j)^2 \ge \sum_j g_j^2$. Any permutation visits both sides of every gap and must cross each gap at least once. Summing the jump costs therefore gives

$$\mathcal{E}(\boldsymbol{p}) \ge \sum_{j=0}^{m-1} g_j^2 = \mathcal{E}(\boldsymbol{p}^{\downarrow}).$$

Repeated equal values add zero-length jumps; if all values are equal, both sides are zero. This argument does not require a competing permutation to have the same endpoints. Monotonicity of Φ gives the joint claim, and the uniform-grid consequence follows immediately. For that grid with the endpoints fixed, the equality characterisation in Theorem E2 makes every nonlinear competitor have strictly larger energy; its terminal loss is also no smaller. Thus $\lambda > 0$ gives uniqueness. □

Within this fixed collection, there is no need to sacrifice one criterion for the other: the decreasing arrangement minimises both. Reversing it has the same squared-change cost, so the dynamics determine why the decreasing direction is preferred.

The result explains two steps in designing linear decay. First, equal increments minimise the squared-change cost for fixed endpoints and step count. Then, on that equally spaced collection, the exchange condition selects the decreasing ordering for the terminal error. The scope paragraph below distinguishes the larger class in which intermediate values can change.

Scope beyond a prescribed grid. Allowing intermediate values to change is a larger comparison class. Linear interpolation still minimises squared changes, but the preceding sorting result does not establish its optimality for terminal training error over that larger class. A fitted error model and its approximation errors would have to be assessed separately.

O.5 Interpretation for the proposed HolderPO schedule

The theory gives two reasons for linear decay. Equal changes in order minimise the squared-change cost for fixed endpoints and step count. Under the exchange condition, using higher orders earlier minimises the terminal model error for a fixed collection. For equally spaced values used once each, both conclusions select the same linear decreasing schedule. This does not compare all possible choices of intermediate values.

Applying the ordering result to HolderPO requires a model of one common training error whose updates satisfy the exchange condition. The model should include the effects of the learning rate, clipping, normalisation, buffer reuse, and optimizer when they affect that error. The concentration and variance results in Sections 3.2 and 3.3, together with the endpoint comparisons in Theorem 3, help describe these effects; the exchange condition is the additional comparison required here.

The squared-change argument is independent of the dynamics, and Eq. (83) connects it to an upper bound on token-weight changes caused by the order. These results explain the schedule's shape and its ordering within the stated classes. Endpoint values, training duration, and usage counts remain design choices outside the fixed-grid comparison.

Relation to the experiments. Tables 3 and 7 show that linear decay achieves the highest average score among the tested shapes: 55.2%, compared with 52.6% for linear increase and 51.1–52.9% for the nonlinear alternatives. This supports linear decay as an effective design in the tested setting. The present analysis supplies two complementary reasons for this choice: equal changes minimise the squared-change cost, and decreasing order minimises terminal model error under the exchange condition. The shape experiments vary both the ordering and the values visited, so they do not directly test the fixed-collection ordering guarantee.

P Token Reweighting and Improving Direction

We hold the current policy, old policy, and evaluation objective F fixed, with $v = \nabla F(\theta)$. For a fixed rollout its advantage also stays fixed. The covariance criterion isolates reweighting; the complete update comparison additionally includes its scale, clipping, and second moment. These results connect concentration to the order-selection criteria.

P.1 A covariance criterion for token reweighting

We first study reweighting alone: when does placing more weight on larger ratios increase a useful token contribution? Fix a sequence y_i with $n_i = |y_i|$ and positive token ratios. Write

$$z_{i,t} = \log r_{i,t}(\theta), \qquad s_{i,t} = \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t}), \qquad W_{i,t}(p) = \frac{e^{pz_{i,t}}}{\sum_j e^{pz_{i,j}}}.$$

Here $z_{i,t}$ is the token log-ratio, and the score $s_{i,t}$ is the gradient of that token's log-probability with respect to the model parameters. Both are fixed as we vary p. Let $a_{i,t}$ be any fixed real contribution assigned to token t, and let $S_i(p)$ be its weighted average:

$$S_i(p) = \sum_t W_{i,t}(p) a_{i,t}. \tag{86}$$

To measure progress towards F, choose

$$a_{i,t} = \widehat{A}_i \langle v, s_{i,t} \rangle. \tag{87}$$

The inner product measures how much the advantage-scaled token update points along v. Thus $a_{i,t} > 0$ means a helpful direction and $a_{i,t} < 0$ an opposing direction, including when the advantage is negative. This is a geometric measure of an update's contribution, not a token reward or a measure of its semantic importance.

Lemma F0 (Directional effect of reweighting)**.** *For fixed finite $z_{i,t}, a_{i,t}$ and every finite p,*

$$S_i'(p) = \mathrm{Cov}_{W_i^p}(z_{i,t}, a_{i,t}) = \sum_t W_{i,t}(p)\big(z_{i,t} - \mu_i(p)\big)a_{i,t}, \tag{88}$$

where $\mu_i(p) = \sum_t W_{i,t}(p) z_{i,t}$. Consequently, for finite $p < q$,

$$S_i(q) - S_i(p) = \int_p^q \mathrm{Cov}_{W_i^u}(z_{i,t}, a_{i,t})\, du. \tag{89}$$

In particular, positive covariance at p gives a positive local derivative. Nonnegative covariance throughout $[p, q]$ gives $S_i(q) \geq S_i(p)$.

Proof. Differentiating the normalised exponential weights gives

$$\frac{\partial W_{i,t}(p)}{\partial p} = W_{i,t}(p)\big(z_{i,t} - \mu_i(p)\big).$$

Since $a_{i,t}$ is fixed, differentiating the finite sum defining S_i proves Eq. (88). All weights are smooth for finite p, so integration gives Eq. (89). The sign conclusions follow immediately. □

The weighted covariance measures whether tokens with larger log-ratios also make larger contributions, using W_i^p to average across positions in this sequence. It is calculated from the fixed token values; no noise model or independence assumption is needed.

When increasing the order helps. The same covariance has the pairwise form:

$$\mathrm{Cov}_{W_i^p}(z, a) = \frac{1}{2}\sum_{t,j} W_{i,t}(p) W_{i,j}(p)(z_{i,t} - z_{i,j})(a_{i,t} - a_{i,j}) \tag{90}$$

This follows by expanding the right-hand side. If a larger log-ratio always comes with a contribution at least as large, every pair contributes a nonnegative term. Increasing p then cannot decrease S_i. If at least one pair contributes strictly positively, the covariance is positive for every finite p, since all token weights are positive. When some pairs disagree, the covariance still gives the exact local test: its sign is the sign of $S_i'(p)$.

Why the location of a useful token matters. Consider a nonconstant log-ratio profile with exactly one token contributing 1 and all others contributing 0. If the useful token has a maximum log-ratio, then

$$S_i'(p) = W_{i,t}(p)\big(z_{i,t} - \mu_i(p)\big) > 0.$$

If the useful token instead has a minimum log-ratio, the derivative is negative. The two cases are equally sparse, even for arbitrarily long sequences, but favour opposite changes in p. What matters is where useful contributions occur relative to the ratios.

With Eq. (87), $S_i(p) = \langle v, \widehat{A}_i \sum_t W_{i,t}(p) s_{i,t}\rangle$ measures the alignment of the reweighted contribution with v. The full Holder update also includes the multiplier $\rho_{i,p}$, and its improvement bound depends on the update's second moment. We next include both effects.

P.2 An exact finite comparison of complete updates

We now compare complete updates at the same learning rate $\eta > 0$. Let H_p be the random direction actually used in $\theta + \eta H_p$, including group normalisation, clipping, batch averaging, and any optimiser transformation. Candidates share the same pre-update state and sampling distribution; tokens and rollouts may be dependent, and advantages may have either sign.

Assume that F has an L-Lipschitz gradient on every line segment from θ to a candidate updated parameter, and that $\mathbb{E}\|H_p\|^2 < \infty$. Here L bounds how quickly the objective gradient can change along an update. The utility from Eq. (51) is

$$U(p) = \eta\langle v, \mathbb{E}H_p\rangle - \frac{L\eta^2}{2}\mathbb{E}\|H_p\|^2. \tag{91}$$

The first term rewards movement along v; the second accounts for the size and variability of the complete update. Its second moment includes all interactions between token and rollout contributions. The theorem gives an exact comparison of this one-step improvement lower bound for any two orders.

Theorem F1 (Finite differences of the improvement certificate)**.** *Under the conditions above,*

$$\mathbb{E}[F(\theta + \eta H_p) - F(\theta)] \geq U(p). \tag{92}$$

For any two candidates p, q, construct H_p, H_q on a common probability space with their correct marginal distributions. Then

$$U(q) - U(p) = \eta \mathbb{E}\left\langle v - \frac{L\eta}{2}(H_q + H_p),\ H_q - H_p \right\rangle. \tag{93}$$

This identity does not require continuity or differentiability of the update as a function of the order.

Proof. The smoothness inequality gives, for every realised update,

$$F(\theta + \eta H_p) - F(\theta) \geq \eta \langle v, H_p \rangle - \frac{L\eta^2}{2}\|H_p\|^2.$$

Taking expectations proves Eq. (92). For the difference, expand Eq. (91) and use

$$\|H_q\|^2 - \|H_p\|^2 = \langle H_q + H_p, H_q - H_p \rangle.$$

Combining the linear and quadratic terms proves Eq. (93). Finite second moments make the displayed inner product integrable by Cauchy–Schwarz. The construction of the joint distribution does not alter either candidate's expected utility, so the identity holds for any coupling with the prescribed marginals. □

Reading the comparison. The change $H_q - H_p$ raises U exactly when its expected inner product with $v - (L\eta/2)(H_q + H_p)$ is positive. This balances better alignment with v against the change in the quadratic update cost, including changes caused by clipping.

One way to compare candidates is to compute them on the same evaluation batch. This puts them on a common probability space, as required by the theorem, without requiring independent candidates or tokens. On a fixed buffer, remove expectations to obtain the same identity for deterministic candidate updates.

A positive $U(p)$ guarantees positive expected one-step improvement. Comparing two values of U ranks these guarantees, rather than the actual gains or final training performance. For $F(\theta + u) = F(\theta) + \langle v, u \rangle - (L/2)\|u\|^2$, the bound is exact, so it also ranks the actual one-step gains.

Local utility comparison. If H_p is differentiable in L^2, dividing Eq. (93) by $q - p$ and taking $q \to p$ gives

$$U'(p) = \eta \mathbb{E}\langle v - L\eta H_p, \dot{H}_p \rangle. \tag{94}$$

The L^2 condition permits the expectation-level limit by Cauchy–Schwarz. At clipping switches use the finite comparison; samplewise differentiation alone need not justify differentiation of an expectation.

P.3 The two components of the Holder update derivative

For HolderPO, changing p affects both the common multiplier $\rho_{i,p}$ and the relative token weights. We now separate these two effects. For a fixed sequence, define

$$m_{i,p} = \sum_t W_{i,t}(p) s_{i,t}, \qquad \alpha_{i,p} = \frac{\partial}{\partial p} \log \rho_{i,p}, \qquad h_{i,p} = \widehat{A}_i \rho_{i,p} m_{i,p}.$$

Here $m_{i,p}$ is the weighted mean token score, $\alpha_{i,p}$ measures the sensitivity of $\log \rho_{i,p}$ to p, and $h_{i,p}$ is the sequence's unclipped surrogate gradient. Its advantage and token scores stay fixed. The covariance between log-ratios and score vectors is

$$\mathrm{Cov}_{W_i^p}(z, s) = \sum_t W_{i,t}(p)(z_{i,t} - \mu_i(p)) s_{i,t}.$$

This vector combines the token scores according to whether their log-ratios are above or below the weighted mean $\mu_i(p)$. Subtracting $m_{i,p}$ from every score leaves the expression unchanged because the centred log-ratios have weighted mean zero.

Proposition F2 (Common-scale and reweighting terms)**.** *For every finite order, including the continuous extension at $p = 0$,*

$$\frac{\partial h_{i,p}}{\partial p} = \widehat{A}_i \rho_{i,p} \left[\alpha_{i,p} m_{i,p} + \mathrm{Cov}_{W_i^p}(z, s) \right]. \tag{95}$$

Consequently, with $a_{i,t}$ from Eq. (87)*,*

$$\frac{\partial}{\partial p} \langle v, h_{i,p} \rangle = \rho_{i,p} \left[\alpha_{i,p} S_i(p) + \mathrm{Cov}_{W_i^p}(z, a) \right]. \tag{96}$$

Proof. Since the scores are fixed,

$$\frac{\partial m_{i,p}}{\partial p} = \sum_t W_{i,t}(p)(z_{i,t} - \mu_i(p)) s_{i,t} = \mathrm{Cov}_{W_i^p}(z, s).$$

Apply the product rule to $h_{i,p} = \widehat{A}_i \rho_{i,p} m_{i,p}$ and use $\partial_p \rho_{i,p} = \rho_{i,p} \alpha_{i,p}$ to prove Eq. (95). Taking the inner product with the fixed vector v, and using $\widehat{A}_i \langle v, m_{i,p} \rangle = S_i(p)$, proves Eq. (96). The extension at zero follows from the smooth finite-order identities in Appendix J.3. □

In Eq. (95), $\alpha_{i,p} m_{i,p}$ rescales the current weighted direction, while the covariance changes the relative contributions of the token directions. Theorem A2 gives

$$\alpha_{i,p} = \int_0^1 u \, \mathrm{Var}_{W_i^{up}}(z_{i,t}) \, du \geq 0.$$

A larger common scale amplifies whichever direction the update already has. If $S_i(p) < 0$, it amplifies an opposing contribution; it also changes the quadratic cost in U. Thus the covariance describes reweighting alone, while Eq. (94) combines both effects in the complete utility.

Group updates and clipping. For an unclipped group update, average Eq. (95) across rollouts. For the raw sequence-clipped gradient on a fixed buffer, let $I_{i,p}$ indicate whether rollout i contributes to the gradient. Then

$$H_p = \frac{1}{G} \sum_i I_{i,p} h_{i,p}.$$

On an open interval where every $I_{i,p}$ is constant,

$$\frac{\partial H_p}{\partial p} = \frac{1}{G} \sum_i I_{i,p} \widehat{A}_i \rho_{i,p} \left[\alpha_{i,p} m_{i,p} + \mathrm{Cov}_{W_i^p}(z, s) \right].$$

Substituting this derivative into Eq. (94), under its stated conditions, keeps all interactions between rollout contributions through the full H_p. Use the finite comparison at clipping switches. If an optimiser transforms this gradient, apply the comparison to its transformed update and differentiate that update where appropriate.

Connection to A1 and A2. At the old policy, every $z_{i,t} = 0$, so both derivative terms vanish: all finite orders give the same update, as in Theorem A1. Away from the old policy, A2 bounds the size of the change. If the token score norms are bounded by M, it gives

$$\left\| \frac{\partial h_{i,p}}{\partial p} \right\| \leq |\widehat{A}_i| \rho_{i,p} M \left(\frac{D_i^2}{8} + \frac{D_i}{2} \right), \qquad D_i = \max_t z_{i,t} - \min_t z_{i,t}.$$

Here D_i is the log-ratio range within the sequence. A2 controls how large a change can be; the covariance and utility identities determine whether its direction helps the criterion.

The fixed-state identities do not compare separately trained policies: changing ratios, advantages, scores, or evaluation directions adds further derivative terms.

P.4 Relation to parameter selection and to the main text

The results give a directional interpretation of the token concentration in Section 3.2: increasing p favours larger ratios, and a small increase improves the weighted contribution when those ratios are positively associated with useful update directions. Sparsity, token-weight ESS, or ratio magnitude alone does not determine this association or establish a change in policy entropy or reasoning diversity.

For complete updates, the relevant comparison is $U(p)$ at the shared learning rate. In contrast, the efficiency $\mathcal{R}(p)$ compares directional signal to second moment while allowing a separate step size for each order. Keeping the actual scale and clipping in U connects the present identities directly to order selection in Appendix M.

That appendix establishes selection guarantees for finite candidates, including consistency under its sampling and moment conditions. Evaluation units may be whole groups with dependent tokens; the statistical guarantees require independence across units, whereas the identities here do not. Applying the utility criterion also requires estimates of v and a valid smoothness bound L. The next level of analysis is repeated updating: Appendices N and O provide the corresponding convergence and schedule-comparison conditions.

Relation to the experiments. Our supplementary diagnostics show that $p = 2$ gives high-ratio tail tokens about 1.56 times their uniform share of weight, while $p = -2$ gives low-ratio tail tokens about 1.68 times their uniform share. These measurements confirm that the order changes which token contributions receive greater weight. The present analysis identifies what determines whether this reweighting helps: the association between token log-ratios and contributions along the objective gradient.

Q Robustness Limits under Imperfect Rewards

This appendix bounds how specified errors change token weights and the sequence-clipped group update. Comparisons hold policy parameters and rollouts fixed. The token analysis controls concentration on corrupted positions; the reward analysis includes group normalisation and changes of clipping branch. The relation to the reported random-reward experiment is retained below.

Information required to correct rewards. Changing p reweights available observations; it does not supply missing correctness information. If two environments have identical observable data but opposite hidden correctness labels, any shared decision rule has success probabilities summing to one. Hence additional trusted information or an error model is needed to identify systematic verifier errors. The bounds below quantify specified contamination rather than guarantee its correction.

Q.1 A sharp bound on contaminated token weight

We first ask how much of the total token weight can fall on corrupted contributions. Fix one sequence y_i and the current policy. Write $n = |y_i|$ for its length and $z_t = \log r_{i,t}(\theta)$ for its token log-ratios, with

$$w_t(p) = W_{i,t}(p) = \frac{e^{pz_t}}{\sum_k e^{pz_k}}, \qquad D = \max_t z_t - \min_t z_t.$$

Here D measures the log-ratio spread within this sequence. Let S contain the token positions whose update vectors may be corrupted. The fraction of such positions is $\alpha = |S|/n$; their weight mass $w_p(S)$ is the sum of their Holder weights. We allow S to depend arbitrarily on the ratios, including the worst case where it contains the most favoured tokens. The theorem bounds this mass and gives a condition for keeping it below a chosen threshold τ.

Theorem G1 (Sharp amplification of a contaminated subset)**.** *For finite log-ratios and any finite order p, put $\alpha = |S|/n$. Then*

$$w_p(S) := \sum_{t \in S} w_t(p) \le \frac{\alpha e^{|p|D}}{1 - \alpha + \alpha e^{|p|D}}. \tag{97}$$

For fixed α, D, and p, this bound is sharp over all compatible ratio profiles and subsets. In particular, if $0 < \alpha < \tau < 1$, the condition

$$|p|D \le \log \frac{\tau(1-\alpha)}{\alpha(1-\tau)} \tag{98}$$

is sufficient to guarantee $w_p(S) \le \tau$, and is necessary for such a guarantee uniformly over these profiles and subsets.

Proof. The cases $S = \varnothing$ and $S = \{1, \dots, n\}$ are immediate. Otherwise let

$$b_{\min} = \min_t e^{pz_t}, \qquad b_{\max} = \max_t e^{pz_t}, \qquad b_{\max}/b_{\min} = e^{|p|D}.$$

The total unnormalised weight in S is at most $n\alpha b_{\max}$, whereas its complement has weight at least $n(1-\alpha)b_{\min}$. The ratio $x/(x+y)$ increases with x and decreases with y, so

$$w_p(S) \le \frac{n\alpha b_{\max}}{n\alpha b_{\max} + n(1-\alpha)b_{\min}},$$

which proves Eq. (97). For $p > 0$, equality is attained by putting all tokens in S at the largest log-ratio and all other tokens at the smallest. For $p < 0$, reverse the two assignments. If $p = 0$ or $D = 0$, the weights are uniform and the bound equals α. Finally, solving the bound $\alpha e^{|p|D}/(1-\alpha+\alpha e^{|p|D}) \le \tau$ yields Eq. (98); sharpness gives the worst-case necessity. □

From token weight to update error. Let Z_t and $\widetilde{Z}_t$ be the clean and corrupted update vectors for token t. Compare their weighted sums using the same ratios, weights, and multiplier $c_p \ge 0$:

$$H_p = c_p \sum_t w_t(p) Z_t, \qquad \widetilde{H}_p = c_p \sum_t w_t(p) \widetilde{Z}_t.$$

The common multiplier c_p scales the entire sequence update. Suppose the vectors agree outside S and the error at each position in S has norm at most E_0. Then

$$\|\widetilde{H}_p - H_p\| \le c_p \sum_{t \in S} w_t(p) \|\widetilde{Z}_t - Z_t\| \le c_p E_0 \frac{\alpha e^{|p|D}}{1-\alpha+\alpha e^{|p|D}}. \tag{99}$$

For a fixed advantage, take $c_p = |\widehat{A}_i| \rho_{i,p}$ and include the advantage sign in the token vectors. The bound is the product of three quantities: the sequence multiplier, the largest token error, and the total weight on corrupted positions. It allows dependent and systematically biased errors. This comparison holds the weights and multiplier fixed; reward changes are treated separately below.

Interpretation and example. As in Theorem A2, the product $|p|D$ controls how concentrated the weights can become. Here it controls the worst-case weight on corrupted tokens. If at most 10% of the tokens are corrupted, $|p|D \le \log(9/4)$ keeps their total weight at or below 20%. A known upper bound on α is enough, since the bound increases with α.

The sign of p determines which end of the ratio profile is favoured, so either sign can amplify errors. A larger $|p|D$ permits more amplification, without requiring the actual contaminated mass to increase. Equation (98) therefore controls a worst-case weight allocation; converting it to an update-error bound also uses c_p and E_0 in Eq. (99).

Q.2 Reward errors, group normalisation, and clipping

A reward error acts at the rollout level. Group normalisation then spreads its effect: changing one reward can change every advantage in the group. We now bound the resulting change in the full sequence-clipped group gradient, including changes of clipping branch when an advantage changes sign.

Hold a group of G rollouts fixed. Let $R = (R_1, \dots, R_G)$ be its ideal rewards and $\widetilde{R}$ its corrupted rewards, with corresponding advantages $\widehat{A}_i$ and $\widetilde{A}_i$. For the same candidate order, write

$$\rho_i = \rho_{i,p}(\theta), \qquad m_i = \sum_t W_{i,t}(p) \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t}).$$

Here ρ_i is the sequence aggregate ratio and m_i its weighted mean token score. Both stay fixed because we change only the rewards. To express sequence clipping with $0 < \epsilon < 1$, define

$$\phi_\rho(a) = \begin{cases} \max\{a, 0\}, & \rho \le 1 - \epsilon, \\ a, & 1 - \epsilon < \rho < 1 + \epsilon, \\ \min\{a, 0\}, & \rho \ge 1 + \epsilon. \end{cases} \tag{100}$$

The function ϕ_ρ keeps or removes the advantage according to the clipping branch. Away from a threshold, the sequence's clipped gradient is $\rho\phi_\rho(a)m$. At a threshold we use the clipped-side subgradient for the affected sign, with the same convention in both updates. As usual, advantages stay fixed when differentiating with respect to θ. The key property is that ϕ_ρ cannot enlarge a change in its input advantage: it is 1-Lipschitz. This gives the following bound even when the advantage sign changes.

Proposition G2 (Reward perturbations of a clipped group update)**.** *Define the raw clipped surrogate-gradient contributions*

$$H_p = \frac{1}{G}\sum_{i=1}^{G} \rho_i \phi_{\rho_i}(\widehat{A}_i) m_i, \qquad \widetilde{H}_p = \frac{1}{G}\sum_{i=1}^{G} \rho_i \phi_{\rho_i}(\widetilde{A}_i) m_i.$$

For arbitrary real advantages,

$$\|\widetilde{H}_p - H_p\| \le \frac{1}{G}\sum_{i=1}^{G} \rho_i \|m_i\| \, |\widetilde{A}_i - \widehat{A}_i| \tag{101}$$

$$\le \frac{1}{G}\left(\sum_{i=1}^{G} \rho_i^2 \|m_i\|^2\right)^{1/2} \|\widetilde{A} - \widehat{A}\|_2. \tag{102}$$

The same bounds hold for the unclipped update, with $\phi_\rho(a) = a$.

Suppose additionally that both advantage vectors satisfy

$$\|\widehat{A}\|_2 \le \sqrt{G}, \qquad \|\widetilde{A}\|_2 \le \sqrt{G}. \tag{103}$$

Let $\mathcal{E}$ be the event that at least one reward in the group is corrupted, with $\widetilde{A} = \widehat{A}$ off $\mathcal{E}$. If, on $\mathcal{E}$, $\rho_i \le R_{\max}$ and $\|m_i\| \le M$ for every i, then

$$\|\widetilde{H}_p - H_p\| \le 2R_{\max} M \mathbf{1}_{\mathcal{E}}. \tag{104}$$

Consequently, whenever the expected updates exist and $\Pr(\mathcal{E}) \le \varepsilon_{\mathrm{g}}$,

$$\|\mathbb{E}\widetilde{H}_p - \mathbb{E}H_p\| \le 2R_{\max} M \varepsilon_{\mathrm{g}}, \tag{105}$$

$$\mathbb{E}\|\widetilde{H}_p - H_p\|^2 \le 4R_{\max}^2 M^2 \varepsilon_{\mathrm{g}}. \tag{106}$$

No independence of the errors, rollouts, or token contributions is needed for these inequalities.

Proof. For every fixed ρ, the map ϕ_ρ is the identity, the positive-part map, or the negative-part map. Each is 1-Lipschitz, so

$$|\phi_\rho(a) - \phi_\rho(b)| \le |a - b|.$$

Subtracting the two updates and applying the triangle inequality therefore gives Eq. (101). Cauchy–Schwarz gives Eq. (102). This argument allows an advantage sign change to switch a clipping branch.

Under Eq. (103), $\|\widetilde{A} - \widehat{A}\|_2 \le 2\sqrt{G}$. On $\mathcal{E}$, the first factor in Eq. (102) is at most $\sqrt{G}R_{\max}M/G$. Off $\mathcal{E}$, the update difference is zero. These observations prove Eq. (104). Taking first and second moments gives the remaining bounds, using $\|\mathbb{E}(\widetilde{H}_p - H_p)\| \le \mathbb{E}\|\widetilde{H}_p - H_p\|$. □

The advantage bound from group normalisation. Condition (103) follows from the usual group normalisation. Let $\overline{R}$ be the group mean and s_R the population standard deviation:

$$\overline{R} = \frac{1}{G}\sum_i R_i, \qquad s_R^2 = \frac{1}{G}\sum_i (R_i - \overline{R})^2, \qquad \widehat{A}_i = \frac{R_i - \overline{R}}{s_R + \varepsilon_A},$$

Here $\varepsilon_A \geq 0$ is the stabilising constant; assign zero advantages if the denominator is zero. Otherwise,

$$\|\widehat{A}\|_2^2 = \frac{G s_R^2}{(s_R + \varepsilon_A)^2} \leq G$$

Thus the squared advantages sum to at most G, including the zero-group case. The same holds for corrupted rewards. More generally, any denominator at least as large as s_R gives this bound, without requiring bounded rewards. An additional rescaling of the advantages changes the constant accordingly.

Using the perturbation bounds. Equation (101) gives the error for the observed ratios, scores, and advantage changes. To obtain the explicit probability bounds in Eqs. (105)–(106), the proposition additionally bounds ρ_i by $R_{\max}$ and $\|m_i\|$ by M on corrupted groups. These bounds must be supplied separately: clipping can leave a large-ratio rollout active when its advantage is negative. Alternatively, control the required moments of the weighted expression in Eq. (101) directly.

The proposition bounds the effect of a corrupted group, rather than making the error proportional to the raw reward change. Such proportional control needs a lower bound on the normalisation denominator. For example, without a stabiliser, $(u, -u)$ and $(-u, u)$ have opposite normalised advantages for every $u > 0$, although the reward difference tends to zero with u.

Let ε_{r} bound the error probability of each rollout reward. The probability ε_{g} that at least one reward in the group is corrupted satisfies

$$\varepsilon_{\mathrm{g}} \leq \min\{1, G\varepsilon_{\mathrm{r}}\}.$$

This union bound allows dependent errors. The relevant event is corruption of the group, because one wrong reward can change all its advantages; averaging therefore does not automatically reduce its effect by $1/G$. The bounds apply to the raw clipped gradient. A shared fixed additive regulariser cancels in the comparison, while a subsequent nonlinear optimiser transformation requires a bound for that transformation.

Effect on improvement and alignment. Let $\Delta H = \widetilde{H} - H$ and suppose its mean error is at most b_p, its root second moment at most e_p, and the clean update's root second moment at most B_p. Subtraction of Eq. (51) and Cauchy–Schwarz give

$$|U(\widetilde{H}) - U(H)| \leq \eta\|v\|b_p + L\eta^2 B_p e_p + \tfrac{L\eta^2}{2} e_p^2. \tag{107}$$

The mean error also adds at most $\|v\|b_p$ to the alignment error in Appendix N, when the bounds hold conditionally on the training history. This is a consequence of the update bounds, not an additional corruption-detection guarantee.

Relation to the experiments. Our supplementary experiment replaces 10% of binary rollout rewards with uniformly sampled binary values before advantage normalisation, producing approximately 5% actual label flips. At step 500, the corrupted-reward and clean-reward runs achieve average scores of 54.4% and 54.8%, respectively, when evaluated with the clean verifier. This provides evidence of tolerance to the tested random reward corruption and motivates the group-level perturbation analysis here.

R Quantitative Advantage of Dynamic Scheduling

Remark 2. *The pre-saturation condition $r_{i,t^*}^{p_{\mathrm{high}}} \ll n - 1$ keeps the high-ratio token's amplified mass below the background total. It defines the regime where the stated multiplicative gain is informative; long sequences alone do not verify the condition. The endpoint comparison holds at one policy state and does not identify when saturation occurs in training.*

Proof of Theorem 3. Let $R \coloneqq r_{i,t^*} \gg 1$. For the remaining tokens $t \neq t^*$, since their ratios are constant-bounded, we can denote their sum of p-th powers as $S(p) \coloneqq \sum_{t \neq t^*} r_{i,t}^p = \Theta(n-1)$, which holds uniformly for any p in a bounded interval $[p_{\text{low}}, p_{\text{high}}]$. By definition, the weight of the high-ratio token is:

$$W_{i,t^*}(p) \;=\; \frac{R^p}{R^p + S(p)}.$$

Therefore, the relative amplification of the gradient weight when shifting from p_{stat} to p_{high} is given by:

$$\begin{aligned}\frac{W_{i,t^*}(p_{\text{high}})}{W_{i,t^*}(p_{\text{stat}})} \;&=\; \frac{R^{p_{\text{high}}}}{R^{p_{\text{high}}} + S(p_{\text{high}})} \cdot \frac{R^{p_{\text{stat}}} + S(p_{\text{stat}})}{R^{p_{\text{stat}}}} \\ &=\; R^{\,p_{\text{high}} - p_{\text{stat}}} \cdot \frac{R^{p_{\text{stat}}} + S(p_{\text{stat}})}{R^{p_{\text{high}}} + S(p_{\text{high}})}.\end{aligned}$$

Under the pre-saturation condition $R^{p_{\text{high}}} \ll n-1$, the term $R^{p_{\text{high}}}$ is asymptotically dominated by the denominator's background sum $S(p_{\text{high}}) = \Theta(n-1)$. Since $p_{\text{stat}} < p_{\text{high}}$, we naturally also have $R^{p_{\text{stat}}} \ll n-1$. Consequently, the fractional multiplier is bounded from below by a strictly positive constant $C = \Theta(1)$:

$$\frac{R^{p_{\text{stat}}} + S(p_{\text{stat}})}{R^{p_{\text{high}}} + S(p_{\text{high}})} \;\geq\; \frac{S(p_{\text{stat}})}{R^{p_{\text{high}}} + S(p_{\text{high}})} \;\geq\; C \;>\; 0.$$

Substituting $R = r_{i,t^*}$ back into the expression yields the desired exponential lower bound for the signal amplification:

$$\frac{W_{i,t^*}(p_{\text{high}})}{W_{i,t^*}(p_{\text{stat}})} \;\geq\; C \cdot r_{i,t^*}^{\,p_{\text{high}} - p_{\text{stat}}}.$$

For the second claim, $V(p) = \mathbb{E}[\widehat{A}_i^2 \rho_{i,p}^2]$. The generalised-mean inequality gives $\rho_{i,p_{\text{low}}} \leq \rho_{i,p_{\text{stat}}}$ pointwise when $p_{\text{low}} \leq p_{\text{stat}}$. Squaring, multiplying by $\widehat{A}_i^2 \geq 0$, and taking expectations yields

$$V(p_{\text{low}}) \leq V(p_{\text{stat}}).$$

The weak inequality includes equal orders, constant ratios, and zero advantages, as allowed in the theorem. □

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